

Luminex

Field Service Manual

Luminex[®] 200[™] RoHS



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Luminex® 200™ RoHS Field Service Manual

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Table of Contents

Chapter 1: System Introduction

System Overview	1
Software Overview	1
Symbols Glossary	2

Chapter 2: Regulatory and Safety Considerations

Intended Use	5
Safety Testing and Certifications	5
Regulatory and Safety Labels	5

Chapter 3: Performance Specifications and System Components

Installation Diagram	7
General System Workflow	8
Luminex® 200™ RoHS Environmental Conditions	8
Performance Specifications	8
Luminex® 200™ RoHS Specifications	8
Luminex® XYP™ Environmental Conditions	10
Luminex® XYP™ Specifications	10
Power Specifications	10
Electronics Specifications	11
Luminex® SD™ Environmental Conditions	11
Luminex® SD™ System Specifications	11
System Components	11
Hardware Components	11
Consumable Components	12
Recommended Additional Equipment	12
Laser Servicing Information	13
Schedule of Maintenance	13
Controls Increasing the Level of Accessible Radiation	13
Removable Covers of the Luminex® 200™ RoHS Analyzer	13
Biohazard Cautions	14
Reference Documents	14
Reference Diagrams	14
Luminex® 200™ RoHS Analyzer Compartments	14

Chapter 4: Installing the System

Assemble the System	26
Connect the Luminex 200 RoHS Analyzer and the Luminex XYP	27
Connect the Sheath and Waste Bottles	28
Install the SD System	28
Install the Luminex 200 RoHS Analyzer Sample Probe	31
Adjust the Luminex 200 RoHS Sample Probe Height	32
Install the XYP Reservoir	32

Install the XYP Heater Block	34
Power on System Components	34
Accept the Luminex® xPONENT® Software License Agreement	35
Start Up	35
Shut Down System Components	35

Chapter 5: Maintaining the System

Preventative Maintenance	36
Semi-Annual Maintenance (6-Month PM Kit)	36
Replace the Syringe	37
Luminex® 200™ Analyzer Ventilation Filter	39
Annual Maintenance (12-Month PM Kit)	39

Chapter 6: Service Procedures

Align the Optics for the Analyzer	41
Adjust Flow Rate	46
Remove and Replace Procedures	46
Luminex® 200™ RoHS Analyzer Cover	47
Luminex® 200™ RoHS Bezel	48
Luminex® 200™ RoHS Air Filter (HEPA)	49
Luminex® 200™ RoHS Fuses	50
Luminex® 200™ RoHS AC Power Inlet	51
Luminex® 200™ RoHS Power Supply	52
Luminex® 200™ RoHS Air Compressor	53
Luminex® 200™ RoHS Sensor Distribution Board	55
Luminex® 200™ RoHS Laser Driver Board	56
Luminex 200 RoHS Digital Board	58
Luminex® 200™ RoHS Analog Board	60
Luminex® 200™ RoHS Green Coherent Laser Head	61
Luminex® 200™ RoHS Green Coherent Laser Controller	63
Luminex® 200™ RoHS Optical Platform	64
Luminex® 200™ RoHS Backflush 3-Way Valve	65
Luminex® 200™ RoHS Sheath Fluid Filter	67
Luminex® 200™ RoHS Bubble Trap	68
Luminex® 200™ RoHS Debubbler Valve	69
Luminex® 200™ RoHS Sheath Valve	70
Luminex® 200™ RoHS Bubble Trap Optical Sensor	71
Luminex® 200™ RoHS Pressure Transducer	72
Luminex® 200™ RoHS Pressure Regulator	73
Luminex® 200™ RoHS Check Valves	74
Luminex® 200™ RoHS Syringe Pump	77
Luminex® 200™ RoHS Syringe	78
Luminex® 200™ RoHS Blue LED Assembly	79
Luminex® 200™ RoHS Pickup Arm Position Sensor	80
Luminex® 200™ RoHS Sample 3-Way Valve	81
Luminex® 200™ RoHS Air Solenoid	82
Luminex® 200™ RoHS Bimba Air Cylinder	83
Luminex® 200™ RoHS Red Laser	84

Table of Contents

Luminex® 200™ RoHS L-Block..... 85

Luminex® XYP™ Cover 86

Luminex® XYP™ Fuses 87

Luminex® XYP™ AC Power Inlet..... 88

Luminex® XYP™ Power Supply..... 89

Luminex® XYP™ Air Intake Fan..... 90

Luminex® XYP™ Exhaust Fan..... 91

Luminex® XYP™ Controller Board..... 92

Luminex® XYP™ Top Heater 93

Luminex® XYP™ LED Board..... 94

Luminex® XYP™ Motor Driver Board.....95

Luminex® XYP™ Inner Chassis..... 96

Luminex® SD™98

Remove and Replace the AC Power Inlet.....104

Remove and Replace the Controller PCBA..... 108

Remove and Replace the Sheath Pump 109

Remove and Replace the Sight Glass..... 111

Remove and Replace the Level Sensors 112

Recommended Tools114

Chapter 7: Final Testing Procedures

Perform Final Test After a Service Call118

Calibrate and Verify the System..... 119

Chapter 8: Troubleshooting

Generate and Send the Support Utility File.....120

Luminex® 200™ RoHS General Troubleshooting120

Luminex® XYP™ Troubleshooting.....127

Luminex® 200™ RoHS Installation Troubleshooting130

 Luminex® 200™ RoHS Does Not Connect to LxMaint After
New Driver Installation130

 Mitigation for COM Port Number Issue.....130

 Debug an Luminex® 200™ RoHS XYP Connection Issue . . .
131

 Updated Luminex® 200™ RoHS (3-Port Communications)
Does Not Connect to LxMaint or xPONENT® Upon Initial Instal-
lation..... 131

 Updated Luminex® 200™ RoHS (3-Port Communications)
Does Not Connect to LxMaint or xPONENT® After New Driver
Installation 131

 Mitigation for Windows Driver Issue.....131

 Mitigation for Windows Driver Installation on XP OS..... 132

Luminex SDS Troubleshooting133

 Audible Alarm and Fault LED 133

 Drain the Luminex® SD™ Reservoir..... 133

Chapter 9: Field Definitions

Field Definitions.....134

Statistics Definitions..... 135

Statistics Column Definitions136

Chapter 10: Decontaminate the System

Decontaminate the System 138

Chapter 11: Fixture Storage and Shipping

Storage Conditions 139

Shipping 139

Chapter 12: Checklist

Install checklist..... 140

PM Checklist 140

Service Checklist140

Chapter 1: System Introduction

This manual provides information on setting up, troubleshooting and servicing the Luminex® 200™ RoHS system. Chapters will walk you through complete system operation, with procedures presented as step-by-step instructions and examples given as needed.

System Overview

The Luminex® 200™ RoHS system is a compact analysis unit consisting of an analyzer, a computer, software, and reagents. The analyzer integrates optics, fluidics, and advanced signal processing in a 17-inch by 20-inch footprint. The computer uses Luminex software and proprietary hardware to control the analyzer.

Photodiodes and a photomultiplier tube receive fluorescent signals from Luminex microspheres. The analyzer digitizes the waveforms and delivers the signals to a digital signal processor (DSP). Proprietary algorithms work with the DSP to greatly increase the sensitivity.

The Luminex® XYP™ is intended to be used only with the Luminex 200 RoHS analyzer and current software. With the Luminex XYP, a 96-well microtiter plate can be processed without having to handle each sample individually. The Luminex XYP works with 96-well plates that are no thicker than 0.75 inches (19 mm).

The Luminex® SD™ system is also intended to be used exclusively with the Luminex 200 RoHS analyzer and current software. It lets users run samples continuously without refilling sheath bottles. The Luminex SD automatically draws sheath fluid from a non-pressurized bulk sheath container to constantly maintain a reservoir of pressurized sheath fluid.

Software Overview

xPONENT® software provides complete control of the Luminex® 200™ RoHS system and performs the data analysis. The software requires a dedicated PC. The PC accompanying your system is preloaded with xPONENT software; however, Luminex supplies a software DVD if you need to reinstall the software. Unauthorized additional software is prohibited and can result in improper operation of the system.

Symbols Glossary

You will encounter these symbols throughout this manual. They represent warnings, conditions, identifications, instructions, and regulatory agencies.

Symbol	Description	Symbol	Description	Symbol	Description
† 	Burn Hazard / Hot Surface	† 	Puncture/Pinch Point Warning	† 	Hand Crush/Force From Above
5019* 	Protective earth; protective ground To identify any terminal which is intended for connection to an external conductor for protection against electric shock in case of a fault, or the terminal of a protective earth (ground) electrode.	5.1.1§ 	Manufacturer / Date of manufacture Indicates the medical device manufacturer, as defined in EU Directives 90/385/EEC, 93/42/EEC and 98/79/EC. Indicates the date when the medical device was manufactured.	0434B ‡ 	Caution To indicate that caution is necessary when operating the device or control close to where the symbol is placed, or to indicate that the current situation needs operator awareness or operator action in order to avoid undesirable consequences.
5.4.3§ 	Consult instructions for use Indicates the need for the user to consult the instructions for use.	5.4.1§ 	Biological risks Indicates that there are potential biological risks associated with the medical device.	5041 † 	Caution, hot surface To indicate that the marked item can be hot and should not be touched without taking care.
 	Waste Electrical and Electronic Equipment (WEEE) Separate collection for electrical and electronic equipment	5.1.5§ 	Batch Code Indicates the manufacturer's batch code so that the batch or lot can be identified.	5032* 	Alternating Current To indicate on the rating plate that the equipment is suitable for alternating current only; to identify relevant terminals.
5.5.1§ 	<i>In vitro</i> diagnostic medical device Indicates a medical device that is intended to be used as an in vitro diagnostic medical device.	5.1.6§ 	Catalog(ue) Number Indicates the manufacturer's catalogue number so that the medical device can be identified.	5.1.7§ 	Serial number Indicates the manufacturer's serial number so that a specific medical device can be identified.
	UL Mark Representative product samples meet requirement set by UL for electrical safety.	5.1.4§ 	Use-by date Indicates the date after which the medical device is not to be used.	W004†† 	Warning; Laser Beam To warn of a laser beam

Symbol	Description	Symbol	Description	Symbol	Description
5009† 	Stand-by Power on/off To identify the switch or switch position by means of which part of the equipment is switched on in order to bring it into the stand-by condition, and to identify the control to shift to or to indicate the state of low power consumption. Each of different states of power consumption may be indicated using a corresponding color.	** 	Ionizing Radiation Shall be used to signify the actual or potential presence of ionizing radiation (including gamma and X-rays, alpha and beta particles, high-speed electrons, neutrons, protons and other nuclear particles, but not sound waves and other types of electromagnetic waves).	‡ 	TÜV SÜD NRTL Certification Mark TÜV SÜD America is an OSHA-recognized Nationally Recognized Testing Laboratories (NRTL) capable of providing electrical safety certification services to North American requirements for Medical Devices, and Laboratory Test & Measurement Equipment.
# 	Conformite Europeenne (EU CE Marking of Conformity) CE conformity marking	5.3.7§ 	Temperature Limit Indicates the temperature limits to which the medical device can be safely exposed.		

* ANSI/AAMI/ISO 15223-1:2016, Medical devices—Symbols to be used with medical device labels, labeling, and information to be supplied—Part 1: General requirements.

† IEC 60417:2002 DB, graphical symbols for use on equipment. (General I (QS/RM))

‡ ISO 7000: Fifth edition 2014-01-15, graphical symbols for use on equipment - registered symbols. (General I (QS/RM))

§ Council Directive 98/79/EC on In Vitro Diagnostic Medical Devices (IVDMD) (1998)

|| 61010-1: 2010, Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 1: General requirements [including: Corrigendum 1 (2011)]

IEC 60825-1:2014 Safety of Laser Products –Part 1: Equipment classification and requirements

** ISO 3864-1:2011, Graphical symbols -- Safety colors and safety signs -- Part 1: Design principles for safety signs and safety markings

‡‡ DIRECTIVE 2012/19/EU OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 4 July 2012 on waste electrical and electronic equipment (WEEE)

TABLE 1. Common Abbreviations

A	ampere	LX200	Luminex 200 RoHS
ac	alternating current	m	meters
A/D	analog/digital	M	male
Al	aluminum	mA	milliampere
Amp	ampere	max	maximum
APD	avalanche photo diode	ml	Milliliters
C	Celsius	mm	millimeters
CD	compact disc	MP	microtiter plate
CL1	classification 1 channel	ms	millisecond
CL2	classification 2 channel	mW	milliwatts
cm	centimeters	nm	nanometers
CV	coefficient of variance	OD	outer diameter
CW	continuous wave	PC	personal computer
D	deep	PCB	printed circuit board
DB	connector type	PCBA	printed circuit board assembly
dB	decibels	PMT	photomultiplier tube
dc	direct current	PN	part number
DD	doublet discriminator	psi	pounds per square inch
DSP	digital signal processor	QTY	quantity
F	Fahrenheit	RP1	reporter channel
F	female	RS	connector type (recommended standard)
fl	focal length	RTD	resistance temperature detector
ft	feet	SS	stainless steel
ga	gauge	Temp	temperature
H	high	TSB	technical service bulletin
HEPA	high efficiency particulate arrestor	μL	microliters
HV	high voltage	UL	Underwriters' Laboratories
Hz	Hertz	V	volts (dc)
ID	inner diameter	V~	volts (ac)
Kg	kilograms	W	wide
L	long	w/	with
lbs	pounds	w/o	without
lg	large	XYP	Luminex® XY Platform
LED	light emitting diode		

Chapter 2: Regulatory and Safety Considerations

- Safety**
- The non-observance of safety instructions may result in serious personal injuries and material damages.
- Always wear personal protective equipment suitable for the activity.
 - Follow all warnings marked on the instrument.
 - Follow all biological safety instructions
 - Follow all safety, warnings, cautions, and notes

Intended Use

The Luminex® 200™ instrument is a clinical multiplex test system intended to measure and sort multiple signals generated in an In Vitro Diagnostic assay from a clinical sample. This instrumentation is used with a specific assay to measure multiple similar analytes that establish a single indicator to aid in diagnosis. The device includes a signal reader unit, raw data storage mechanisms, data acquisition software and software to process detected signals.

Safety Testing and Certifications

The Luminex® 200™ has been tested and complies with the safety requirements for the United States and Canada and is marked with the MET label or the TUV label.

The Luminex 200 RoHS complies with the European Union (EU) safety requirements and therefore may be marketed in the Europe Single Market. The European Union compliance label appears on the back of the Luminex 200 RoHS instrument.

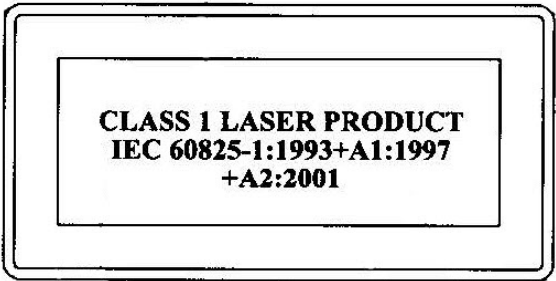
Regulatory and Safety Labels

The following labels appear on the back of the Luminex® 200™ instrument.

FIGURE 1. Laser Warning Label

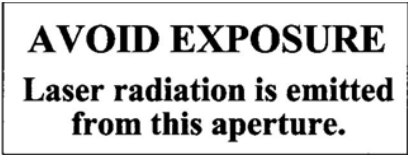


FIGURE 2. Laser Class Label



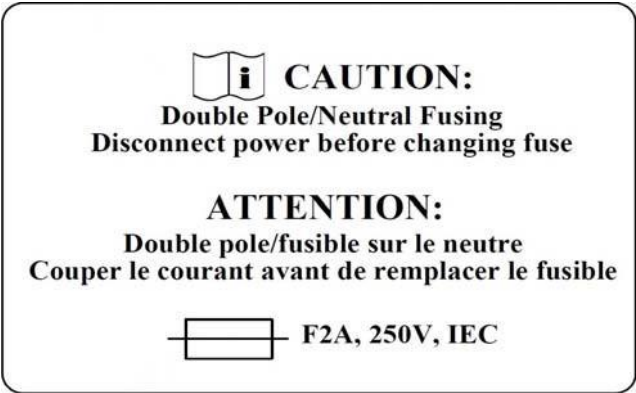
The following label appears above the laser apertures located inside the optics enclosure inside the Luminex 200.

FIGURE 3. Avoid Exposure Label



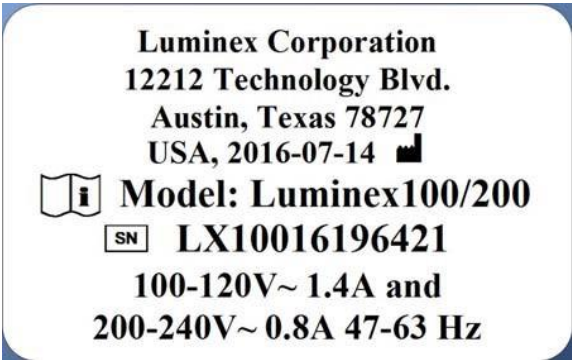
The following fuse caution label appears on the back of the Luminex 200.

FIGURE 4. Fuse Caution Label



The following voltage label appears on the back of the Luminex 200 instrument. The label displays the Luminex 200 serial number, model number, power requirements, and manufacturer's information.

FIGURE 5. Serial Number and Voltage Label



Chapter 3: Performance Specifications and System Components

Installation Diagram

Notes:

1. The Luminex 200 System consists of 5 major components:

- 2 Clearance for cooling and intake/exhaust, and cable clearance for power switch access.

3. Total countertop system weight: 168.67 lb. (76.67 kg.) (sheath fluid box on floor, not shown 46.45 lb. (21.1 kg.).

- Total system weight 215.12 lb. (97.78 kg.).

4. Line cords should be plugged into outlets with protective earthing only (reference user manual). 

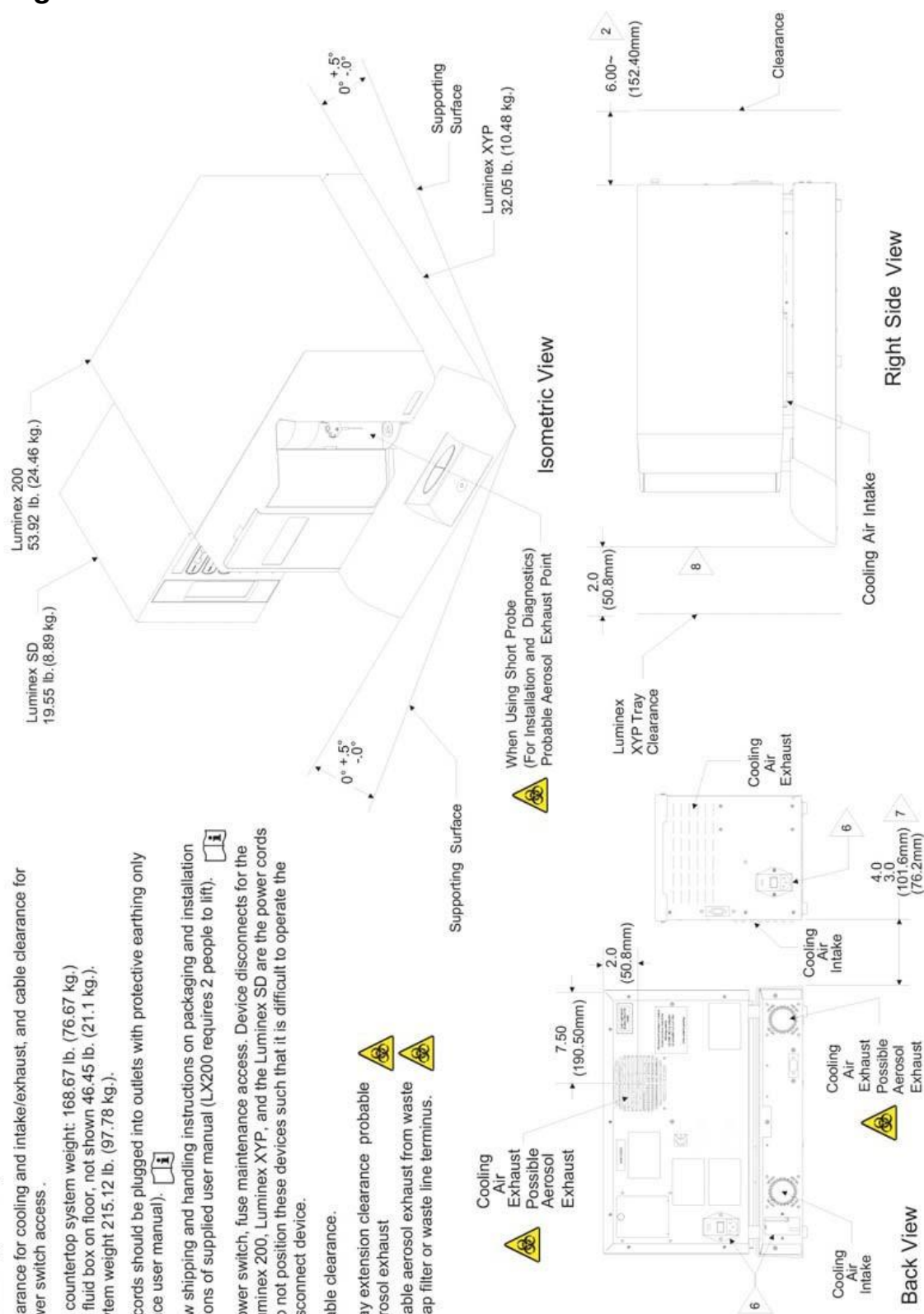
5. Follow shipping and handling instructions on packaging and installation instructions of supplied user manual (LX200 requires 2 people to lift).

- Power switch, fuse maintenance access. Device disconnects for the Luminex 200, Luminex XYP, and the Luminex SD are the power cords. Do not position these devices such that it is difficult to operate the disconnect device.

- 7 Cable clearance.

- 8 Tray extension clearance probable aerosol exhaust

9. Probable aerosol exhaust from waste bottle cap filter or waste line terminus.



General System Workflow

The Luminex® 200™ RoHS system analyzes immunoassays, complex genetic analyses, and enzymatic assays in one simple format. The reactants (antibodies, oligonucleotides, substrates, etc.) of these assays are anchored to the surfaces of uniquely fluorescent microscopic beads (microspheres). The unique fluorescent emission spectra of a given microsphere identifies each of the assays performed simultaneously on a single sample.

The Luminex 200 RoHS analyzes these microspheres individually in a flow stream. Every microsphere is accurately classified to its own subset, based on its fluorescent signature. In addition, the Luminex 200 RoHS scans each microsphere for the presence of a reporter fluorescence that quantifies the assay at the bead's surface.

Luminex® 200™ RoHS Environmental Conditions

- Indoor use only
- Operating temperature: 15°C to 30°C (59°F to 86°F)
- Humidity: 20% to 80%, non-condensing
- Altitude: Operation up to 2400 m (7874 ft.) above mean sea level
- Shipping and storage: The allowable shipping and storage temperature and humidity ranges are 0°C to + 50°C and 20% to 80% non-condensing, respectively

Performance Specifications

Luminex® 200™ RoHS Specifications

General System Specifications

- Physical dimensions: 43 cm (17 inches) W x 50.5 cm (20 inches) D x 24.5 cm (9.5 inches) H
- Weight: maximum of 25 kg (60 lbs.)
- UL installation category: UL Installation Category II, as defined in Annex J of UL 61010A-1
- Pollution degree: UL Pollution Degree 2, as defined in Section 3.7.3.2 of UL 61010A-1
- Input voltage range: 100 to 120 V~ ± 10%, 1.4 Amp, and 200 to 240 V~ ± 10%, 0.8 Amp, 47-63 Hz.
- AC inlet fuse: 3 Amp, 250 V~, Fast Acting

Accuracy and Precision Specifications

- Sample uptake volume: ± 5%
- Classification of xMAP® microspheres: > 80%
- Misclassification of xMAP microspheres: ≤ 2% - can vary by xMAP microsphere product lines. Refer to the specific product information sheet for further details.
- Temperature control: 0°C to +2°C of target
- Internal sample carry over: < 0.9%
- Soluble background fluorescence emission at 575 nm automatically subtracted from fluorescence intensity values

PC Specifications

xPonent 4.2 PC Minimum Specifications

- Windows® 7 Professional SP1 U.S. English 32-bit or 64-bit, or Windows XP Professional SP3 U.S. English 32-bit operating system
- 2.66 GHz Core 2 Duo with 4 GB RAM
- 80 GB hard drive
- System Monitor with 1280 x 1024 screen resolution
- USB communications link for fast data transfer
- Automatic transfer of assay templates and new reagent information into the system via large capacity read/write CD

- Installation: < 4 hours
- System calibration: < 10 minutes
- System controls: < 10 minutes
- Barcode reader entry of sample IDs
- Automatic post-analysis
- Analyze one 96-well plate/hour depending on manufacturer's kit
- Up to 100 xMAP® microsphere sets per sample
- Sheath flow rate: 90 µL/sec ± 5 µL
- Sample injection rate into detector area: 1 µL/sec ± 0.05 µL
- System warm up: 30 minutes. Systems that remain inactive for at least four hours will require warm up to restart the lasers. After acquiring samples, running system calibrators, running system controls, and warming up the analyzer, the system resets the four-hour internal clock.

xPonent 4.3 PC Minimum Specifications

- Windows® 10, 64-bit, Version 1709
- 3.0 GHz Intel Core i5
- 8 GB RAM
- 1 TB hard drive space
- Display Settings:
 - SXGA 1280x1024 with 32 bit color
 - 96 DPI font
 - Default Windows 10 theme

Sensitivity Specifications

- Detect 1000 fluorochromes phycoerythrin (PE) per xMAP® microsphere
- Reporter channel dynamic range: 3.5 decades of detection

Power Specifications

- Input voltage range: 100 V to 240 V~, 1.5 Amps, 47 Hz to 63 Hz

Optics Specifications

- Reporter laser: 532 nm, maximum output of 500 mW, system specifications of 12.08 mW to 14.47 mW, frequency-doubled diode; mode of operation, continuous wave (CW)
- Classification laser: 635 nm, maximum output of 25 mW, system specifications of 7.5 mW to 9.5 mW, diode; mode of operation, continuous wave (CW)
- Reporter detector: Photomultiplier tube, detection bandwidth of 565 nm to 585 nm
- Classification detector: Avalanche photo diodes with temperature compensation
- Doublet discrimination detector: Avalanche photo diodes with temperature compensation

Fluidics Specifications

- Sheath flow rate 90 µL ± 5 µL/second
- Cuvette: 200 micron square flow channel
- Sample injection rate: 1 µL/second ± 0.05 µL
- Sample uptake volume: 20 µL to 200 µL



Use of non-Luminex® approved sheath fluid shall constitute “improper use” and may void warranty rights provided by Luminex and/or its authorized partner.

Electronics Specifications

- Reporter channel detection: A/D resolution 14 bits
- Communications interface: DB9 (Analyzer P1) to USB (PC)

Signal Processing

- Signal modes: linear, with logarithmic or linear display option
- Dynamic range: 70 dB

Operating Specifications

- Air pressure: 7.5 ± 1.5 PSI after stabilization (6-9 PSI)
- Sheath fluid pressure: 7.5 ± 1.5 PSI after stabilization (6-9 PSI)
- Ambient Temperature (15-30°C)
- Manufacturing PSI (6-9 PSI)
- Calibration PSI (6-9 PSI)
- PSI Tolerance % (5%)
- Firmware Revision
- High voltage: 210.00 volts ± 5 v
- Green laser offset (split): 25 μ s to 50 μ s
- Flow rate: 10.2 to 11.4 mL per 2 minutes
- CVs: see *Table 2*.
- DD, CL1, and CL2 Temp: 15°C to 35°C
- DD A/D (Offset) 5-150 (Lasers on)
- CL1, CL2, and RP1 A/D (Offset) 5-35 (Lasers on)

TABLE 2. Optical Specifications

	Peak	Mean	CVs
DD	8500 - 13500	N/A	< 16.0%
CL1	N/A	$\pm 4\%$ of target	< 6.0%
CL2	N/A	$\pm 4\%$ of target	< 7.5%
RP1	N/A	$\pm 4\%$ of target	< 10.0%

NOTE: Use DD, CL1, and CL2 spec with L100-CAL1 beads. Use RP1 spec with L100 - CAL2 beads, as indicated on the Certificate of Quality. Peak values are based on a DD target value of 10,000. The peak values should be compensated, should the DD target value change.

Luminex® XYP™ Environmental Conditions

- Operating temperature: 15°C to 30°C (59°F to 86°F)
- Humidity: 20% to 80%, non-condensing
- Altitude: designed to operate at 2400 m (7874 ft.) or less above mean sea level

Luminex® XYP™ Specifications

General Specifications

- Physical dimensions: 44 cm (17.25 inches) wide, 60 cm (23.5 inches) deep, 8 cm (3 inches) high
- Weight: 15 kg (33lbs)
- UL installation category: UL Installation Category II, as defined in Annex J of UL 3101-1
- Pollution degree: UL Pollution Degree 2, as defined in Section 3.7.3.2 of UL 3101-1
- Heater operating range: 35°C to 55°C (95°F to 131°F) with a tolerance of 0°C to +2°C

Power Specifications

- Input voltage range: 100 V to 240 V~, 1.8 Amps, 47 Hz to 63 Hz

Electronics Specifications

- Communications interface: DB9 (Analyzer P3) to DB9 (XYP P1)

Luminex® SD™ Environmental Conditions

- Ambient temperature: 15°C to 30°C (59°F to 86°F)
- Humidity: 20% to 80%, non-condensing
- Altitude: designed to operate at up to 2400m (7874 feet) above mean sea level

Luminex® SD™ System Specifications

General Specifications

- Physical dimensions: 20 cm (8 inches) W x 30 cm (11.75 inches) D x 24.75 cm (9.75 inches) H
- Weight: 9 kg (20 lbs)
- UL installation category: UL Installation Category II, as defined in Annex J of UL 61010A-1
- Pollution degree: UL Pollution Degree 2, as defined in Section 3.7.3.2 of UL 61010A-1

Power Specifications

- Input voltage range: 100 V to 240 V~ ± 10%, 0.4 Amps, 47 Hz to 63 Hz
- AC inlet fuse: 2 Amp, 250 V~, Time Lag

Electronics Specifications

- Communications interface: DB9 (Analyzer P2) to DB9 (SDS P1)

System Components

- Luminex 200 RoHS analyzer
- xPONENT® software
- Computer
- Monitor
- Keyboard
- Mouse
- Luminex® XYP™
- Luminex® SD™
- Sheath and waste containers
- xMAP® technology reagents

Hardware Components

The Luminex® 200™ RoHS system includes the following hardware:

- Luminex® 200™ RoHS analyzer
- Computer (PC), monitor, and accessories
- Luminex® SD™
- Power cables
- Two long sample probes
- XYP Reagent Reservoir
- Probe Shield
- Heater block
- Sheath Fluid container

- Empty Sheath container
- Waste container
- Sheath fluid line
- Air line
- Sheath fluid intake line
- Analyzer to PC Communications: RS232 to USB 2.0 communication cable
- SDS/XYP to Analyzer Communications: CANBUS communication cables
- Sample probe height alignment kit
- Automated Maintenance Plate (AMP)
- Waste bottles

Consumable Components

xMAP® Technology Reagents

- Calibration Kit (CAL)
- Verification Kit (VER)
- xMAP® Sheath Fluid

Required Laboratory Reagents

The following items are necessary to run the Luminex® 200™ RoHS system, but are not provided by Luminex.

- 10% bleach solution

NOTE: Luminex recommends 0.6% sodium hypochlorite with no additives.

- 70% isopropanol or 70% ethanol solution
- Mild detergent
- Distilled water



Isopropanol and ethanol are flammable liquids. Keep them in a well-ventilated area and away from heat, open flames, and sparks.

Remove these liquids from the system when they are not in use.

Recommended Additional Equipment

- Uninterruptible Power Supply (UPS) - Should provide at least 1050 Watts for at least 45 minutes, and be UL listed and CE marked when used internationally
- Surge Protector - Should be used if a UPS is not used. Should be six outlets, rated for 1500 Watts, UL listed, CSA Certified, and CE marked when used internationally
- Printer - HP LaserJet 2300 or available equivalent
- Vortex - Speed range 0 rpm to 3200 rpm or equivalent
- Bath Sonicator - Operating frequency 55 kHz or equivalent

Laser Servicing Information



xPONENT® software has a feature that allows you to program the analyzer to warm up at any given time. If the **Warm Up** command is sent to the analyzer while the optical cover is removed, it can create a safety hazard to the Service Engineer or other personnel in the area. To disable the software feature, either close xPONENT while servicing the optics or disable the feature in the software.

This warning does not apply to LxMaintenance since the feature does not exist in this software application.



Use of controls, adjustments, or procedures other than those specified herein may result in hazardous radiation exposure.

Schedule of Maintenance

No specific maintenance is required to keep the lasers in compliance with federal regulations.

Controls Increasing the Level of Accessible Radiation

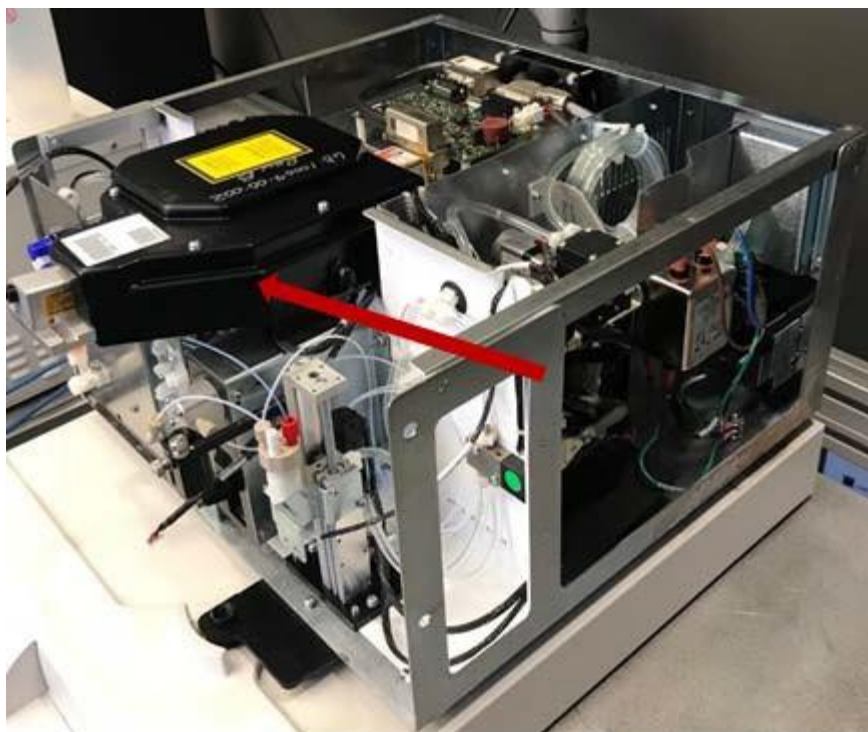
None are required.

Removable Covers of the Luminex® 200™ RoHS Analyzer

There are three removable covers of protective housing: the analyzer cover, the analyzer bezel cover, and the optics platform cover (which is exposed when the analyzer cover is removed).



FIGURE 6. Optics Platform Cover



Biohazard Cautions

Any specimen or reagent that contains human or animal blood or serum may also contain hazardous infectious agents. Always follow appropriate biosafety procedures when handling these products and containers.

Observe all local, state, and federal biohazard handling regulations when disposing of biohazardous waste material.

Observe these biohazard cautions at all times:

- Always check the unit for leaks before servicing.
- Always wear proper biohazard safety clothing, including lab coat, gloves, and safety glasses.
- Wipe down the outside of the analyzer, Luminex® XYP™, and Luminex® SD™ system with a solution of 10% to 20% bleach in water.
- Wash and sanitize the analyzer with a solution of 10% to 20% bleach in water before servicing. See the “*Maintaining the System*” for detailed information on decontaminating the analyzer.

Reference Documents

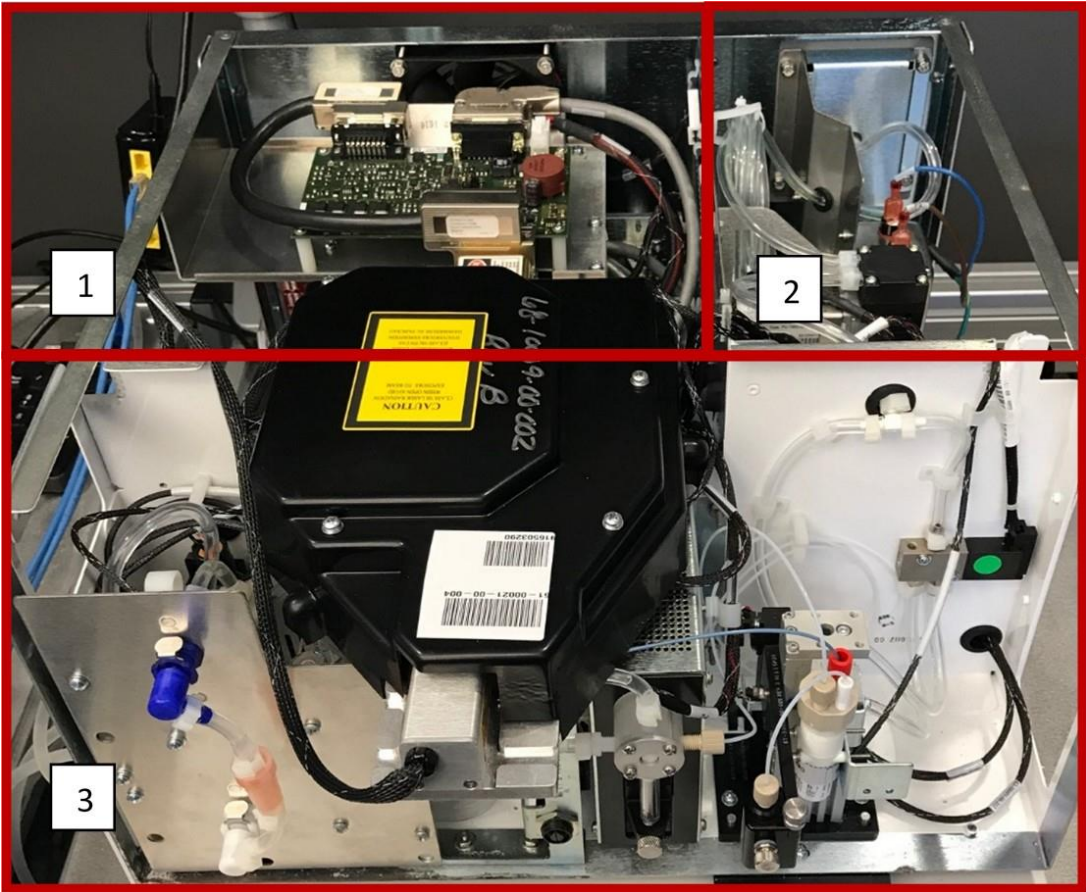
- *Inspection, Measuring and Test Equipment Operating Procedure*, Luminex® Corporation, document number 00093
- *Field Service Operating Procedure*, Luminex Corporation, document number 00014
- *Training Operating Procedure*, Luminex Corporation, document number 00091
- FDA 21 CFR Part 820.000
- ISO 9001:2000 Quality management systems - Requirements

Reference Diagrams

Luminex® 200™ RoHS Analyzer Compartments

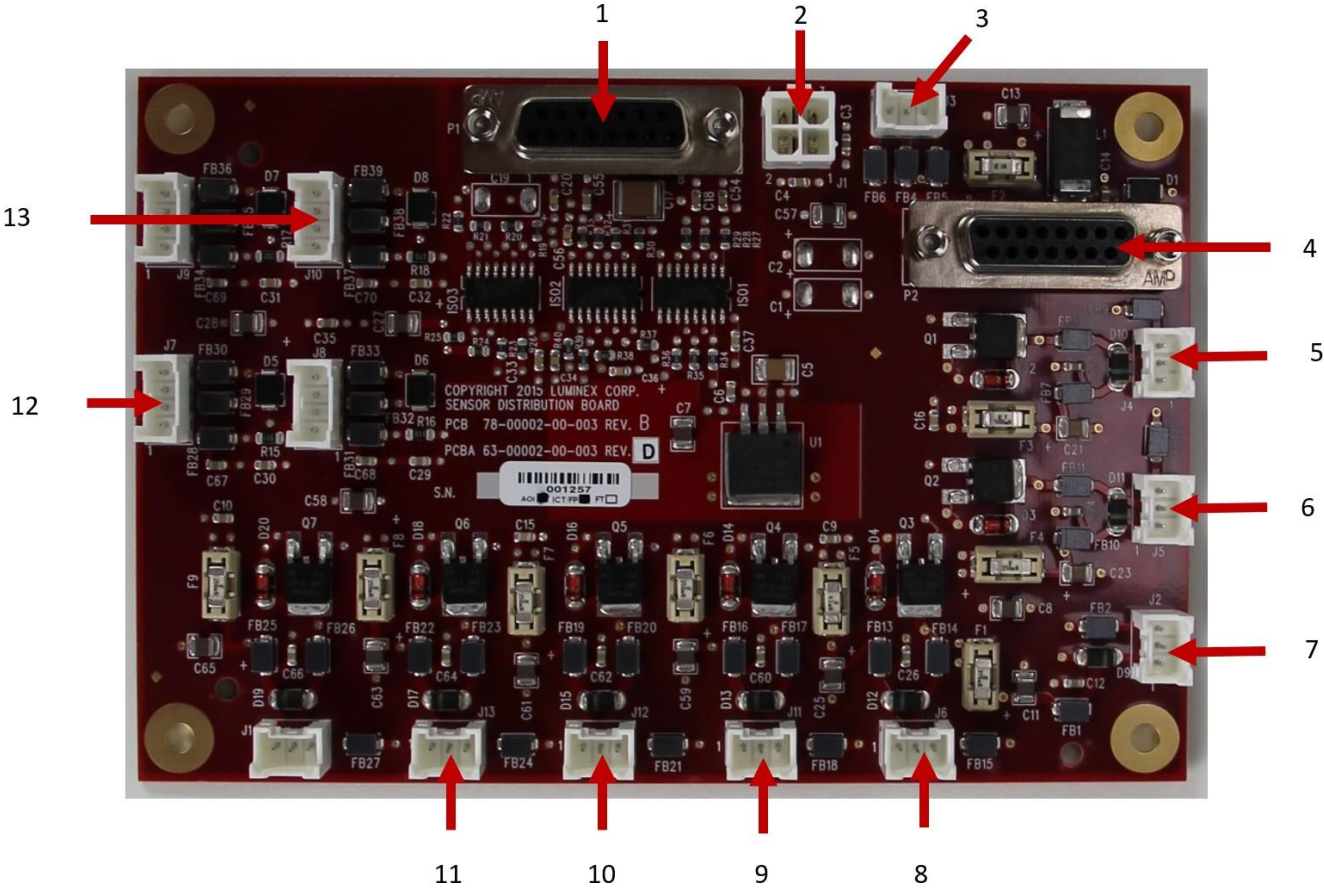
The inside space of the analyzer is divided into three compartments: the main compartment, the power supply compartment, and the fluidics compartment.

FIGURE 7. **Luminex® 200™ RoHS Analyzer Compartments**



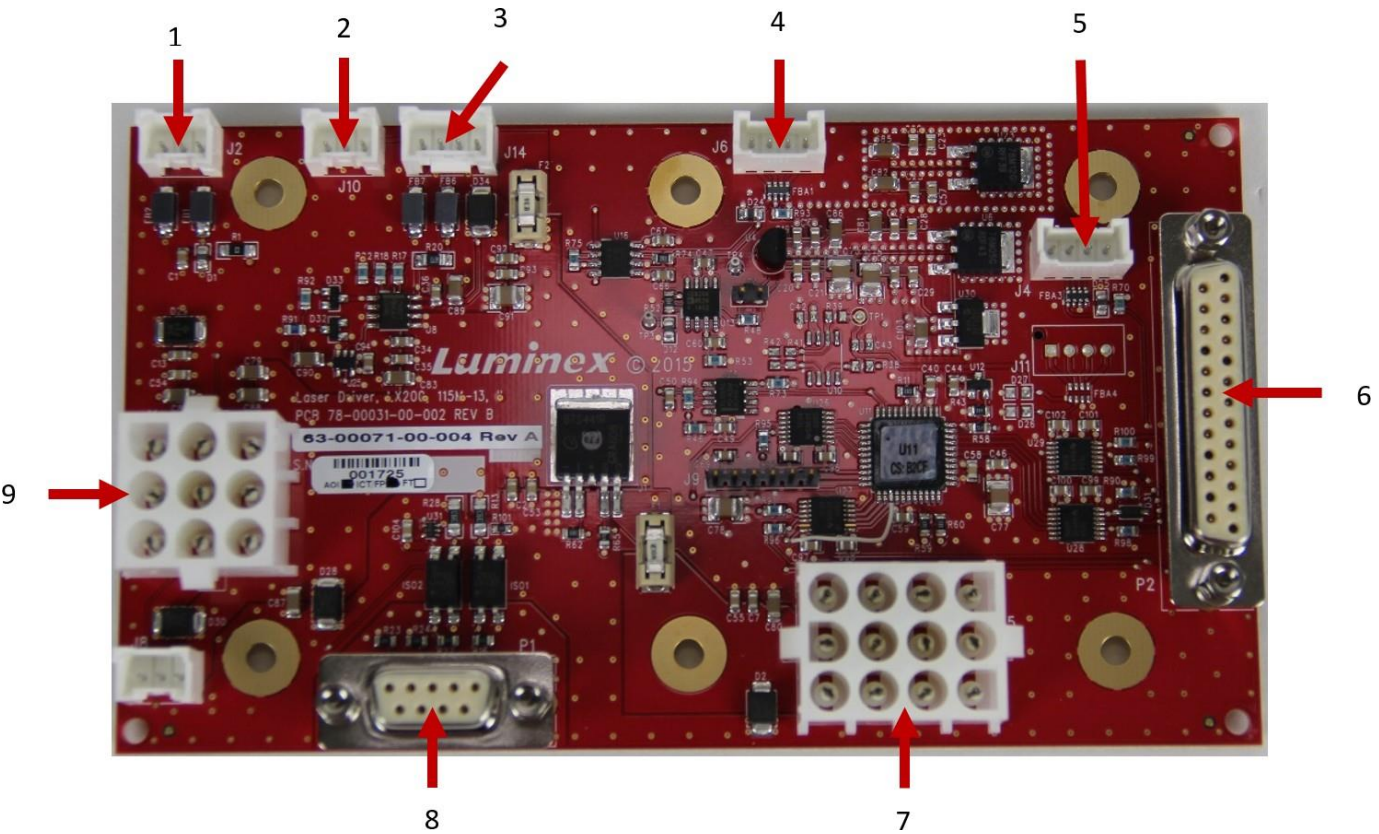
1	Main Compartment
2	Power Supply Compartment
3	Fluidics/Pneumatics Compartment

FIGURE 8. Luminex® 200™ RoHS Sensor Distribution Board



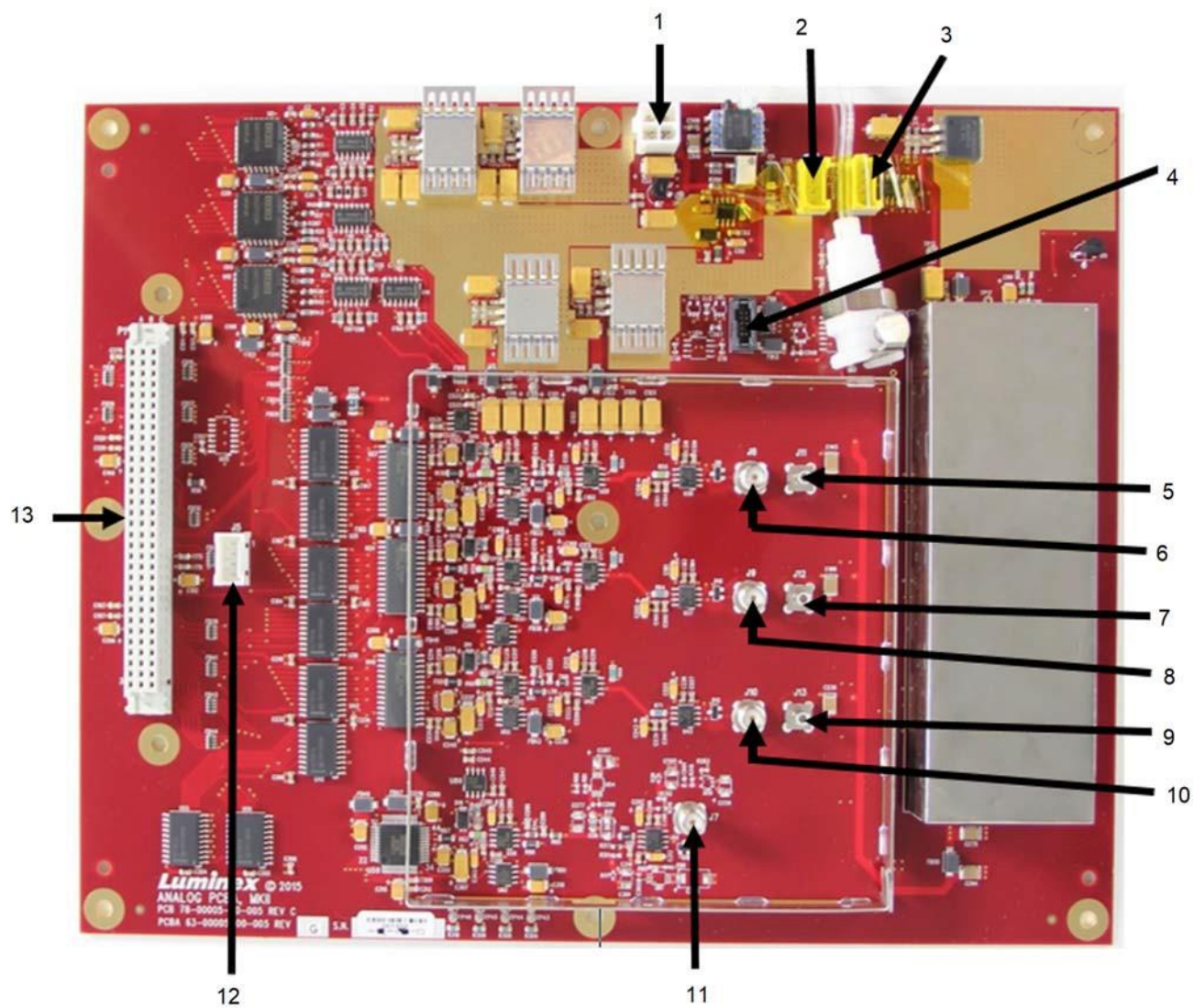
1	To Digital Board P1	8	To Air Solenoid
2	Power	9	To Sample Valve
3	To Digital Board J3	10	To Backflush Valve
4	To Syringe Pump	11	To Debubbler Valve
5	To Compressor	12	To Sheath Valve
6	To Sheath Valve	13	To Sheath Level Sensor
7	To Fan		

FIGURE 9. Luminex® 200™ RoHS Laser Driver Board



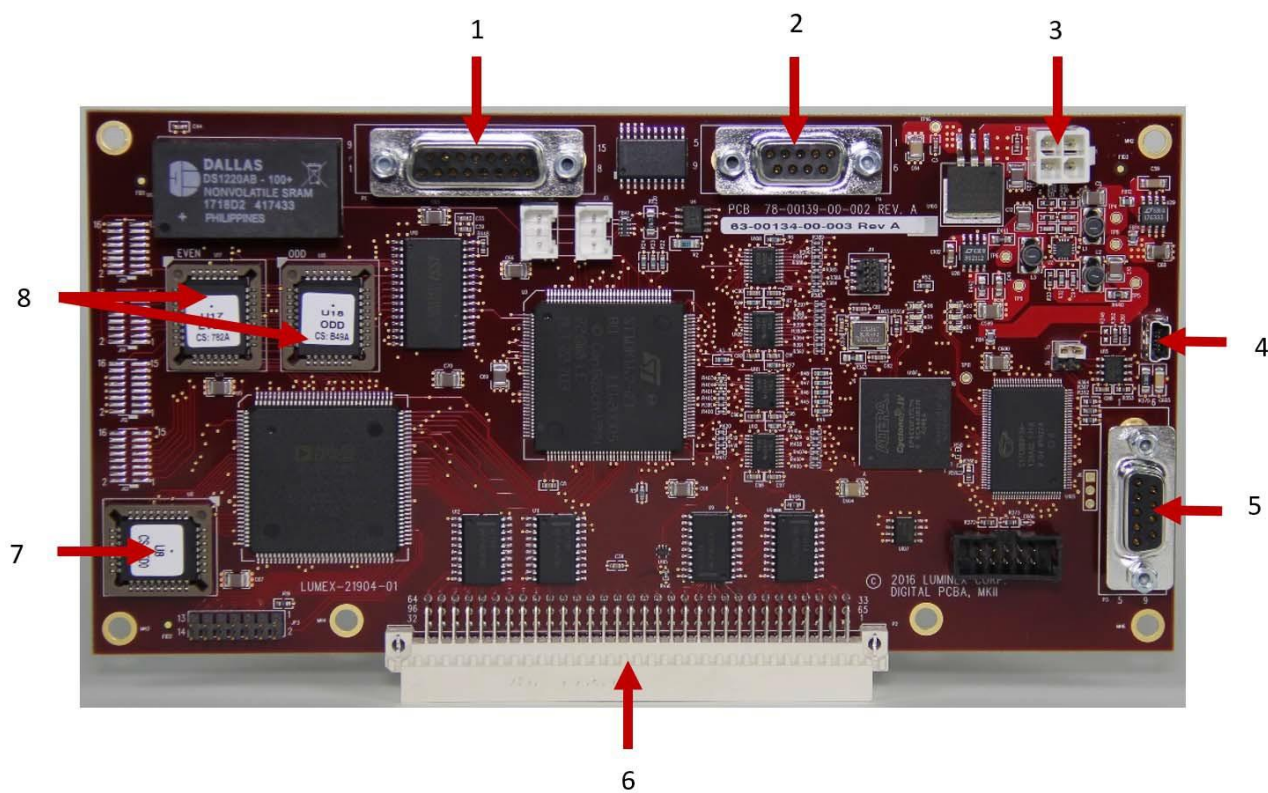
1	To LED Assembly P1	6	To Green Laser Controller P1
2	Red Laser Return	7	To Green Laser Power P1
3	To Red Laser	8	To Digital Board P2
4	To Laser Power Monitor P3	9	Power
5	To Laser Power Monitor P1		

FIGURE 10. Luminex® 200™ RoHS Analog Board



1	Power	8	APD HV (CL1)
2	To PMT P1	9	APD Signal (CH3)
3	To Pressure Transducer P2	10	APD HV (CL2)
4	APD Temperature Sensor	11	PMT Signal (CH5)
5	APD Signal (CH1)	12	Multi-Laser Monitor
6	APD HV (DD)	13	To Digital Board
7	APD Signal (CH2)		

FIGURE 11. Luminex® 200™ RoHS Digital Board



1	To Distribution Board	5	To DB9 Panel
2	To Laser Driver Board	6	To Analog Board
3	Power	7	DSP
4	USB to PC	8	Firmware

FIGURE 12. XYP Motor Driver PCBA

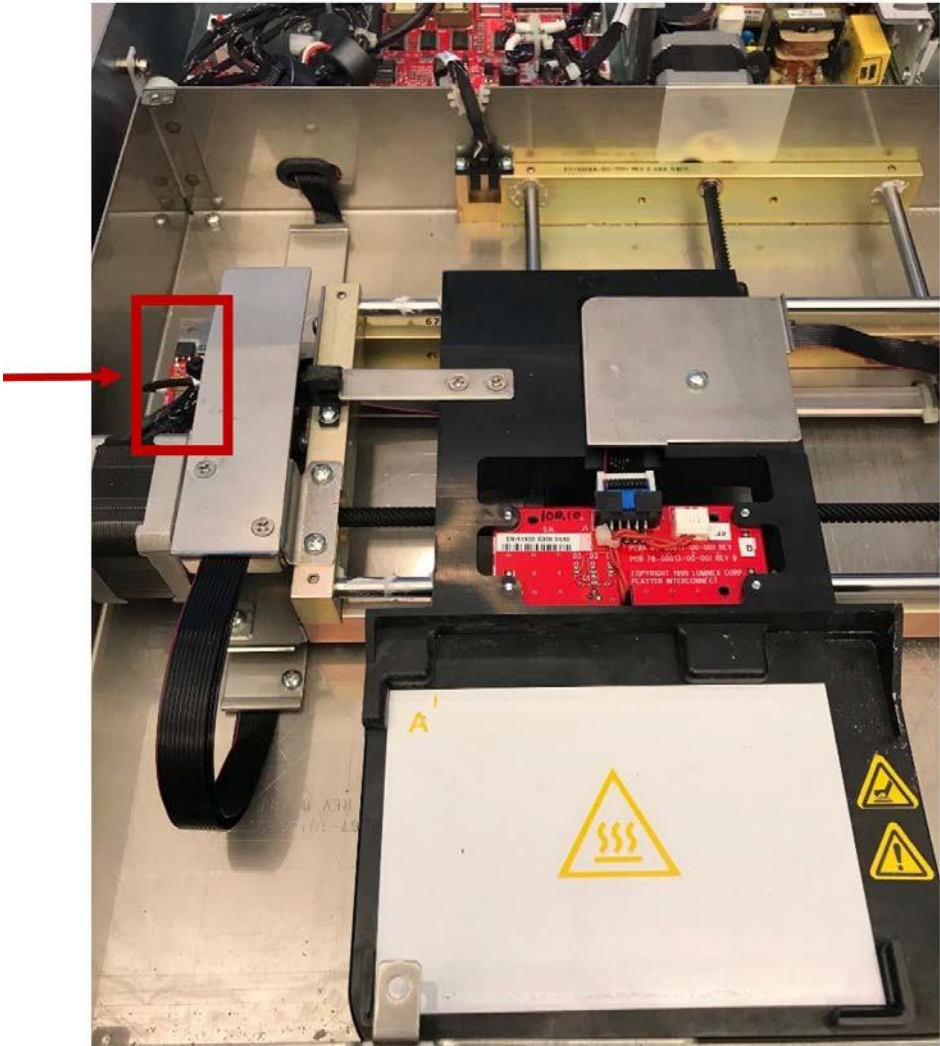


FIGURE 13. XYP Controller PCBA

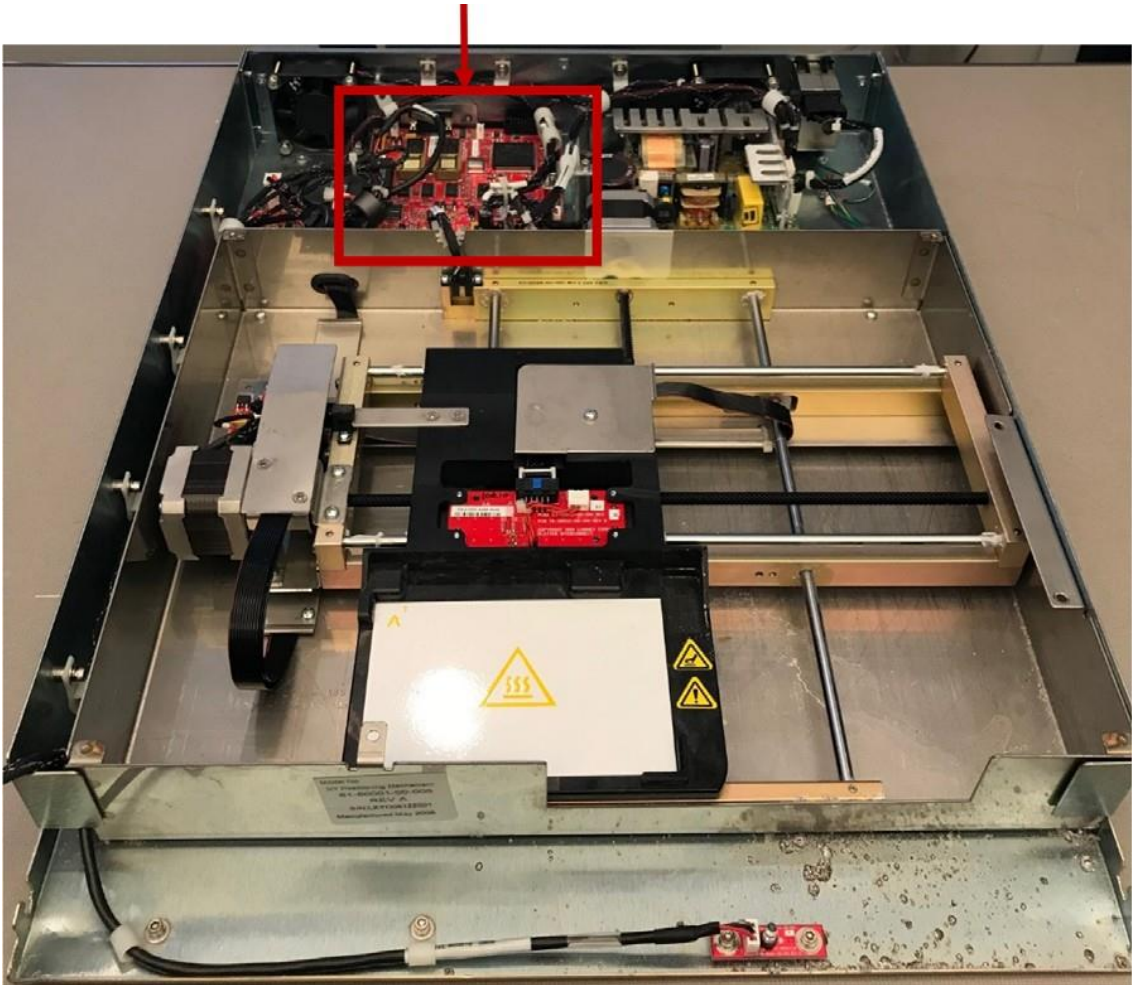
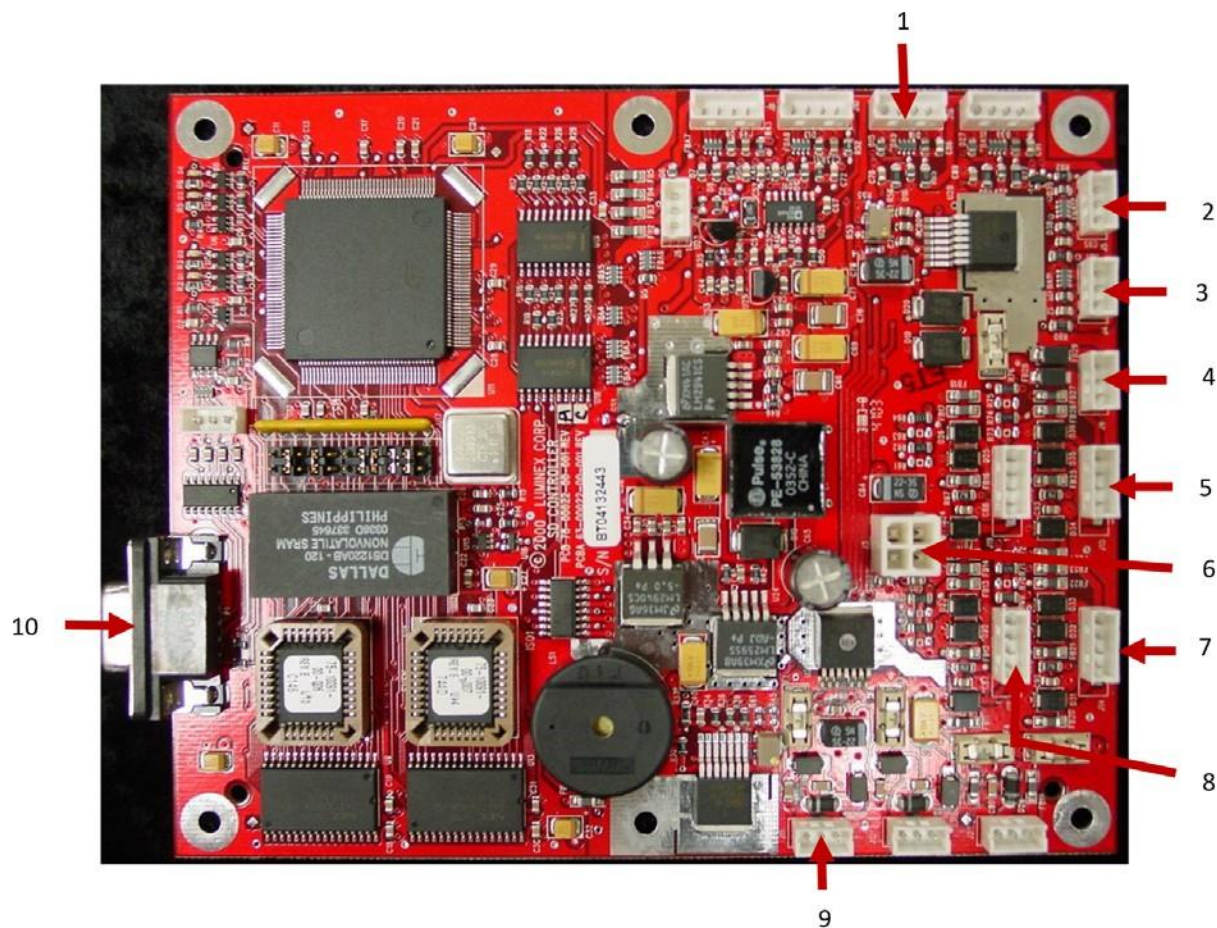


FIGURE 14. SD Controller



1	Prime Control	6	Power
2	Fault LED	7	Normal Fill Sensor (Middle)
3	Power LED	8	Auto Fill Sensor (Lower)
4	Overflow Valve	9	Reservoir Fill Pump
5	Overflow Sensor (TOP)	10	To Analyzer P2

FIGURE 15. Luminex® XYP™ Interconnect Diagram

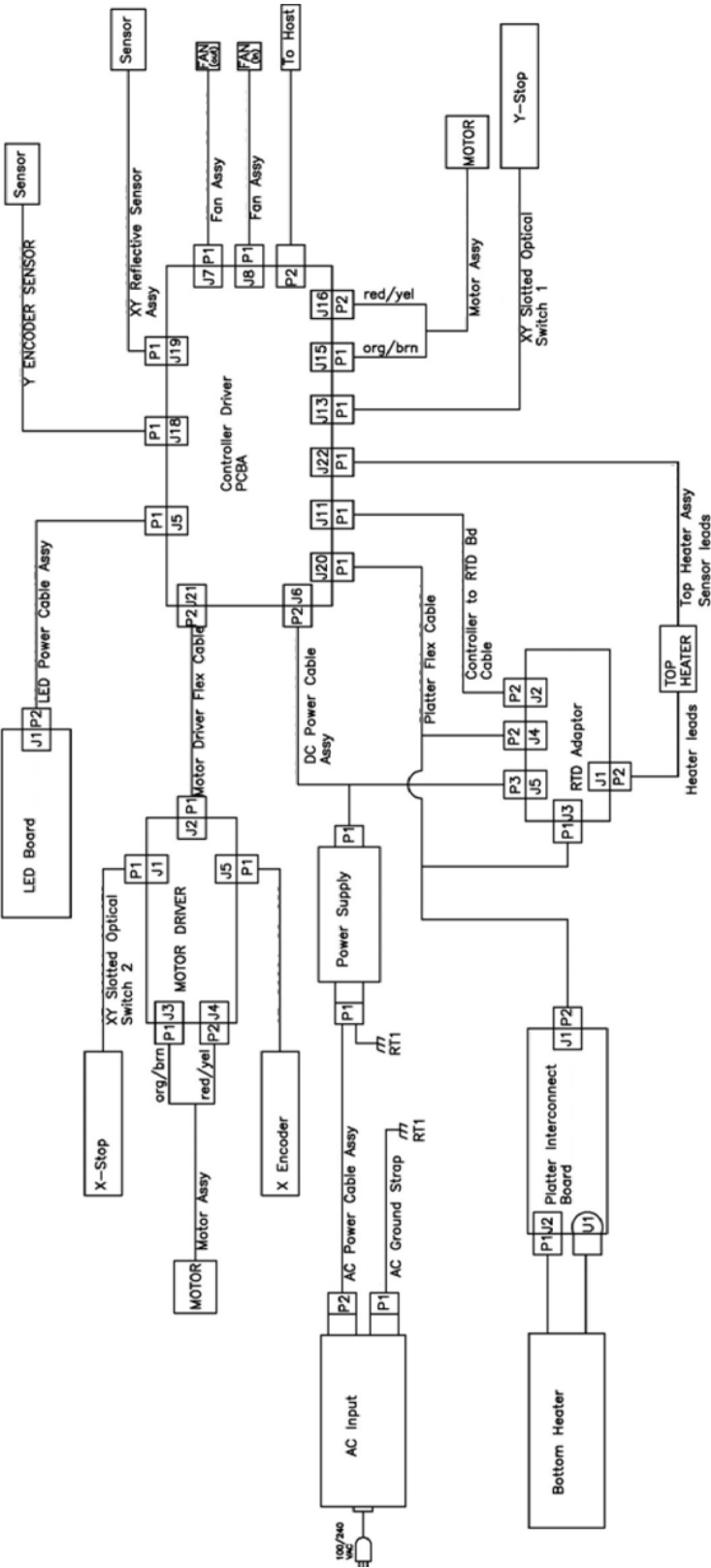
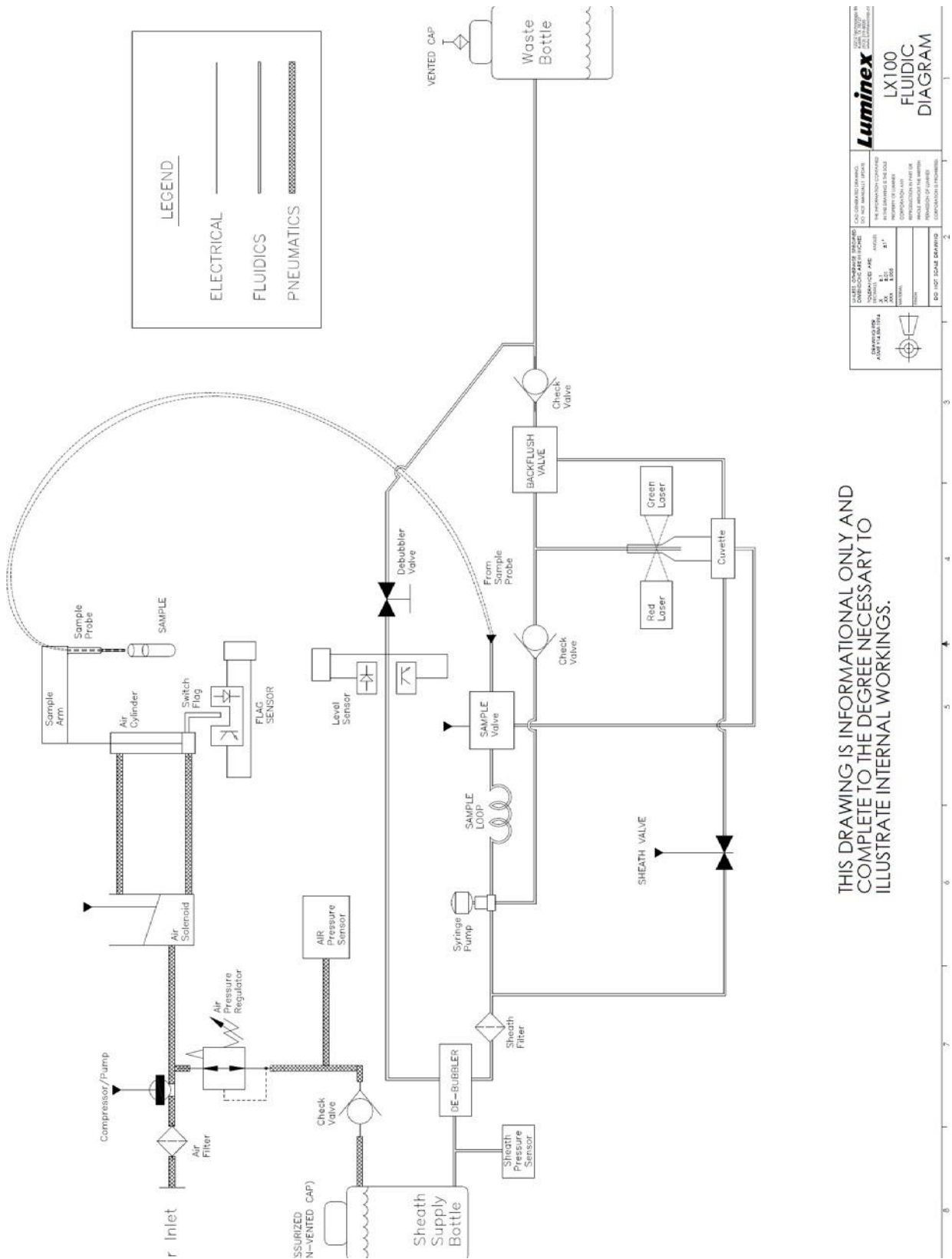


FIGURE 16. Luminex 200 Fluidics Diagram





Chapter 4: Installing the System

Safety

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Always wear personal protective equipment suitable for the activity.
- Follow all warnings marked on the instrument.
- Follow all biological safety instructions
- Follow all safety, warnings, cautions, and notes

NOTE: If damage is observed or an incorrect part/incomplete part, stop the installation and contact Luminex Technical Support for assistance

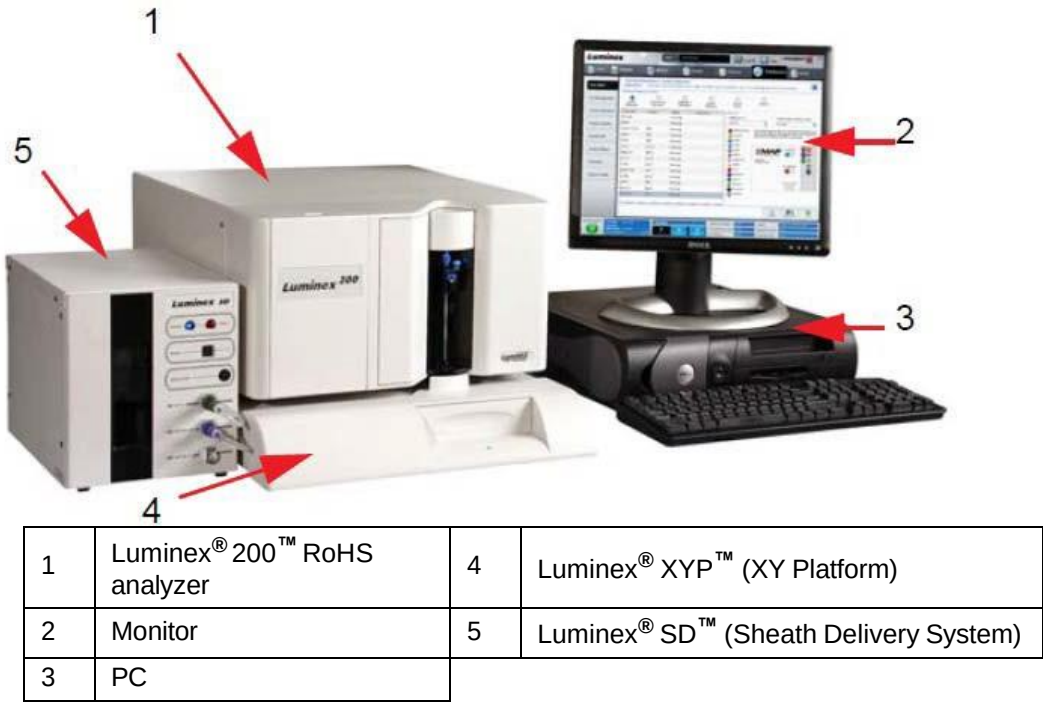
To perform the initial system installation for any new Luminex® 200™ RoHS system, you must be a Luminex®-Certified Installer. The conventions in this manual assume you already have a basic understanding of computers, xMAP® technology, and Microsoft® Windows® software.

This chapter describes how to connect the Luminex® 200™ RoHS system components. *Figure 16*, also provides clearances and other related information.

Before you connect the system components, make sure the facility you are in complies with all system and safety requirements listed in “*System Introduction*”.

Ensure you place the Luminex® 200™ RoHS system in a place that will minimize temperature fluctuations. *Figure 18*, shows the components of the system and how they should be placed.

FIGURE 18. Luminex® 200™ RoHS System Setup



Assemble the System

The following sections describe how to connect the Luminex 200 RoHS system components. An Installation Diagram (*Figure 16*) provides clearances and other related information. Each of these steps can be found on the page number shown in parentheses:

1. Set up the PC (Install the PC per the manufacturer’s instructions.).
2. Install the User Documentation (See “*Connect the Luminex 200 RoHS Analyzer and the Luminex XYP*”).
3. Connect the Luminex 200 RoHS analyzer and Luminex XYP to the PC (See “*Assemble the System*”).

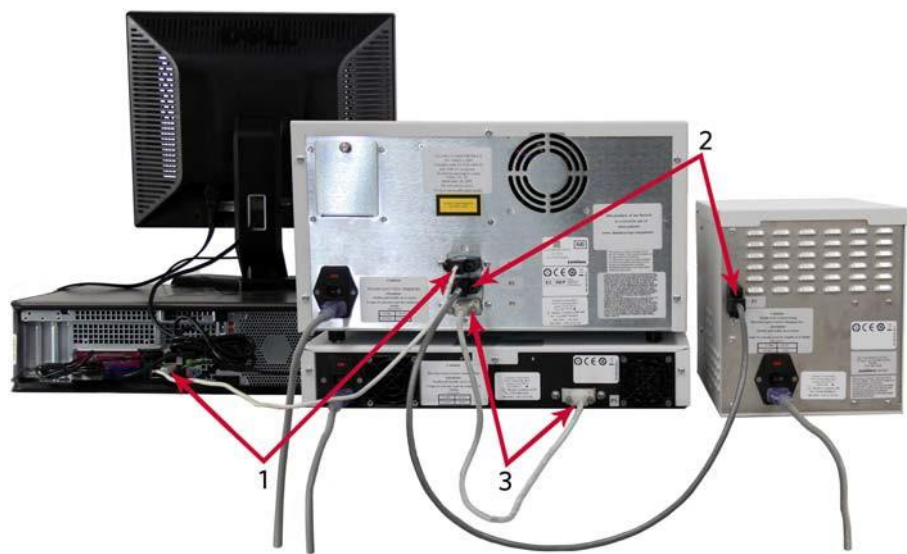
4. Install the XYP (See *"Install the Luminex 200 RoHS Analyzer Sample Probe"*).
5. Power on the System Components.
6. Accept the Software License Agreement.
7. Adjust the Sample Probe Vertical Height (See *"Adjust the Luminex 200 RoHS Sample Probe Height"*).
8. Install the Luminex XYP Reservoir (See *"Install the XYP Reservoir"*).
9. Calibrate and verify the system (See *"Calibrate and Verify the System"*).
10. Install the Luminex SD system (See *"Install the XYP Heater Block"*).
11. Install the Luminex XYP Heater Block (See *"Install the XYP Heater Block"*).

Connect the Luminex 200 RoHS Analyzer and the Luminex XYP

To connect and power on the system:

1. Install the PC and monitor according to the instructions provided by the PC manufacturer. Place the monitor on top of the PC.
2. Unpack the Luminex® 200™ RoHS analyzer and the Luminex® XYP™ and Luminex® SD™ (optional).
3. Place the Luminex XYP on a clean, flat surface to the left of the PC.
NOTE: Do not place the PC or monitor on top of the Luminex 200 RoHS analyzer or the Luminex XYP.
4. Remove the blue shipping pin. Leave the black knob in the Luminex XYP.
5. Place the Luminex 200 RoHS analyzer on top of the Luminex XYP. Align the black knob on the Luminex XYP with the hole in the adapter plate on the bottom front of the Luminex 200 RoHS analyzer.
NOTE: Ensure that the Luminex 200 RoHS analyzer and Luminex XYP power switches are both in the Off position.
6. Attach the power cord to the power input module of the Luminex XYP. Attach the serial cable to the Luminex XYP. Do not plug the Luminex XYP into the power outlet.
7. Using *Figure 19*, as a reference, attach the power cord to the input module of the Luminex 200 RoHS analyzer, then attach the USB cable to P1 on the analyzer. Do not plug the power cord into the power outlet.
NOTE: The device disconnects for the Luminex 200 RoHS and Luminex XYP are the power cords. Position the devices so that unplugging the power cords is easy.
8. Connect the Luminex 200 RoHS analyzer USB cable to the PC.
9. Connect the Luminex XYP (P1) to the Luminex 200 RoHS analyzer (P3).
10. Connect the keyboard USB cable and mouse USB cable to the PC.
11. Plug the Luminex XYP, Luminex 200 RoHS analyzer, PC, and monitor power cords into approved outlets.
NOTE: Luminex strongly recommends using an uninterruptible power supply (UPS) to protect the system from power variations. Choose a UPS that can provide 1050 Watts for at least 45 minutes. The UPS should be UL listed (or equivalent), CSA certified, and CE marked when used internationally.

FIGURE 19. Luminex® 200™ RoHS Connection Diagram (



1.	USB Communication Cable (P1)
2.	Luminex® SD™ Cable (P2)
3.	Luminex® XYP™ (P3)

Connect the Sheath and Waste Bottles

To connect the sheath and waste bottles:

1. Connect the color-coded tubing connectors to the Luminex® 200™ RoHS analyzer.
 - Blue tubing = Sheath
 - Green tubing = Air
 - Orange tubing = Waste
2. Fill the sheath bottle with Sheath Fluid to the fill line found at the air intake fitting.
3. Place the sheath and waste bottles to the left of the analyzer, on the same surface as the Luminex® XYP™.
4. Connect the blue connector to the sheath (bottom) fitting on the sheath bottle. Connect the green connector to the air (top) fitting on the sheath bottle. Connect the orange connector (middle) to the waste fitting on the waste bottle. Verify that the waste tubing is not elevated more than 12 inches above the Luminex 200 RoHS.

NOTE: To maintain a stable flow rate, do not move the waste bottle or waste line during system operation.

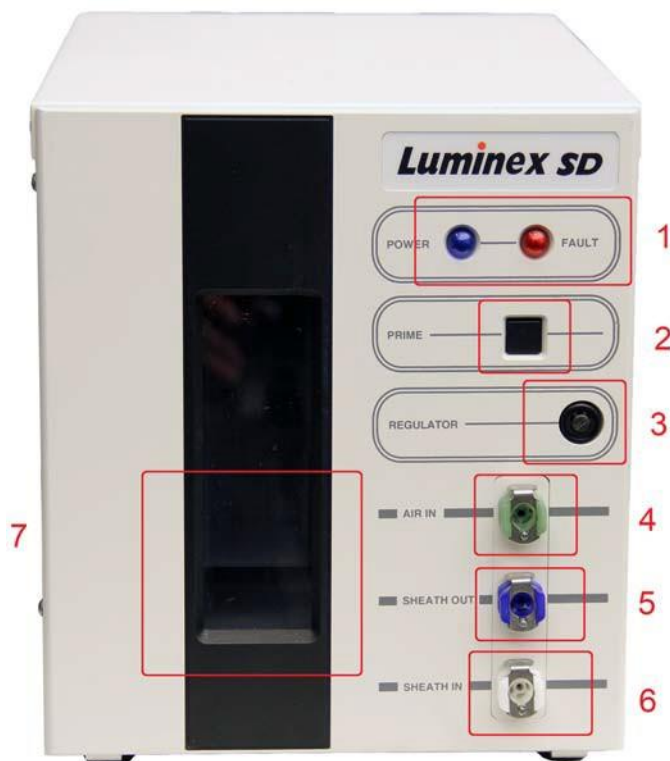
Install the SD System

1. Place the Luminex SD system near the Sheath Fluid connection on the left side of the Luminex® 200™ RoHS analyzer. Do not place the Luminex SD on top of the Luminex 200 RoHS analyzer. Using *Figure 15*, as a reference, make the following connections:
 - a. Connect the Sheath Fluid line (blue fitting) to the **Sheath Out** connector on the front of the Luminex SD system.
 - b. Connect the air line (green fitting) to the **Air In** connector on the front of the Luminex SD system.
 - c. If you are using the Luminex SD system waste line, connect the waste line tubing to the waste connector on the left side of the Luminex 200 RoHS analyzer (orange fitting). Place the other end of the waste line into a large waste receptacle. Cut off any excess tubing. Be sure that the waste receptacle is level with the Luminex 200 RoHS analyzer, or no more than 3 feet below it.

NOTE: Once the system is installed, do not permanently raise or lower the waste line. If you must permanently reroute the waste line, contact Luminex Technical Support. You can move the waste line temporarily for cleaning or other maintenance.

- NOTE:** The device disconnect for the Luminex SD is the power cord. Do not position these devices so that they will be difficult to disconnect.
2. Connect the Sheath Fluid intake line (white fitting) to the **Sheath In** connector (located on the front of the Luminex SD system).

FIGURE 20. **Luminex® Sheath Delivery System**



1	Power and Fault indicators	5	Sheath Out connector (blue)
2	Prime button	6	Sheath In connector (white)
3	Regulator adjust	7	Reservoir window
4	Air In connector (green)		

3. Lower the stainless steel filter end of the Sheath Fluid line to the bottom of a full 20-liter box of Luminex xMAP® Sheath Fluid. Secure the cap on the Sheath Fluid box. Place the Sheath Fluid container so that the cap is on the top.
- NOTE:** Ensure that the Sheath Fluid box is placed at least three feet below the Luminex SD system. If the Sheath Fluid box is not placed at least 3 feet below the Luminex SD system, it will overfill.
4. Using *Figure 14*, as a reference, attach the power cord to the input module (located on the back of the Luminex SD system). Then connect the CANBUS (short) cable to P1 on the back of the Luminex SD system and to P2 on the back of the analyzer.
5. Plug the power cord into the power outlet.

FIGURE 21. Back of SD System, P1 Port



- 6. Turn on the power to the Luminex SD system. It will automatically prime, stopping when the reservoir is 2/3 full.
- 7. Open the center access door on the Luminex 200 RoHS analyzer. Use a screwdriver to turn the regulator fully clockwise (about three to six full turns). The regulator must be fully open to work properly. The regulator is shown in *Figure 22*.
NOTE: Make sure you turn the regulator as far as it will go. Do not force the regulator past the fully open position.
- 8. Once the regulator is fully open, turn the screwdriver counterclockwise by 1/4 turn.

FIGURE 22. Luminex® 200™ RoHS Analyzer Regulator Adjustment



1	Luminex® 200™ RoHS Analyzer Regulator
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- 9. Click **Prime**. The **Prime** cycle will run. During this time, use a screwdriver to adjust the regulator, turning counterclockwise. Turn it until the sheath pressure within specifications (6 PSI to 9 PSI). The system should stabilize at this sheath pressure.

- The air pressure should be the same as the sheath pressure (6 PSI to 9 PSI), within 0.1 PSI. If the **Prime** cycle ends before you have completed the adjustment, select **Prime** again and continue to adjust the regulator.
10. **Prime** the system again. Observe the sheath pressure during the **Prime** cycle; the pressure should be within 0.1 PSI of the originally recorded pressure.
- NOTE:** Do not re-route or lengthen the waste tubing after it has been installed by a Luminex® Certified Installer. If the waste tubing length is extended or rerouted, the flow rate must be readjusted. The maximum recommended waste tubing length is 12 feet, which is the length of the tubing provided with the Luminex SD system.
- NOTE:** The waste tubing may be shortened, if necessary.

Install the Luminex 200 RoHS Analyzer Sample Probe

To install the Luminex® 200™ RoHS sample probe:

1. Verify that the Luminex® 200™ RoHS analyzer and Luminex® XYP™ power switches are both off.
2. Remove the light housing by grasping and firmly pulling it out. The light housing remains attached by a wire harness; place it on top of the analyzer.
3. Unscrew the cap from the Cheminert® fitting.

FIGURE 23. Sample Probe and Cheminert® Fitting



1	Cheminert® Fitting	3	Thumbscrew	5	Sample Probe
2	Sample Probe Holder	4	Adjustment Slide		

4. Insert the sample probe into the sample probe holder. Insert the Cheminert fitting into the sample probe holder, turning it clockwise. Be sure that the threads are correctly aligned. Hand tighten only.
5. Push the light housing back into place.
6. Plug the Luminex XYP and the Luminex 200 RoHS analyzer into approved outlets.
7. Power on the system components. The first time xPONENT® starts, you have to acknowledge the license agreement. Read and accept the agreement to continue.

Adjust the Luminex 200 RoHS Sample Probe Height

To adjust the sample probe height:

NOTE: Before anything else, verify that the microtiter plate is not warped. Warped plates can lead to incorrect probe height adjustment.

NOTE: You may place alignment discs or sphere in any well, as long as the well is properly designated in xPONENT® software.

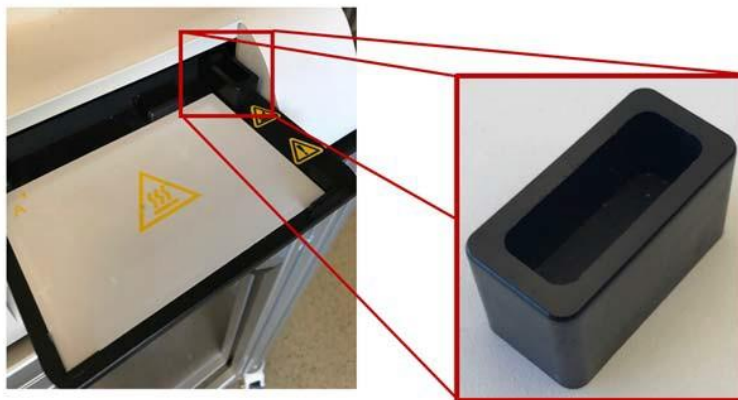
1. In a 96-well microtiter plate with an overall height of no more than 0.75 inches (19 mm), place the appropriate alignment tool in the plate:
 - **For a standard plate with flat-bottom wells** - Stack two of the larger (5.08 mm diameter) alignment discs together and place them into the selected well.
 - **For a filter bottom plate** - Stack three of the larger (5.08 mm diameter) alignment discs together and place them into the selected well.
 - **For a half-volume plate with flat-bottom wells** - Stack two of the smaller (3.35 mm diameter) alignment discs together and place them into the selected well.
 - **For a round-bottom (U-bottom) plate** - Stack two of the smaller (3.35 mm diameter) alignment discs into the selected well.
 - **For a plate with conical wells (V-Bottom)** - Place one alignment sphere into the selected well.
2. Click **Eject** in xPONENT to eject the plate holder. Place the 96-well microtiter plate on the Luminex® XYP™ plate holder, so that position A1 is in the top left corner.
3. Verify that the correct well location is selected in xPONENT, and that you are using the appropriate number of alignment discs or sphere. Click **Retract** in xPONENT to retract the plate.
4. Loosen the front thumbscrew on the probe holder by one-third to one-half turn. Pull the probe upward until it touches the top of the adjustment slide. Tighten the thumbscrew.
5. Use xPONENT to lower the sample probe.
6. Loosen the front thumbscrew. Gently push the probe down until it just touches the top of the alignment discs or sphere.
7. Tighten the front thumbscrew.
8. Using xPONENT, raise the sample probe.

Install the XYP Reservoir

You will use the Luminex® XYP™ reservoir to perform the Luminex® 200™ RoHS analyzer maintenance functions. To install the reservoir:

1. Click **Eject** in xPONENT to eject the plate holder.
2. Insert the reservoir into the upper-right corner of the plate holder.
3. Click **Retract** in xPONENT to retract the plate holder.

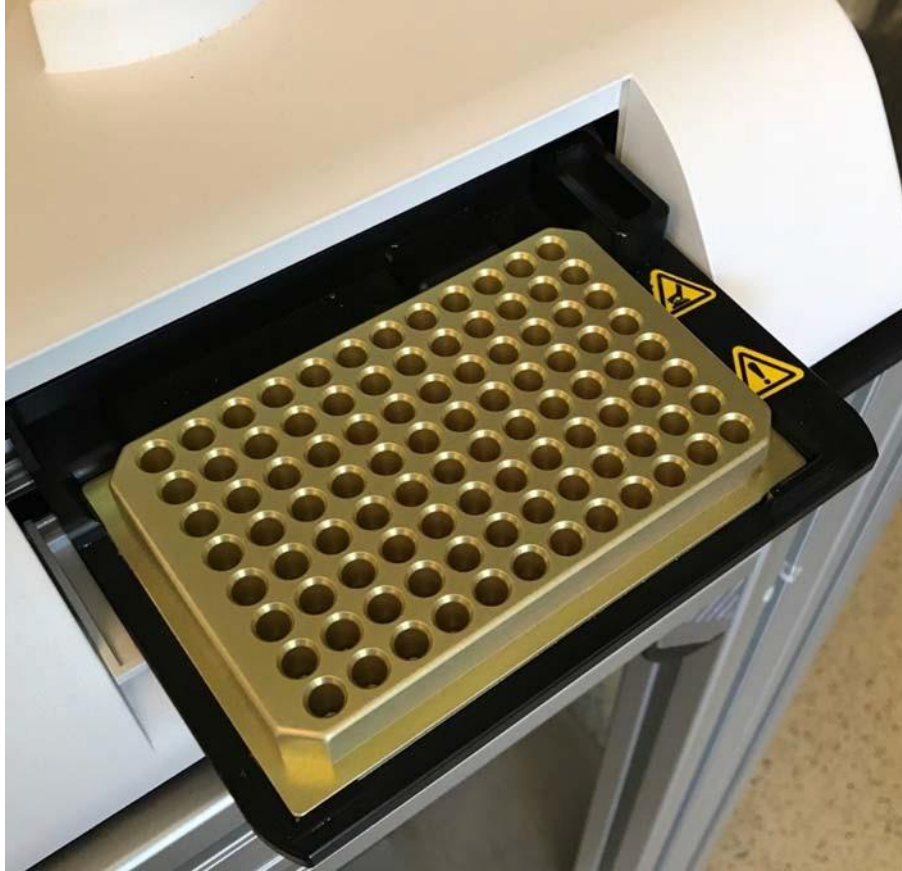
FIGURE 24. Luminex® XYP™ Reservoir



Install the XYP Heater Block

Use the Luminex® XYP™ heater block as required by the assay kit instructions. This is an optional step. When the heater block is not in use, store the heater block in the bracket located inside the left access door of the Luminex® 200™ RoHS analyzer.

FIGURE 25. Luminex® XYP™ Heater Block



To install the Luminex XYP heater block:

1. Click **Eject** in xPONENT® to eject the plate holder.
2. Insert the Luminex XYP heater block on the plate holder.
3. Click **Retract** in xPONENT to retract the plate holder.
4. Turn the heater on and set the temperature.
 - a. In xPONENT®, navigate to **Maintenance > Probe and Heater**.
 - b. Select **Enabled** to enable the heater.
 - c. In the field, enter the desired value in degrees Celsius. The temperature range is 35°C to 60°C in 0.5 increments.

NOTE: For new system installations, perform the preliminary commands before running samples.

Power on System Components

- The Luminex® 200™ RoHS analyzer (switch on back)
- The Luminex® XYP™ (switch on back)
- The PC and monitor (switches on front)
- Luminex® SD™ (if installed) (switch is on the back)

NOTE: If Calibration or Controls fail when performing Final Test After Service Call, minor optical alignment may be required to correct it

Accept the Luminex® xPONENT® Software License Agreement

The first time the xPONENT® software starts up, you must acknowledge the License Agreement. Read the agreement and select **Accept** to continue.

Start Up

To start the system:

1. Prime the system until the waste line outside the analyzer appears free of all significant air pockets.
2. Perform two **Wash** cycles using sheath fluid.
3. Perform two **Sanitize** cycles, using a 10% to 20% bleach solution.

Shut Down System Components

To perform the shutdown procedure:

1. Sanitize with a 10% to 20% household bleach solution.
2. Soak with distilled water.
3. Release the pressure by loosening the sheath fluid container cap.
4. Turn the analyzer off.
5. Turn the XYP off.
6. Turn the SD (if installed) off.

Chapter 5: Maintaining the System

Safety

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Always wear personal protective equipment suitable for the activity.
- Follow all warnings marked on the instrument.
- Follow all biological safety instructions
- Follow all safety, warnings, cautions, and notes

Preventative Maintenance

Note: The Luminex 200 System should have preventative maintenance on an annual basis (every 12 months). Semi-annual Preventative maintenance (every 6 months) is highly recommended but not required.

Semi-Annual Maintenance (6-Month PM Kit)

Replace Luminex® 200™ Analyzer Air Intake Filter

NOTE: Hold on to the tubing. Do not allow the tubing to fall inside the instrument.

1. Turn off the analyzer, but keep it plugged in.
2. On the back of the Luminex 200 analyzer, in the upper left corner, remove the screw at the top of the panel and open the panel door.
3. Grasp the tubing and pull the filter 3 to 4 inches from the unit. See *Figure 26*.

FIGURE 26. Grasping the Tubing



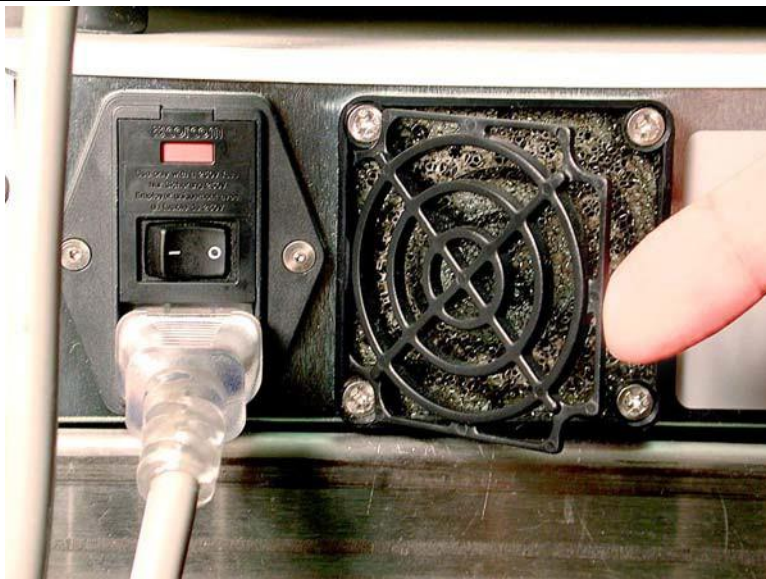
4. Remove the filter with one hand, and hold the tubing with the other hand.
5. Connect a new filter to the tubing and position the filter inside the panel.
6. Reattach the panel door to the unit using the thumb screw.
7. Plug in and power on the Luminex 200 analyzer.

Replace Luminex® XYP™ Instrument Air Intake Filter

1. Turn off the XYP, but leave it plugged in.
2. On the back of the Luminex XYP instrument, to the left side, gently remove the screen from the Luminex XYP instrument filter.

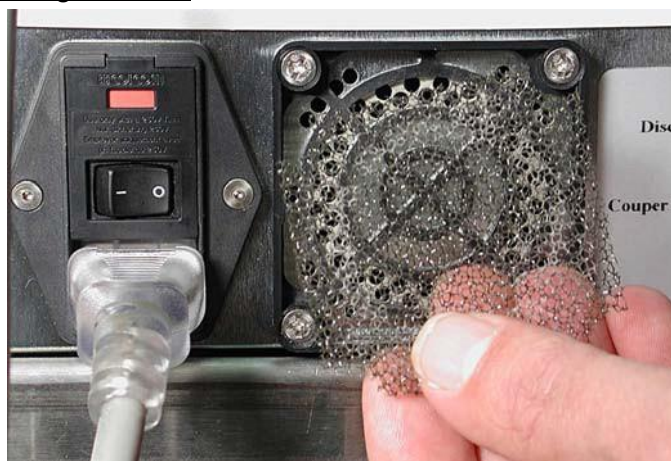
NOTE: Do not remove the screws.

FIGURE 27. Removing the Screen



3. Replace the filter and replace the screen.

FIGURE 28. Removing and Replacing the Filter



4. Plug in and turn on Luminex XYP instrument power.

Replace the Syringe



The syringe arm does not deactivate when changing the plunger; injury could result if the system is not unplugged.

Remove the Syringe

1. Turn off and unplug the analyzer.
2. Open the door on the front center of the analyzer to gain access to the syringe. The syringe is the glass cylinder with a metal plunger inside, shown in *Figure 29*.



3. At the base of the syringe, loosen the thumbscrew by turning it counter-clockwise six quarter-turns.
4. Using your two thumbs, manually push down on the base of the syringe. The base of the plunger comes out.
5. If the base of the plunger did not come out of the arm when it cycled, loosen the thumbscrew a little more and very lightly attempt to lift the plunger away from the base.
6. Unscrew the syringe from the top of its housing. Remove the syringe.

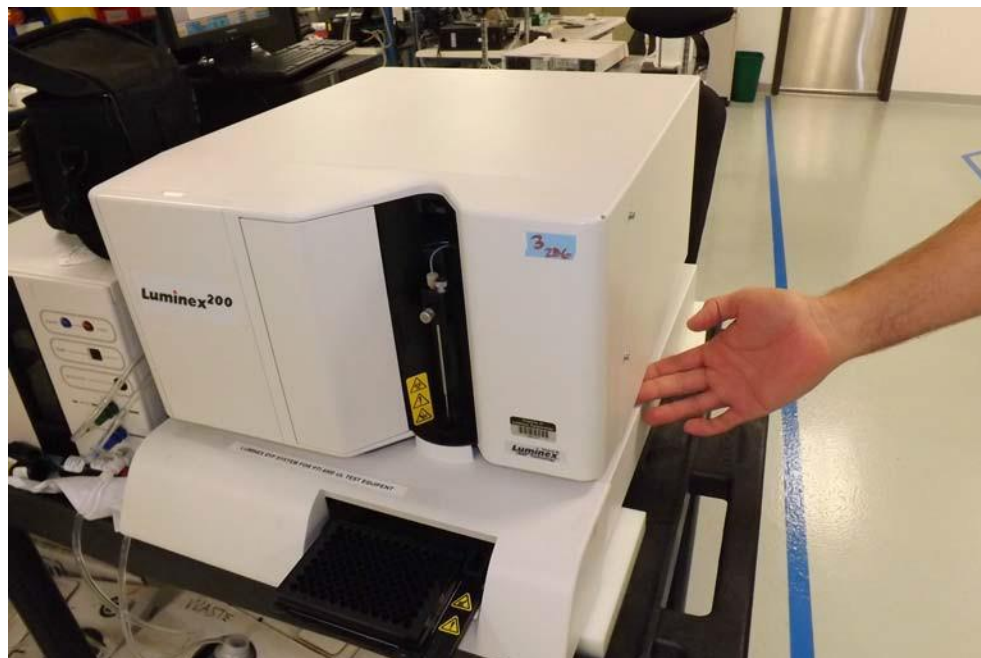
Replace the Syringe

1. Screw the top of syringe into place.
2. Grab the base of the syringe plunger and gently pull down on it until it is fully seated into the opening of the arm.
3. Tighten the thumbscrew at the base of the syringe completely. If the thumbscrew does not go in as far as it did before, reposition the plunger and try again.
4. Plug in and turn on the analyzer. The syringe will return to its home position before the analyzer begins its normal start up process.
5. **Prime** the system twice, watching for any leaks in the syringe area.
6. When the **Prime** finishes, close the analyzer door.

Luminex® 200™ Analyzer Ventilation Filter

1. Turn off and unplug the analyzer.
2. While facing the Luminex 200 analyzer, place your index finger up under the right side of the analyzer (in the space between the Luminex 200 analyzer and the Luminex® XYP™ instrument). When you feel the filter, push the filter toward the left of the analyzer. See *Figure 30*.

FIGURE 30. Luminex® 200™ Analyzer Ventilation Filter



3. Remove the filter from the left side of the Luminex 200 analyzer.
4. Clean the filter with a vacuum or with distilled water. Stand the filter upright to air dry.
5. Re-install it with the arrows facing up once dry. The filter should click into place.
6. Plug in and turn on the Luminex 200 analyzer power.

Annual Maintenance (12-Month PM Kit)

NOTE: Perform all semi-annual maintenance procedures prior to annual maintenance.

Note: The Luminex 200 System should have preventative maintenance on an annual basis (every 12 months).

Replace the Sheath Filter

1. Turn off the analyzer, but leave it plugged in.
2. Open the left door on the Luminex 200 analyzer. Disconnect the filter by pushing down on the metal clamps on each connection. See *Figure 31*.

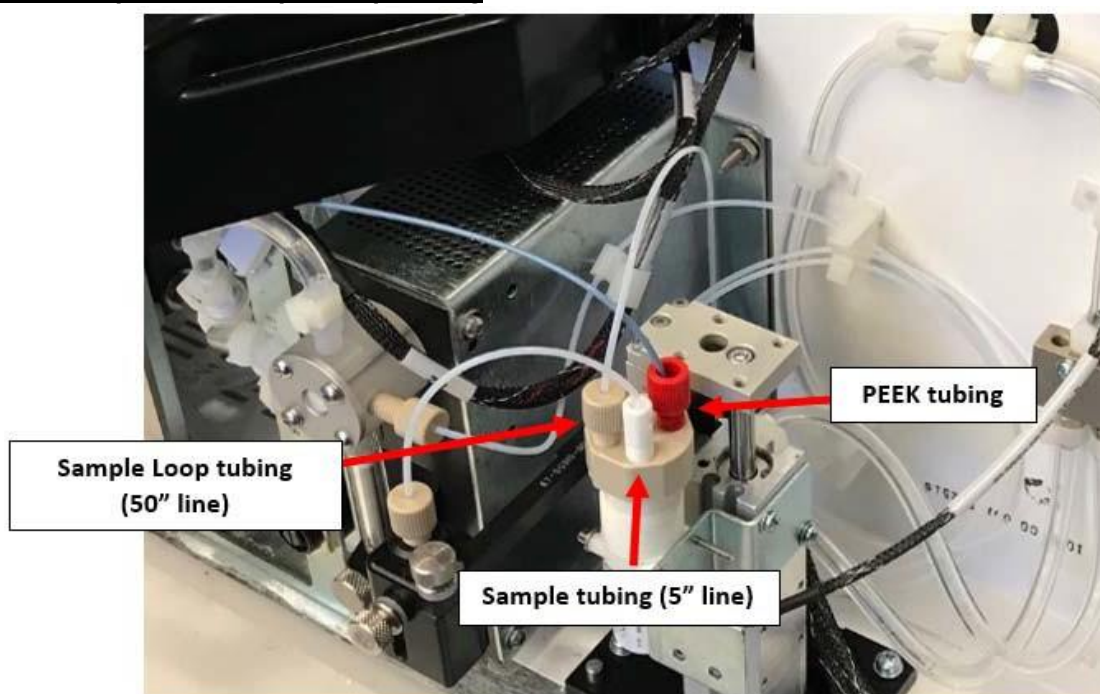
FIGURE 31. Sheath Filter



3. Connect the new sheath filter, matching up the color-coded fittings. The arrow on the sheath filter should be pointing up.
4. Reconnect the sheath fluid tubing.
5. Plug in and power on the Luminex 200 analyzer.
6. Close the left analyzer door.
7. Prime twice.

Remove and Replace the PEEK, Sample, and Sample Loop Tubing

FIGURE 32. PEEK, Sample, and Sample Loop Tubing



1. Turn off the analyzer, but leave it plugged in.
2. Unscrew the Sample Loop tubing (50" line) from the sample valve and right side of the syringe valve.
NOTE: Notice the Sample Loop tubing is looped in circles and held by a bracket secured to the chassis. When replacing the tubing, ensure it is looped around the bracket, as originally configured.
3. Remove the Sample Loop tubing.

4. Put the new Sample Loop tubing into place, ensuring the loops are placed onto the bracket, as shown above.
5. Hand-tighten one fitting into the syringe valve and the other fitting into the sample valve. Do not over tighten.
6. Unscrew the red PEEK tubing fitting from the sample valve and the brown fitting from the base of the injection needle.

NOTE: DO NOT discard the PEEK tubing fittings. They will be re-used.

7. Remove the PEEK tubing.
8. Put the new PEEK tubing through the red fitting and hand-tighten it onto the sample valve. Do not over tighten.
9. Route the other end of the PEEK tubing over the syringe pump behind the tubes, as shown above.
10. Put the other end of the PEEK tubing through the brown fitting, so approximately 1/4" is sticking out of the end.
11. Place the end of the tubing into the base of the injection needle and hand-tighten the brown fitting. Do not over tighten.
12. Unscrew the white Sample tube (5" line) fitting from the sample valve and the brown fitting from the probe holder.
13. Hand-tighten the new Sample tube to the sample valve and probe heater.

Chapter 6: Service Procedures

Safety

The non-observance of safety instructions may result in serious personal injuries and material damages.

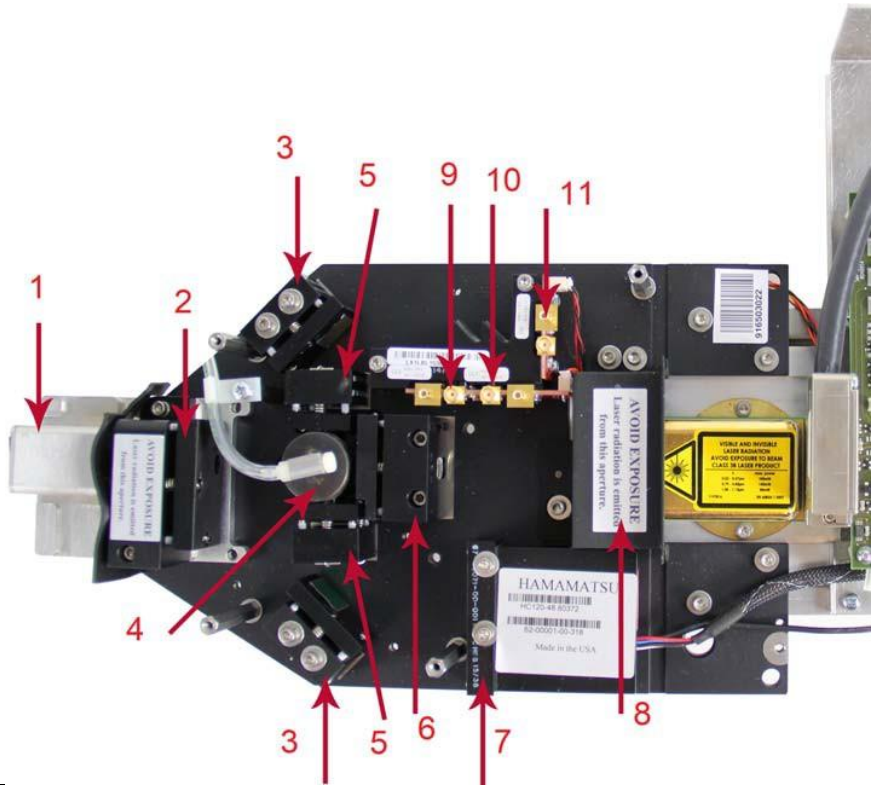
- Always wear personal protective equipment suitable for the activity.
- Follow all warnings marked on the instrument.
- Follow all biological safety instructions
- Follow all safety, warnings, cautions, and notes



NOTE: For all service procedures, an anti-static strap must be worn at all times to prevent electro-static discharge to the instrument.

Align the Optics for the Analyzer

Optics alignment should be performed if the analyzer does not meet optical specifications. See “Common Abbreviations” for more information. This procedure involves servicing the laser components of the Luminex® 200™ RoHS analyzer.



1	Red Laser (635 nm)	7	RP1 (PMT)
2	Red Laser Lens Assembly	8	Green Laser (532 nm)
3	Dichroic Mirror	9	CL2 (APD)
4	Cuvette	10	CL1 (APD)
5	4 mm Lens	11	DD (APD)
6	10 mm or 15 mm Lens		



Laser radiation when open. Avoid direct exposure to beam.



Wear laser safety glasses before removing the optics cover. Wear an ESD strap at all times when working on optics.

Perform a Visual Inspection

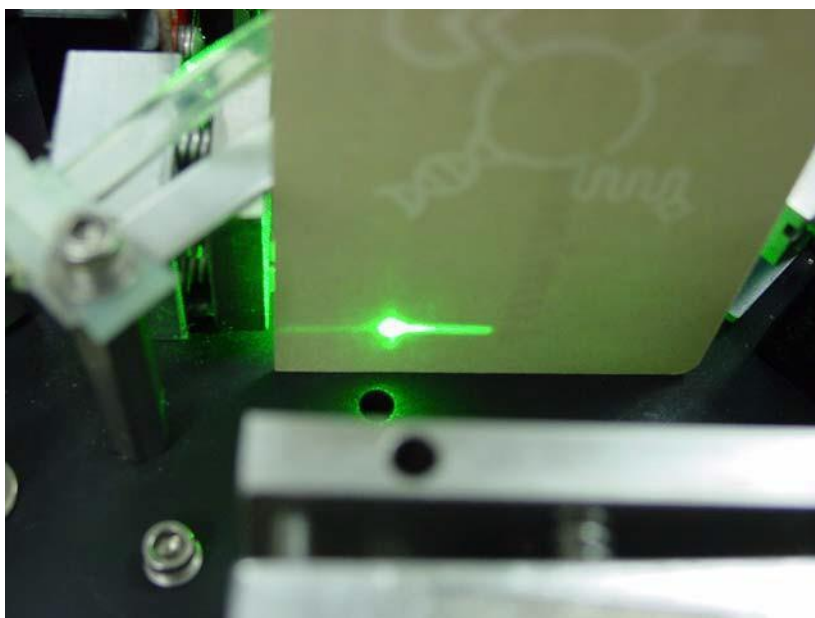
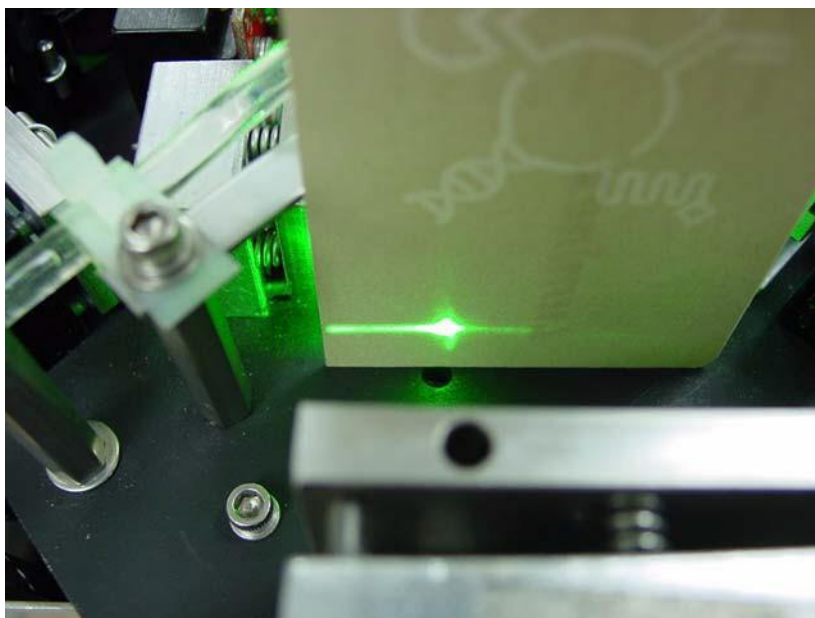
1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Check the cuvette holder base for visible defects such as burrs or scratches. Ensure it is clean.
3. Look at the lenses. They should appear clear, with no smudges or excessive glue.

Coarse-Align the Green Laser

NOTE: The red and green lasers pass through the cuvette. The vertical position of the green laser is coarsely positioned so that the green laser is above the red laser at the red laser head.

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the Optics Platform cover.
3. Turn the green laser on and the red laser off.
4. Using the vertical adjustment screw on the 10 mm lens assembly, align the beam to match the center height of both 4 mm lens assemblies.
5. Place a piece of paper in front of the red laser near the cuvette.
6. Sweep the green laser beam from left to right by turning the horizontal screw on the 10 mm lens until you see a comet pattern display on the paper.
7. When you can turn the screw and the comet pattern shifts from one direction to the other within one-half a turn, you have found the center of the cuvette.
8. Balance the pattern until the amount of light on each side is equal.

NOTE: The amount of visible pattern remaining after balancing will vary.



9. Remove the paper from in front of the red laser and place it in front of the PMT.
10. Fill a sample tube with laser alignment solution and run the dye as a sample.
11. Using the vertical adjustment screw on the 10 mm lens assembly, move the beam up and down along the cuvette until you see a dot display on the paper in front of the PMT. (Ensure the shape and size of the dot are accurate. If the dot is too large or small, use the vertical screw on the 4 mm lens of the reporter channel to adjust it to the correct size.)
12. Use the vertical and horizontal screws on the dichroic mirror to bring the dot into the opening of the PMT.
13. Perform several washes to remove the dye from the system.
14. Replace the Optics Platform cover.
15. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.
16. Turn the red laser back on.
17. Calibrate and verify the results.

Fine-Adjust the Green Laser

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the Optics Platform cover.
3. Turn on the green laser.
4. Attach the oscilloscope channel A probe to the DD test point and the channel B probe to the RP1 test point on the analog board. Set the scope to trigger on DD (channel A).
5. Set the oscilloscope's horizontal time scale to 10 μ s per division and the vertical scale to 500 mV per division.
6. Fill a sample tube with alignment microspheres and run it as a sample.
7. Adjust the 10 mm lens assembly's horizontal, and vertical screws to obtain the highest amplitude waveform.
8. Adjust all three dichroic mirror screws for the highest amplitude.
9. Stop making adjustments when further changes will not improve the peak height or consistency on the oscilloscope.
10. Replace the Optics Platform cover.
11. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.
12. Calibrate and verify the results.

Coarse-Align the Red Laser

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the Optics Platform cover.
3. Turn the red laser on and the green laser off.
4. Using the vertical screw on the red laser housing, move the laser beam until you see it go into the opening of the green laser head.
5. Either place a piece of paper between the 10 mm lens and cuvette, or use the front surface of the 10 mm lens, to view the beam.
6. Sweep the red laser from left to right by turning the horizontal screw on the red laser housing.
7. Continue turning the screw until you see a comet pattern display on the paper.
8. When you can turn the screw and the comet pattern shifts from one direction to the other within one-half a turn, you have found the center of the cuvette.
9. Balance the pattern until the amount of light on each side is equal. (The amount of visible pattern remaining after balancing can vary.)
10. Remove the paper placed in front of the 10 mm lens, if applicable.
11. Replace the Optics Platform cover.
12. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.
13. Calibrate and verify the results.

Fine-Adjust the Red Laser

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the Optics Platform cover.
3. Turn on the red and green lasers.
4. Attach the oscilloscope channel A probe to the DD test point and the channel B probe to the RP1 test point on the analog board. Set the scope to trigger on RP1 (channel B).
5. Set the oscilloscope's horizontal time scale to 10 μ s per division and the vertical scale to 500 mV per division.
6. Fill a sample tube with alignment microspheres and run it as a sample.
7. Slowly adjust the vertical screw on the red laser housing until a signal is seen on the oscilloscope. (You should be able to find a signal for the red laser while the laser beam is somewhere inside the opening of the green laser head.)
8. Adjust the horizontal and vertical screws to obtain the highest amplitude waveform.
9. Adjust all three dichroic mirror screws for the highest amplitude.
10. Stop making adjustments when further changes will not improve the peak height or consistency on the oscilloscope.
11. Replace the Optics Platform cover.
12. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.
13. Calibrate and verify the results.

Perform the Green Laser Offset Adjustment

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the Optics Platform cover.
1. Attach the oscilloscope channel A probe to the DD test point and the channel B probe to the RP1 test point on the analog board. Set the scope to trigger on DD (channel A).
2. Fill a sample tube with alignment microspheres and run a sample using the maintenance software.
3. The waveform peaks on the oscilloscope should be separated by 25 to 50 μ s with the red laser signal to the left of the green laser signal. If this is not occurring, adjust the 10 mm vertical adjustment screw and move the signal in the desired direction until there is a small amount of signal left on the oscilloscope.
4. Adjust the green side dichroic mirror's vertical screw to restore the signal.
5. If the laser offset is not correct, repeat the previous steps until it is.
6. Once the proper offset is achieved, adjust the horizontal and vertical alignment screws on the 10 mm lens to maximize the signal.
7. Adjust the horizontal and vertical alignment screws on the dichroic mirror (reporter channel) to maximize the signal.
8. Replace the Optics Platform cover.
9. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.
10. Calibrate and verify the results.

Performing Focal Turns

NOTE: DO NOT adjust the focal distance unless necessary.

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the Optics Platform cover.
3. Attach the oscilloscope channel A probe to the DD test point and channel B to the RP1 test point on the analog board.
4. Set the oscilloscope's horizontal time scale to 10 μ s division and vertical scale to 500 mV per division. .
5. Fill a sample tube with alignment microspheres and run a sample using the maintenance software.
6. Prior to making any adjustments, make note of the current amplitude on the scope for RP1.
7. While the sample is running, turn all three screws on the 10 mm lens assembly (or the red laser lens assembly for the same side) in the same direction and the same amount (3 to 4 full turns). If you keep the turn ratio consistent from screw to screw, you will see some signal change upon completion of the last turn.
8. Re-adjust all three screws for maximum peak amplitude and peak consistency.
9. Make a note of the peak amplitude. If it has increased by way of the focal point adjustment, repeat the process. Continue adjusting the distance in the same direction until the amplitude begins to fall, then return to the distance that produces the highest amplitude.
10. If the first distance adjustment causes the peak amplitude to lower, return to the original starting point and begin adjusting the distance in the opposite direction.
11. When the distance is found that gives the highest peak amplitude, adjust all three screws for maximum peak amplitude and consistency.
12. Finish by adjusting all three screws on the dichroic mirror for maximum peak amplitude and consistency.
13. Replace the Optics Platform cover.
14. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.
15. Repeat steps 7 - 12 for the red laser lens assembly and red side dichroic mirror, if necessary.
16. Calibrate and verify the results.

Perform the Final Optics Alignment Test

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the Optics Platform cover.
3. Perform Optical Channel Verification:
4. Create a long sample. Under Region Map Setup, click Deactivate All. Set the uptake volume to 200 microliters with a total count of 1,000,000.
5. Set the CL1 gain value to 90 in the service software and select the box for Disable Temp. Comp.
6. With your vortexed alignment microspheres as a sample, click Acquire Well

7. Click on Tools and open a DotPlot. Select the Decaying DotPlot button.
8. When you see the bead population, cancel the acquisition.
9. Right-click in the DotPlot, and select Region > Create. Place the crosshairs in the middle of the bead population, then left-click to create a region.
10. Click Acquire Well to run the alignment microspheres a second time.
11. While the sample is running, view the real-time %CV's in the service software.

NOTE: Ensure %CV's are within specification. If they are not, perform optical alignment, then repeat these steps.

Adjust Flow Rate

1. Create a long sample with an uptake volume of 200 microliters and a "total count" of 1,000,000.
2. Disconnect the waste tubing from the analyzer and connect a ~10" tubing with a male orange quick-disconnect to the waste outlet.
3. Place the 10" tube in a beaker or container and perform a Wash to remove air from the tube.
4. Run the long sample of DI water or sheath fluid.
5. While the sample is running, collect waste flow in a 15 mL graduated tube for 2 minutes. The volume collected should be within the target range of 10.4 mL to 11.2 mL in 2 minutes (5.2 mL to 5.6 mL per minute).
6. Make pressure adjustments as needed to get the flow rate within specifications.

NOTE: The air pressure should be within specification (6.0 to 9.0 PSI).

Remove and Replace Procedures

Always decontaminate the system before performing maintenance procedures. Use an ESD strap when working on electronic components.

Safety

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Always wear personal protective equipment suitable for the activity.
- Follow all warnings marked on the instrument.
- Follow all biological safety instructions
- Follow all safety, warnings, cautions, and notes

Replace the Sheath Delivery System with Sheath Bottles

You may need to replace the Sheath Delivery (SD) system with sheath bottles for maintenance or troubleshooting.

1. Replace With the SD still connected, perform a Warm Up via the software.
2. When the pressure has stabilized, note the value of the sheath pressure in the software.
3. Open the access door on the analyzer. Use a screwdriver to turn the regulator approximately five full turns to the left (counterclockwise).
4. Shut down the SD and disconnect it from the analyzer.
5. Attach the sheath bottle (preferably full) to the analyzer.
6. Open and close the sheath bottle cap in order to release any remaining pressure within the system.
7. Perform a **Warm Up** via the software, if the compressor has turned off.
8. When the pressure has stabilized, note the value of the sheath pressure in the software.
9. If the pressure is equal to the value noted earlier (within +/-0.1psi), then the setup is complete. If not, continue to the next step. After each adjustment, you will have to release the pressure and let it build back up again in order to obtain an accurate pressure reading.

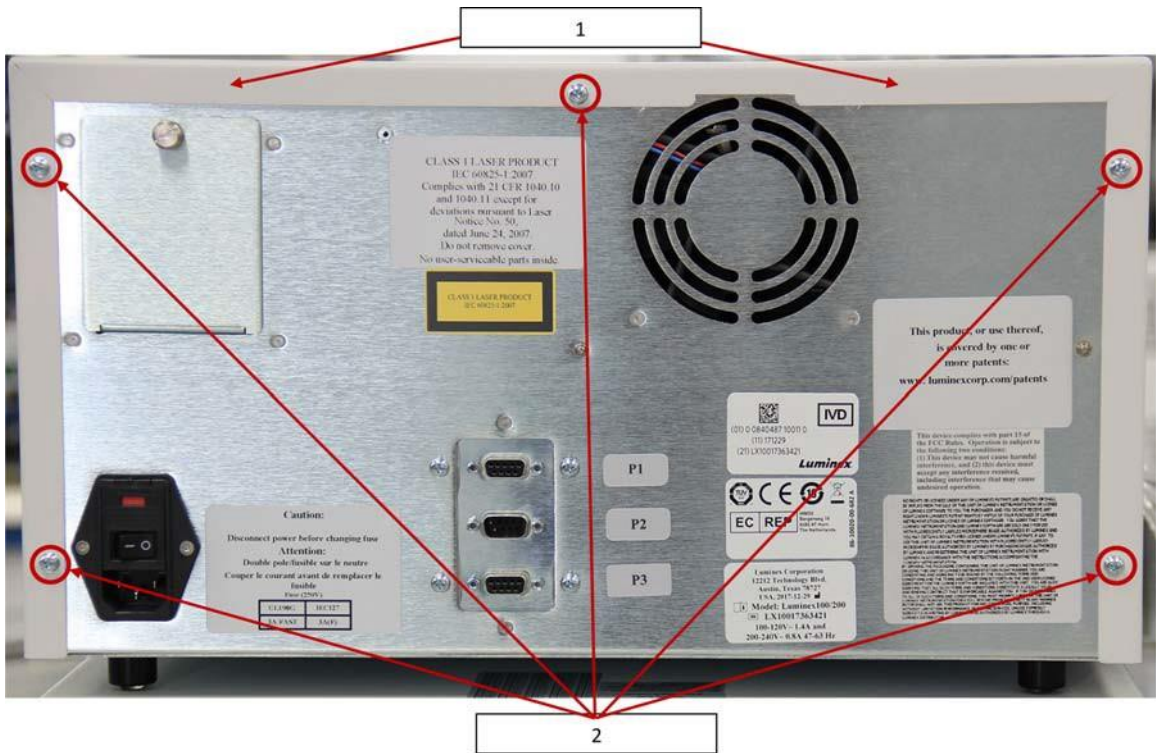
NOTE: Any time the compressor turns off while making adjustments, perform another **Warm Up** or manually turn on the Sheath/Air Pump to maintain pressure in the system.

10. On the analyzer, turn the regulator clockwise (right) to increase the pressure or counterclockwise to decrease the pressure. There is no exact measurement of pressure increase or decrease for each turn of the regulator. Try one full turn in the required direction to start.
11. Release the pressure by opening and closing the sheath bottle cap.
12. Repeat the following steps until you obtain the original Sheath pressure reading noted in step 2, within +/- 0.1 psi.

NOTE: Do not move the sheath bottle or sheath line while the system is in use.

- a. On the analyzer, turn the regulator clockwise (right) to increase the pressure or counterclockwise to decrease the pressure. There is no exact measurement of pressure increase or decrease for each turn of the regulator. Try one full turn in the required direction to start.
- b. Release the pressure by opening and closing the sheath bottle cap.

Luminex® 200™ RoHS Analyzer Cover



1	Analyzer Cover
2	5 x Phillips head screws

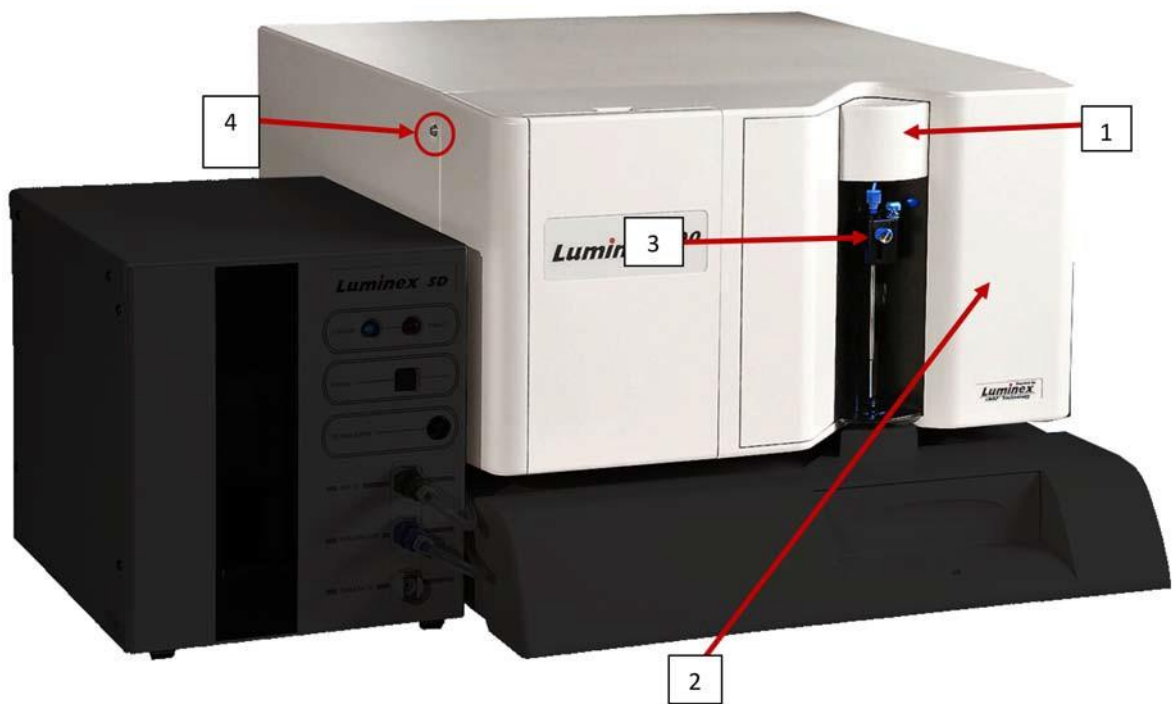
Remove the Analyzer Cover

- 1. Turn off the analyzer but leave it plugged in.
- 2. Remove the five screws on the rear of the analyzer and the four screws on the sides of the analyzer (two on each side).
- 3. Push the cover toward the back of the analyzer.
- 4. Lift the cover off the analyzer.

Replace the Analyzer Cover

- 1. Set the analyzer cover toward the rear of the analyzer.
- 2. Make sure the alignment flanges on the top of the analyzer are correctly aligned.
- 3. Slide the cover toward the front of the analyzer into place.
- 4. Replace the five screws.

Luminex® 200™ RoHS Bezel



1	LED Light Housing	3	Sample Probe Holder
2	Analyzer Bezel	4	Phillips Head screws (only 1 of 4 shown)

Remove the Bezel

1. Turn off the analyzer, but leave it plugged in.
2. Detach the blue LED light housing and pull it forward to reveal the quick-disconnect fitting. Disconnect the LED housing and set it aside.
3. Unscrew the brown Cheminert® fitting and remove it from the probe housing.
4. Loosen the probe adjustment screw one-third to one-half of a turn. (Do not loosen it more than one-half of a turn.)
5. Slide the probe adjustment screw to its highest point and tighten it.
6. Remove the probe.
7. Unscrew the probe housing mounting screw and remove the probe housing from the arm.
8. Using a Phillips screwdriver, remove the four screws on the outside of the analyzer.
9. Pull forward on the bezel to remove it.

Replace the Bezel

1. Slide the bezel onto the front of the analyzer.
2. Route the LED cable through the circular hole inside the bezel. This is easiest when the bezel is lined up with and resting on the bottom of the analyzer chassis.
3. Attach the light to the quick-disconnect fitting and install the light into the bezel.
4. Replace the screws on the bezel.
5. Reinstall the probe housing.
6. Install the probe.
7. Replace the brown Cheminert® fitting in the probe housing.
8. Adjust the probe height.

Luminex® 200™ RoHS Air Filter (HEPA)

FIGURE 33. HEPA Air Filter (behind door)



Remove the Air Filter (HEPA)

1. Turn off the analyzer but leave it plugged in.
2. Unscrew the thumb screw on the back of the analyzer.
3. Remove the door covering the filter.
4. Pull the filter toward you.
5. Hold on to the tubing and cut it off as close as possible to the filter. Do not let the tubing fall inside the machine.

Replace the Air Filter (HEPA)

1. Push a new filter onto the tubing, making sure that the tubing goes about 1/2 inch onto the connector.
2. Place the filter back into its original position.
3. Replace the door covering the filter.
4. Replace and tighten the thumb screw.

Luminex® 200™ RoHS Fuses

FIGURE 34. Luminex 200 Fuses



Remove the Fuses

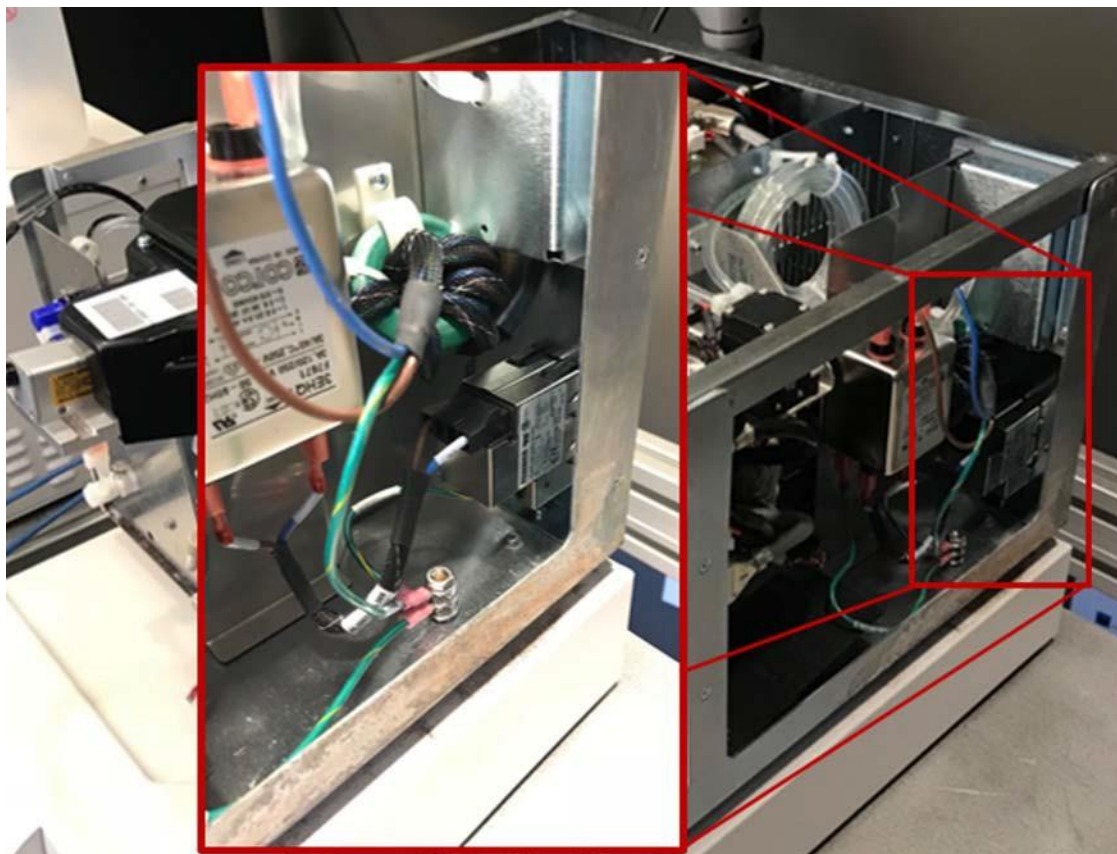
1. Turn off and unplug the analyzer.
2. Locate the slot on the outside of the power inlet. It is at the top, right above a cut-out through which you can see the red fuse holder.
3. Insert a flathead screwdriver in the slot and pry the cover out.
4. Pull the cover forward.
5. Remove the red fuse holder.
6. Remove the old fuses from the fuse holder.

Replace the Fuses

1. Insert the new fuses in the fuse holder. Refer to the label directly to the right of the power inlet for information about the correct replacement fuses.
2. Push the fuse holder back into place.
3. Snap the cover back into place.
4. Plug in and turn on the analyzer.

Luminex® 200™ RoHS AC Power Inlet

FIGURE 35. AC Power Inlet



Remove the AC Power Inlet

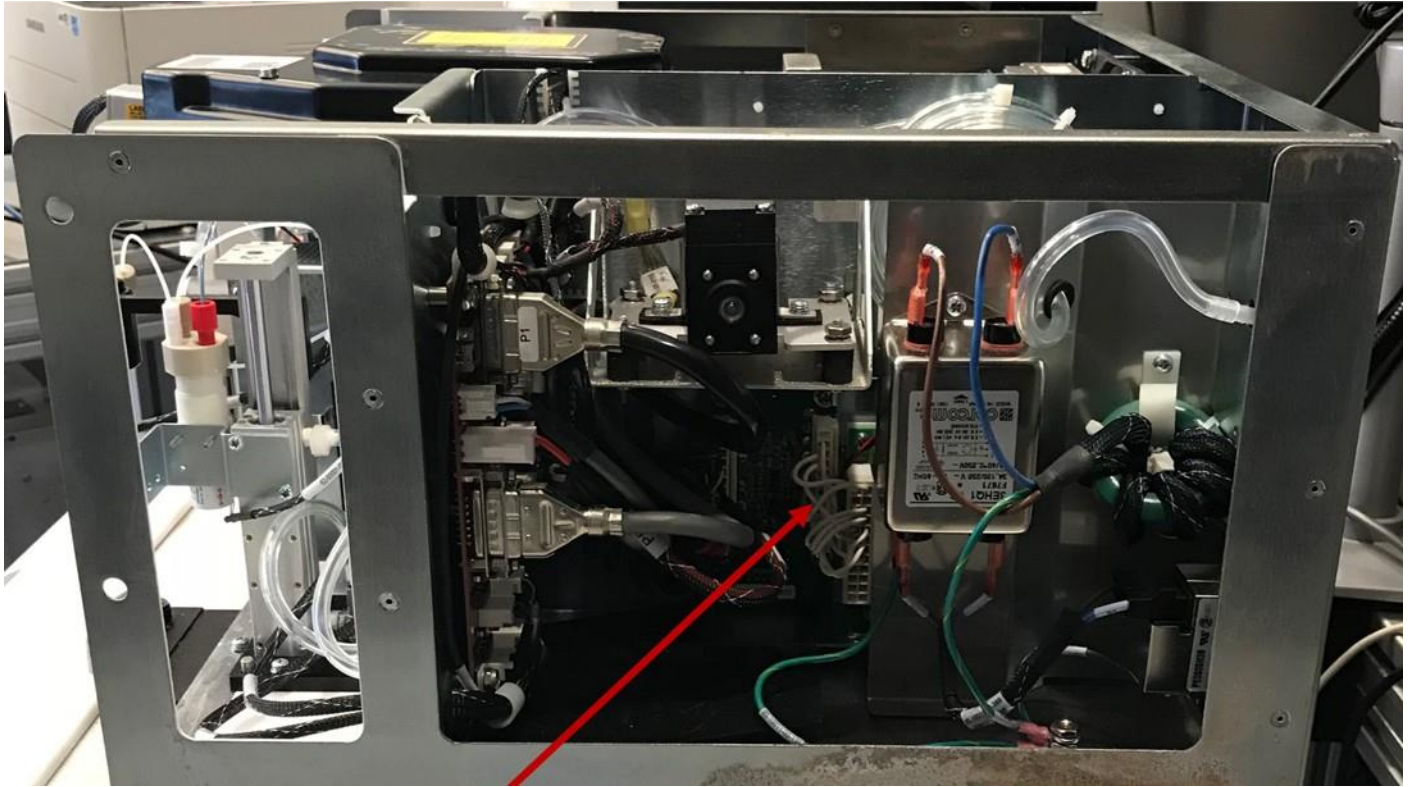
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *"Remove the Analyzer Cover"*.
3. Disconnect the plug from the back of the AC power inlet.
4. Remove the nut securing the green wire from the chassis ground stud.
5. Remove the screws from the AC power inlet on the back of the analyzer.
6. Using pliers, push down on the clips holding the power inlet in place. Push down on both clips at the same time, while lifting up on the bottom of the power inlet. Push firmly toward the outside of the analyzer.

Replace the AC Power Inlet

1. Push the new power inlet into place until the clips catch.
2. Replace the screws from the AC power inlet on the back of the analyzer.
3. Replace the nut securing the green wire to the chassis ground stud.
4. Reconnect the plug going into the back of the power inlet.
5. Plug in and test the analyzer.
6. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Power Supply

FIGURE 36. Power Supply (located behind the barrier)



Remove the Power Supply

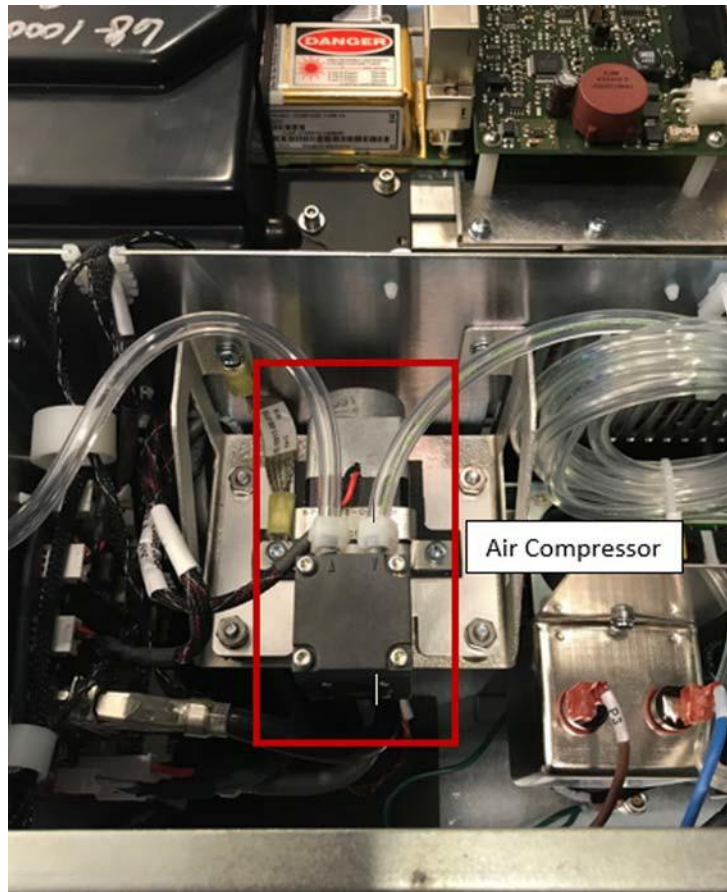
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *"Remove the Analyzer Cover"*.
3. Disconnect the AC power going into the power supply.
4. Disconnect the cable on the front of the power supply (the one that goes toward the front of the analyzer).
5. Remove the green laser controller shelf.
 - a. Remove the screw from the corner air deflector, then remove the corner air deflector from the shelf.
 - b. Remove the two screws from the fan air deflector, then remove the fan air deflector from the shelf.
 - c. Remove the two screws holding the shelf lockdown plate on the inner wall of the chassis that support the shelf. Note the position of the shelf in reference to the opening on the chassis wall.
 - d. Remove the lockdown plate.
 - e. Remove the two screws on the outer edge of the chassis that support the shelf in place.
 - f. Remove the two accessible cables connected to the controller (not the ribbon cable).
 - g. Release the ribbon cable from the cable clamp. Note the way that the ribbon cable is folded and positioned for replacement of the shelf.
 - h. Remove the two screws that secure the shelf to the air duct under the heat sink of the laser head.
 - i. Remove the shelf by pushing in on the outer edge of the shelf just below the two mounting screw holes near the outer edge of the chassis.
 - j. Gently lift the shelf. When the bottom end of the side you are lifting clears the chassis, you can completely remove the shelf. It may be helpful to gently tilt the shelf from side to side as you attempt to lift it out of the chassis.
 - k. Turn the shelf, with the controller, upside down and lay it on top of the optical platform cover.
6. Remove the four screws holding the power supply in place.
7. Lift the power supply out.

Replace the Power Supply

1. Put the new power supply into place and hold it there.
2. Replace the four screws.
3. Replace the green laser controller box shelf in its original position.
 - a. Gently lower the shelf, with the controller, back into the chassis. Make sure to insert the shelf into the correct openings on the chassis wall.
 - b. Replace the two screws that secure the shelf to the air duct under the heat sink of the laser head.
 - c. Replace the two accessible cables connected to the controller.
 - d. Replace the two screws on the outer edge of the chassis that hold the shelf in place.
 - e. Replace the shelf lockdown plate, then replace the two screws that hold the shelf lockdown plate onto the inner wall of the chassis.
 - f. Replace the fan air deflector and the two screws that hold it in place.
 - g. Replace the corner air deflector and the screw that holds it in place.
4. Reconnect the cable on the front of the power supply.
5. Reconnect the AC power going into the power supply.
6. Plug in and test the analyzer.
7. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Air Compressor

FIGURE 37. Air Compressor



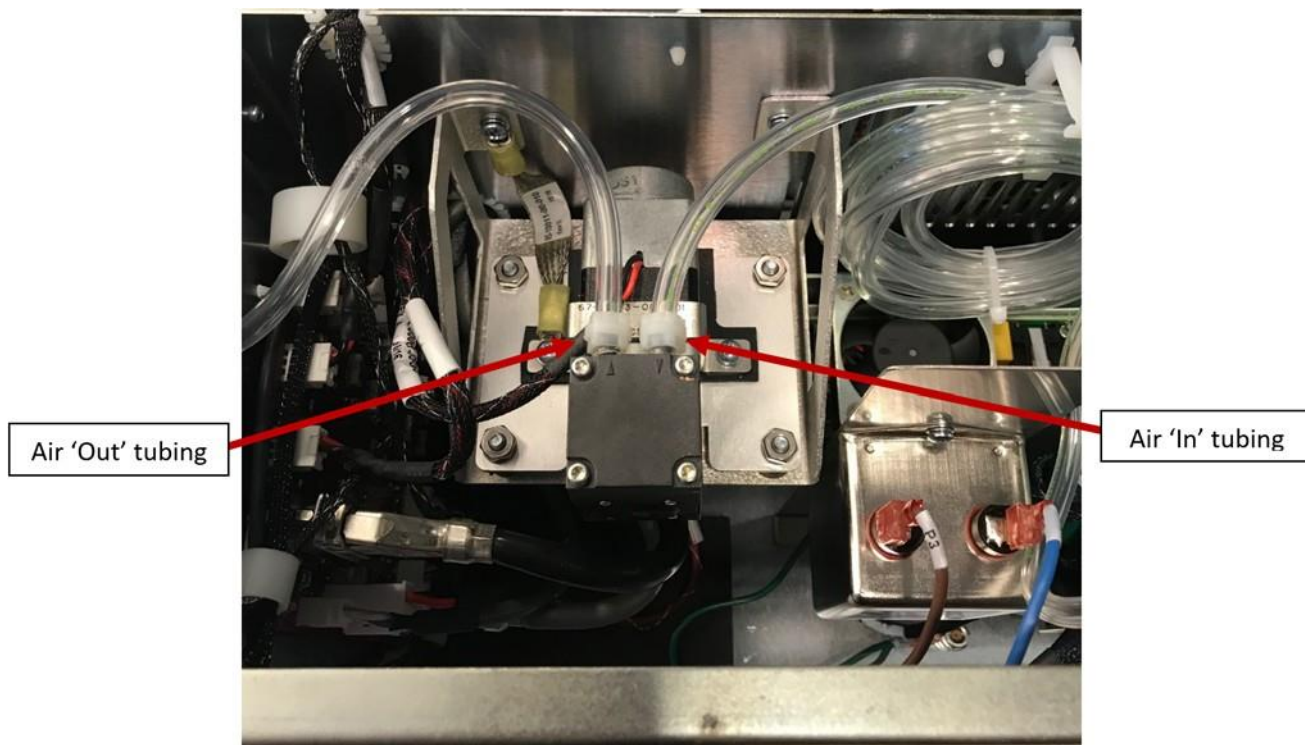
Remove the Air Compressor

1. Turn off and unplug the analyzer.

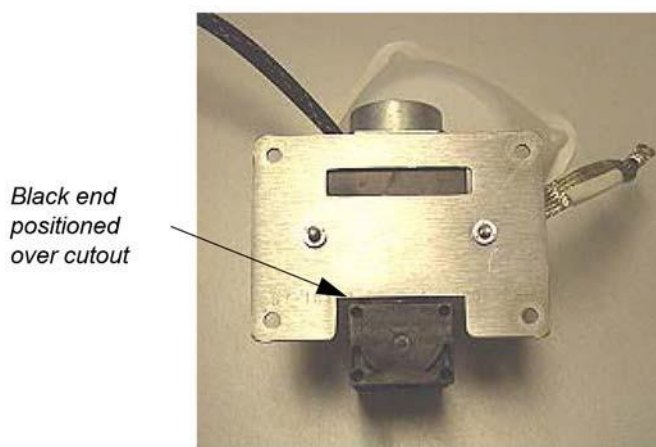
2. Remove the analyzer cover. See *"Remove the Analyzer Cover"*.
3. Detach the air "in" and air "out" tubing from the air compressor.
4. Unscrew the #6 socket caps and remove the U-bracket.
5. Remove the two screws securing the U-bracket and detach the wire, making note of its orientation.
6. Disconnect the air compressor wire from J4 on the distribution board.
7. Lift out air compressor.

Replace the Air Compressor

1. Install the new air compressor on top of the rubber gasket on the mounting plate.



2. Install the U-bracket and attach the grounding wire.
3. Replace the two screws that secure the U-bracket.

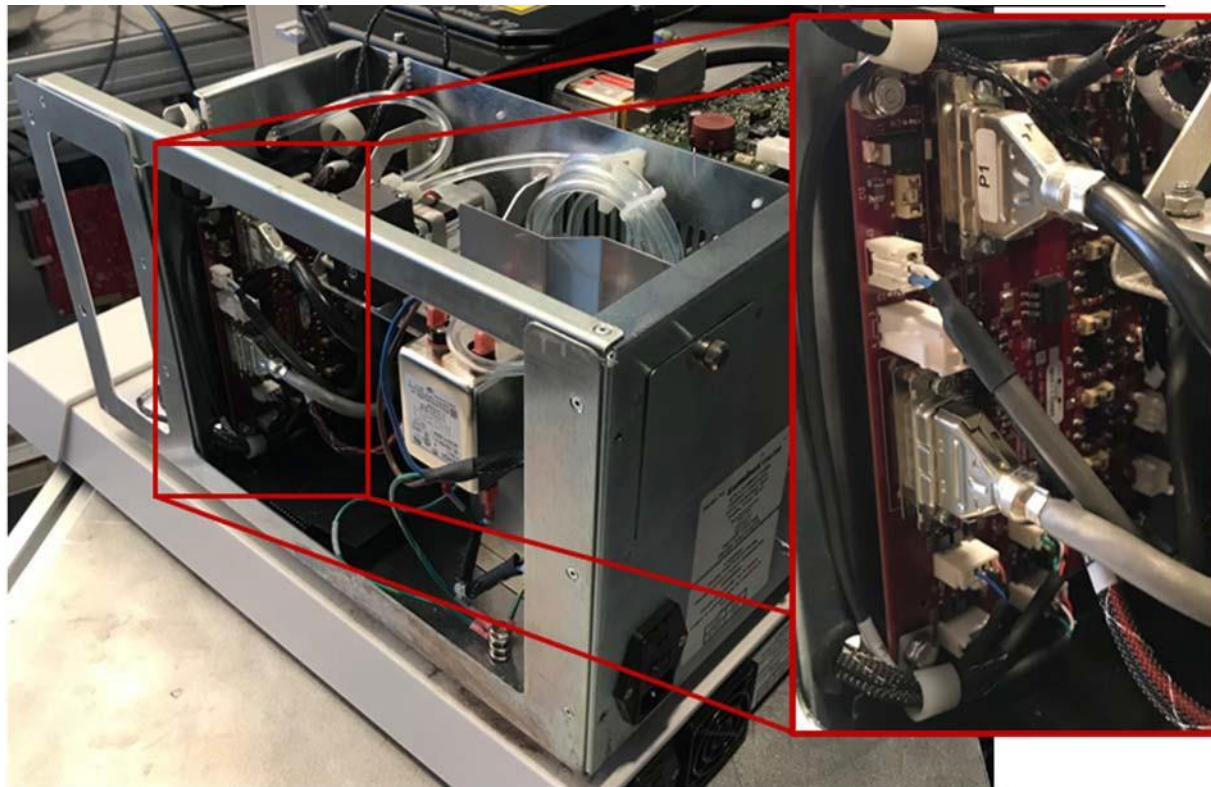


4. Connect the air compressor wire to J4 on the distribution board.
5. Connect the end of air "out" tubing to the compressor. The air "out" tube is attached on the left when viewing the compressor from the perspective.

6. Connect one end of the provided 76 inches length of tubing to the air compressor. The air “in” tube is attached on the right when viewing the compressor from the perspective.
7. Turn the analyzer on and listen for the air compressor to activate during the start-up sequence.
8. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Sensor Distribution Board

FIGURE 38. Luminex® 200™ RoHS Sensor Distribution Board



Remove the Sensor Distribution Board

1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the line in and line out tubing from the compressor.
4. Disconnect the compressor from the distribution board at J4.
5. Supporting the shelf bracket for the compressor, remove the three screws that secure the shelf to the chassis.
6. Remove the shelf bracket and compressor.
7. Remove all of the cables from the distribution board, making note of their locations on the board as you remove them.
8. Remove the four nuts and washers at the corners of the board.
9. Remove the cable clamps.
10. Pull out the air flow dampener positioned at the bottom of the distribution board and put it to the side.
11. Pull out the distribution board.

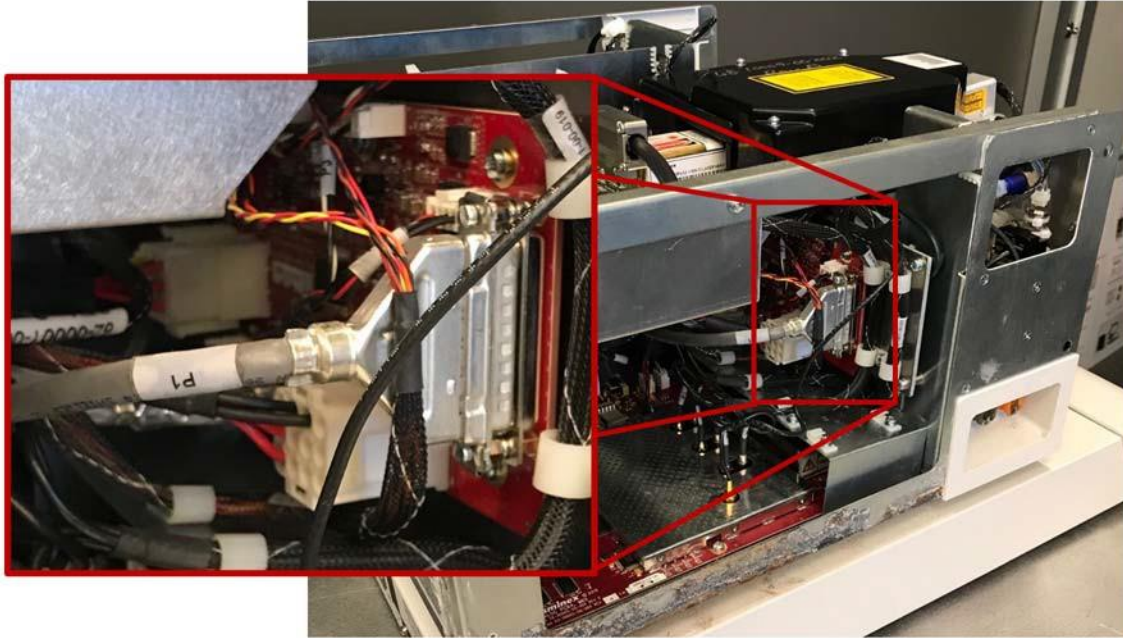
Replace the Sensor Distribution Board

1. Hold the new distribution board in place. Position the top cable clamps and fasten the top two nuts.
2. Slide the air flow dampener back into place.

3. Position the bottom cable clamps and fasten the two bottom nuts.
4. Reinstall all cables to their original positions. See *Figure 8*.
5. Supporting the shelf bracket with the compressor, install the three screws which secure the shelf to the chassis.
6. Connect the compressor to the distribution board at J4.
7. Reconnect the line in and line out tubing from the compressor.
8. Replace the cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Laser Driver Board

FIGURE 39. Luminex® 200™ RoHS Laser Driver Board



Remove the Laser Driver Board

1. Turn off and unplug the analyzer.
2. Remove the front bezel. See *"Remove the Bezel"*.
3. Remove the analyzer cover. See *"Remove the Analyzer Cover"*.
4. Remove the green laser controller shelf.
 - a. Remove the screw from the corner air deflector, then remove the corner air deflector from the shelf.
 - b. Remove the two screws from the fan air deflector, then remove the fan air deflector from the shelf.
 - c. Remove the two screws holding the shelf lockdown plate on the inner wall of the chassis that support the shelf. Note the position of the shelf in reference to the opening on the chassis wall.
 - d. Remove the lockdown plate.
 - e. Remove the two screws on the outer edge of the chassis that support the shelf in place.
 - f. Remove the two accessible cables connected to the controller (not the ribbon cable).
 - g. Release the ribbon cable from the cable clamp. Note the way that the ribbon cable is folded and positioned for replacement of the shelf.
 - h. Remove the two screws that secure the shelf to the air duct under the heat sink of the laser head.
 - i. Remove the shelf by pushing in on the outer edge of the shelf just below the two mounting screw holes near the outer edge of the chassis (step E above).
 - j. Gently lift the shelf. When the bottom end of the side you are lifting clears the chassis, you can completely remove the shelf. It may be helpful to gently tilt the shelf from side to side as you attempt to lift it out of the chassis.

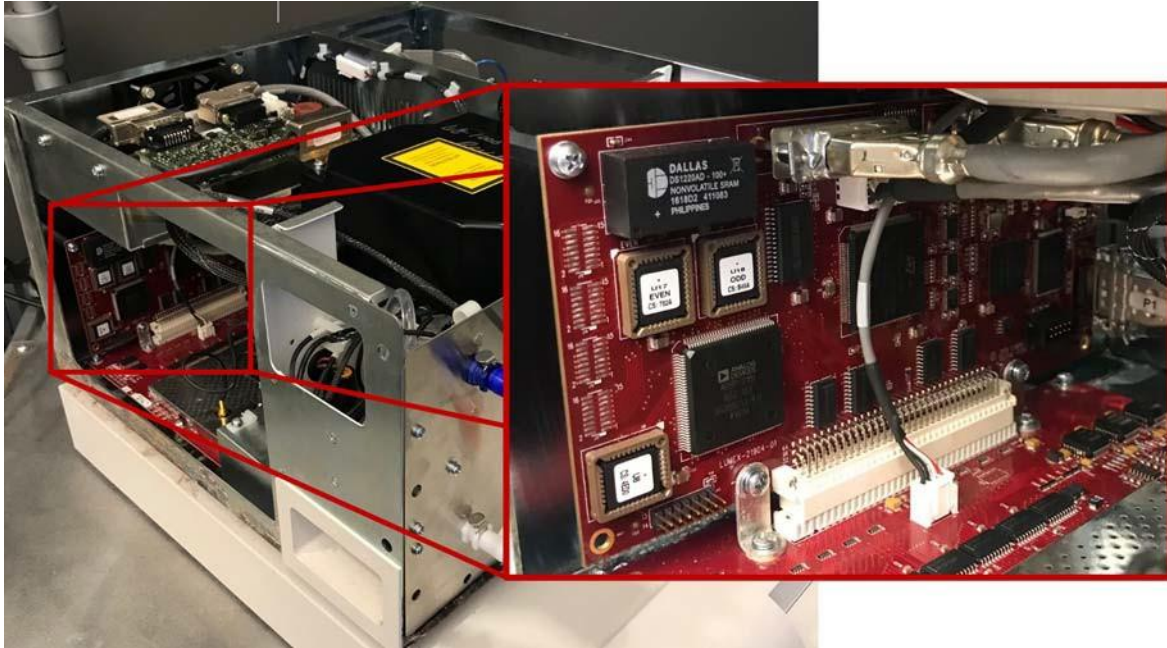
- k. Disconnect the ribbon cable from the controller box. Place the green laser controller shelf to one side.
5. Unscrew the four screws holding down the optics cover and lift the optics cover off of the analyzer.
6. Remove the four screws which secure the platform to the chassis, taking care not to damage the components in the area.
7. Lift the back end of the platform up slightly to enable better access.
8. Remove the flexible temperature cable from the cable clamps on the laser driver board and the high-voltage shield.
9. Remove all cables from the laser driver board.
10. Remove the five screws holding the laser driver board to the chassis, and one screw to the flexible temperature cable clamp.
11. Pull out the laser driver board.

Replace the Laser Driver Board

1. Hold the new laser driver board in place.
2. Replace the six screws that fasten the laser driver board to the chassis.
3. Connect all cables to their original positions.
4. Replace the flexible temperature cable to the cable clamps on the laser driver board and to the high-voltage shield.
5. Align the platform with the mounting holes on the chassis and secure with the four screws removed previously.
6. Install the optics cover. Reconnect the ribbon cable to the controller box.
7. Replace the green laser controller box shelf in its original position.
 - a. Gently lower the shelf, with the controller, back into the chassis. Make sure to insert the shelf into the correct openings on the chassis wall.
 - b. Replace the two screws that secure the shelf to the air duct under the heat sink of the laser head.
 - c. Replace the two accessible cables connected to the controller.
 - d. Replace the two screws on the outer edge of the chassis that hold the shelf in place.
 - e. Replace the shelf lockdown plate, then replace the two screws that hold the shelf lockdown plate onto the inner wall of the chassis.
 - f. Replace the fan air deflector and the two screws that hold it in place.
 - g. Replace the corner air deflector and the screw that holds it in place.
8. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex 200 RoHS Digital Board

FIGURE 40. Luminex® 200™ RoHS Digital Board



Remove the Digital Board

NOTE: Before replacing the digital board, ensure the APD breakdown and gain values can be read on the L-block. If unreadable, write these values down with the serial number, revision, and pressure parameters (manufacturing PSI, calibration PSI, and PSI tolerance %) per the maintenance software.

1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the green laser controller shelf
 - a. Remove the screw from the corner air deflector, then remove the corner air deflector from the shelf.
 - b. Remove the two screws from the fan air deflector, then remove the fan air deflector from the shelf.
 - c. Remove the two screws holding the shelf lockdown plate on the inner wall of the chassis that support the shelf. Note the position of the shelf in reference to the opening on the chassis wall.
 - d. Remove the lockdown plate.
 - e. Remove the two screws on the outer edge of the chassis that support the shelf in place.
 - f. Remove the two accessible cables connected to the controller (not the ribbon cable).
 - g. Release the ribbon cable from the cable clamp. Note the way that the ribbon cable is folded and positioned for replacement of the shelf.
 - h. Remove the two screws that secure the shelf to the air duct under the heat sink of the laser head.
 - i. Remove the shelf by pushing in on the outer edge of the shelf just below the two mounting screw holes near the outer edge of the chassis.
 - j. Gently lift the shelf. When the bottom end of the side you are lifting clears the chassis, you can completely remove the shelf. It may be helpful to gently tilt the shelf from side to side as you attempt to lift it out of the chassis.
 - k. Turn the shelf, with the controller, upside down and lay it on top of the optical platform cover.
4. Remove all cables from the digital board, making note of their installed locations.
5. Remove the two screws from the upper corners of the digital board.
6. Remove the two screws that hold the L-brackets to the analog board, and lift the digital board out.
7. Remove the two screws that hold the L-brackets to the digital board.
8. Remove the L-brackets from the digital board.

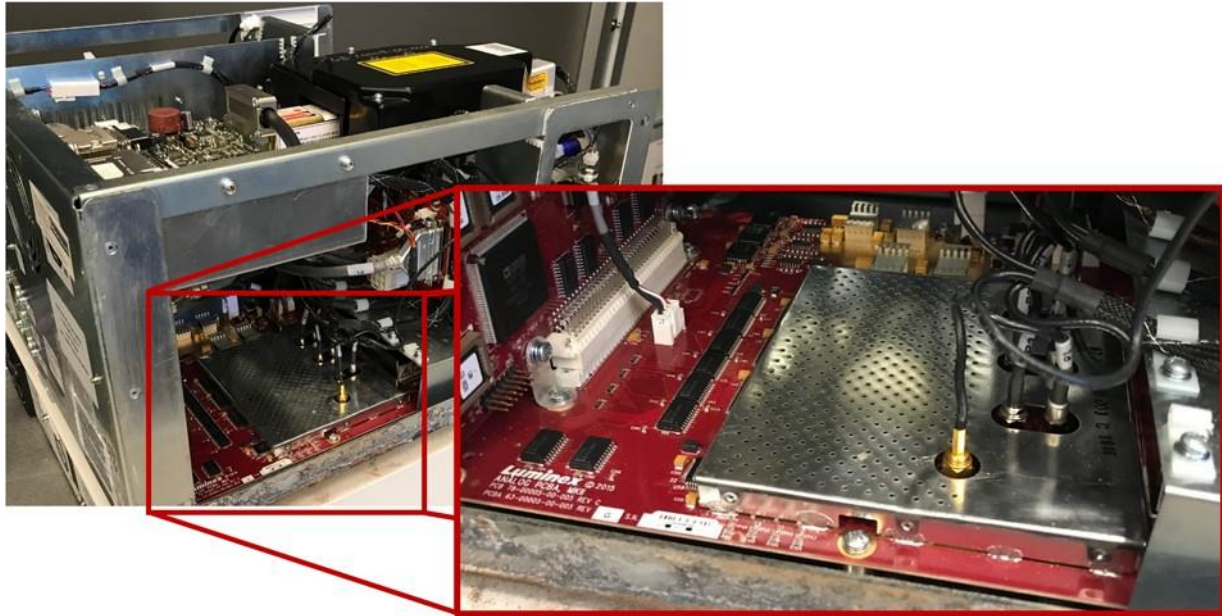
Replace the Digital Board

1. Install the L-brackets onto the new board, using the pieces you removed in steps 5 and 6. Do not fully tighten the screws.
2. Install the digital board.
3. Replace the two screws to the upper corners of the digital board.
4. Use the two screws to reconnect the digital board to the analog board with the L-brackets.
5. Finish tightening the screw you installed in step 1.
6. Connect all cables to their original positions. See *Figure 8*.
7. Replace the controller box shelf to its original position.
 - a. Gently lower the shelf, with the controller, back into the chassis. Make sure to insert the shelf into the correct openings on the chassis wall.
 - b. Replace the two screws that secure the shelf to the air duct under the heat sink of the laser head.
 - c. Replace the two accessible cables connected to the controller.
 - d. Replace the two screws on the outer edge of the chassis that hold the shelf in place.
 - e. Replace the shelf lockdown plate, then replace the two screws that hold the shelf lockdown plate onto the inner wall of the chassis.
 - f. Replace the fan air deflector and the two screws that hold it in place.
 - g. Replace the corner air deflector and the screw that holds it in place.
8. Power up the system.
9. Turn the lasers off using the maintenance software.
10. Unscrew the four screws holding down the optics cover and lift the optics cover off of the analyzer.
11. Using the maintenance software, enter the U-block APD gain (VR), breakdown (VB) values, manufacturing and calibrated PSI, and PSI Tolerance (5%). Remember to select **Write** when putting in the values so that they will be written to the digital board.
12. Enter the system serial number and revision with the maintenance software. Remember to select Write after putting in the values so that they will be written to the digital board.

NOTE: When replacing the digital board, ensure you upgrade the firmware on the Syringe Pump (Rev. D or newer) using the TECAN Firmware Download Tool to avoid introducing disconnect issues. If the Syringe Pump is a Rev. A, B, or C, the pump firmware cannot be upgraded and the pump must be replaced.
13. Replace the optics cover and replace the four screws that hold it in place.
14. Using the maintenance software, run Auto Offset for DD, CL1, CL2, and RP1.
15. Confirm that the digital board has the required firmware and DSP necessary for the appropriate software program. If it does not, use the maintenance software to install the required firmware and DSP.
16. Replace the analyzer cover. See *“Replace the Analyzer Cover”* for reference.

Luminex® 200™ RoHS Analog Board

FIGURE 41. Luminex® 200™ RoHS Analog Board



Remove the Analog Board

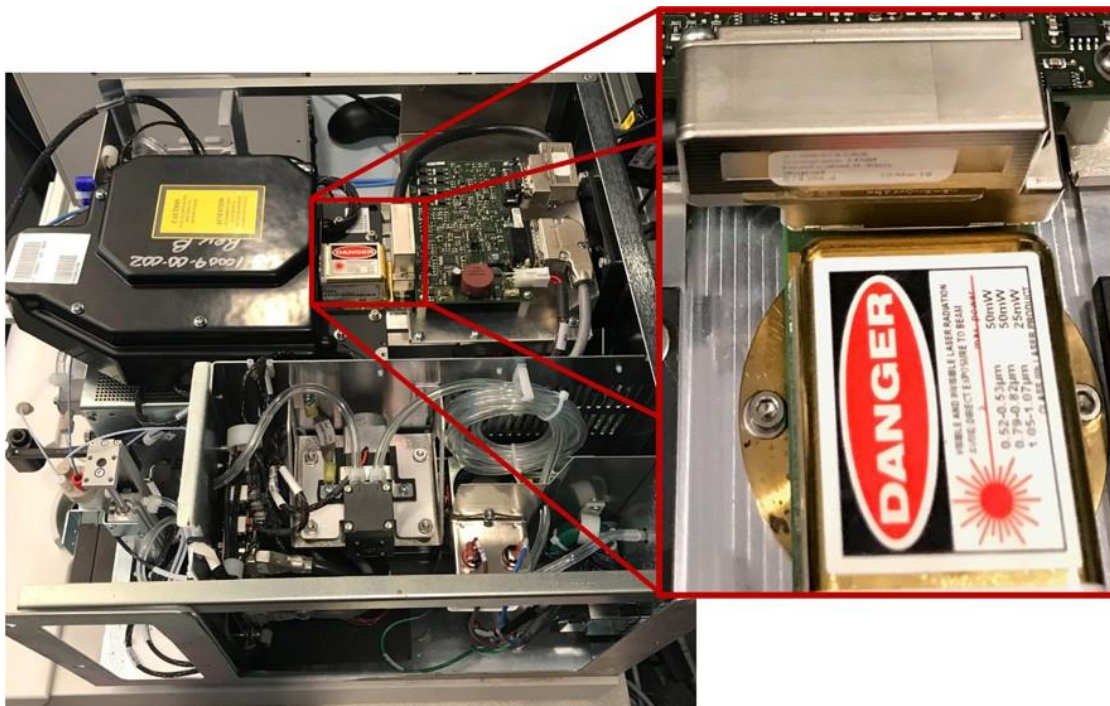
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the green laser controller shelf.
 - a. Remove the screw from the corner air deflector, then remove the corner air deflector from the shelf.
 - b. Remove the two screws from the fan air deflector, then remove the fan air deflector from the shelf.
 - c. Remove the two screws holding the shelf lockdown plate on the inner wall of the chassis that support the shelf. Note the position of the shelf in reference to the opening on the chassis wall.
 - d. Remove the lockdown plate.
 - e. Remove the two screws on the outer edge of the chassis that support the shelf in place.
 - f. Remove the two accessible cables connected to the controller (not the ribbon cable).
 - g. Release the ribbon cable from the cable clamp. Note the way that the ribbon cable is folded and positioned for replacement of the shelf.
 - h. Remove the two screws that secure the shelf to the air duct under the heat sink of the laser head.
 - i. Remove the shelf by pushing in on the outer edge of the shelf just below the two mounting screw holes near the outer edge of the chassis (step D above).
 - j. Gently lift the shelf. When the bottom end of the side you are lifting clears the chassis, you can completely remove the shelf. It may be helpful to gently tilt the shelf from side to side as you attempt to lift it out of the chassis.
 - k. Turn the shelf, with the controller, upside down and lay it on top of the optical platform cover.
4. Remove the digital board. See *“Luminex 200 RoHS Digital Board”*.
5. Remove all cables from the analog board, making note of their locations.
6. Locate the tubing connected to S1 (air pressure sensor). About six inches from the board, there is a quick-disconnect in the tubing. Disconnect it.
7. Remove all cabling from cable clamps on the high-voltage shield.
8. Using a flat-head screwdriver, loosen the two thumb screws holding the high-voltage shield to the chassis. These screws are below the bottom edge of the laser driver board.
9. Pull the high-voltage shield forward and remove it, taking care not to damage the air pressure sensor.
10. Remove the nine screws holding the analog board to the chassis, then lift the analog board out.

Replace the Analog Board

1. Place the new analog board into position.
2. Fasten the nine screws that hold the analog board to the chassis.
3. Connect the tubing from position S1.
4. Replace the high-voltage shield, taking care not to damage the air pressure sensor.
5. Using a flat-head screwdriver, tighten the two thumb screws to install the high-voltage shield to the chassis.
6. Replace the cabling to cable clamps on the high-voltage shield.
7. Connect all cables to their original positions.
8. Replace the digital board. See *"Replace the Digital Board"*.
9. Replace the controller box shelf to its original position.
 - a. Gently lower the shelf, with the controller, back into the chassis. Make sure to insert the shelf into the correct openings on the chassis wall.
 - b. Replace the two screws that secure the shelf to the air duct under the heat sink of the laser head.
 - c. Replace the two accessible cables connected to the controller.
 - d. Replace the two screws on the outer edge of the chassis that hold the shelf in place.
 - e. Replace the shelf lockdown plate, then replace the two screws that hold the shelf lockdown plate onto the inner wall of the chassis.
 - f. Replace the fan air deflector and the two screws that hold it in place.
 - g. Replace the corner air deflector and the screw that holds it in place.
10. Turn on the analyzer.
11. Run Auto Offset for DD, CL1, CL2, and RP1.
12. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Green Coherent Laser Head

FIGURE 42. Luminex® 200™ RoHS Green Coherent Laser Head



This procedure involves servicing the laser components of the Luminex 200 RoHS system.



Laser radiation when open. Avoid direct exposure to beam.

Remove the Green Coherent Laser Head

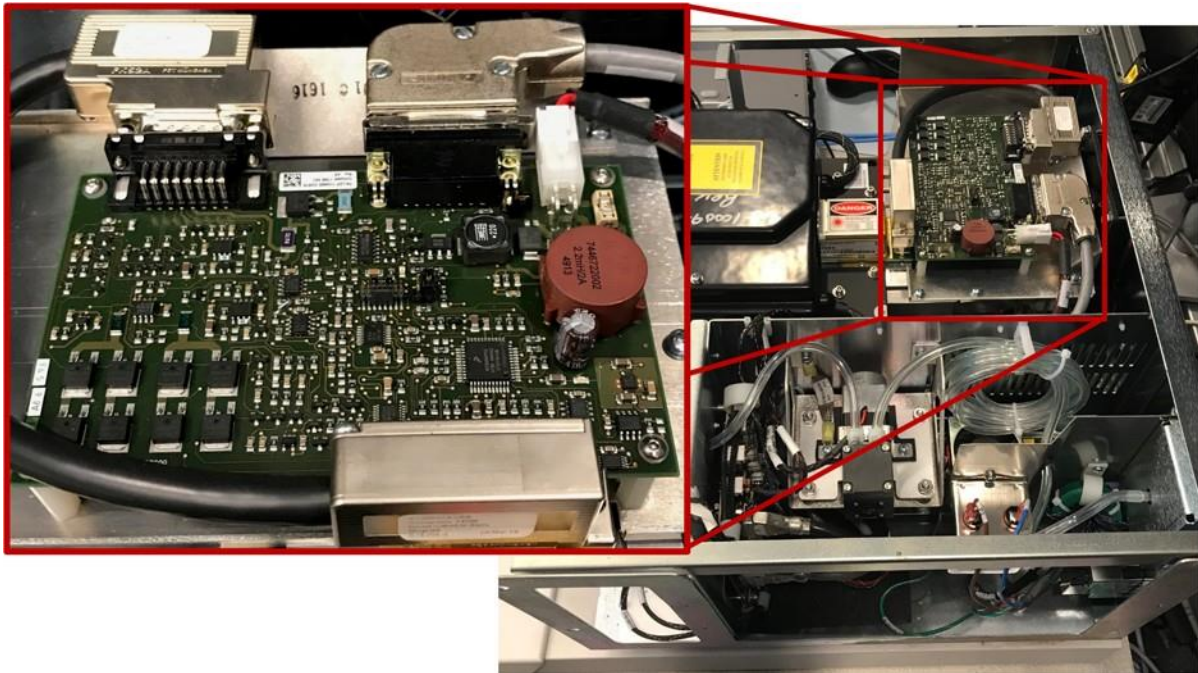
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the bezel. See *“Remove the Bezel”*.
4. Unscrew the four screws holding down the optics cover and lift the optics cover off of the analyzer.
5. Remove the screw which secures the laser head shield to the platform.
6. Remove the two screws holding the laser to the heat sink.
7. Lift out the Coherent laser and disconnect the cable from the laser head.
8. Remove the thin conductive film positioned between the laser head and the heat sink. Place the laser in an antistatic bag.

Replace the Green Coherent Laser Head

1. Install the thin conductive film between the laser head and the heat sink.
2. Connect the cable to the laser head and place it on top of the heat sink.
3. Replace the two screws that hold the laser to the heat sink.
4. Reinstall the laser head shield.
5. Using the maintenance software, reset the Green Laser time to 0 seconds. Select **Write** to write the value to the digital board.
6. Turn on the analyzer.
7. Complete the optical alignment procedure. See *“”*.
8. Replace the optics cover and replace the four screws that hold it in place.
9. Replace the bezel. See *“Replace the Bezel”*.
10. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Green Coherent Laser Controller

FIGURE 43. Luminex® 200™ RoHS Green Coherent Laser Controller



This procedure involves servicing the laser components of the Luminex 200 RoHS system.



Laser radiation when open. Avoid direct exposure to beam.

Remove the Green Coherent Laser Controller

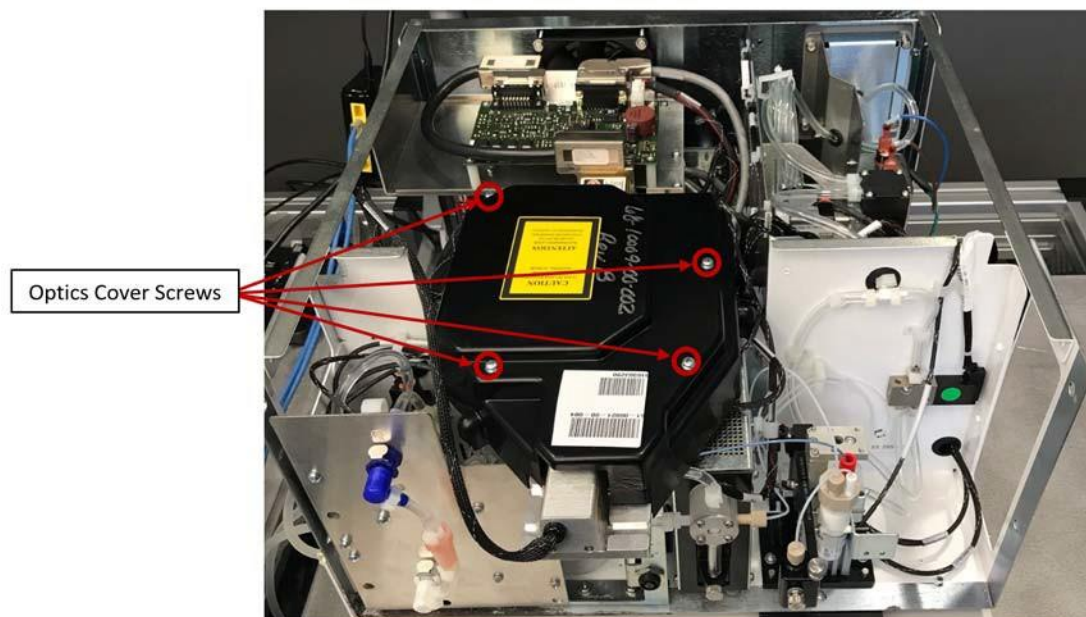
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the cables from the Coherent controller.
4. Remove the four screws holding the controller to the standoffs.
5. Lift out the Coherent controller. Place the controller in an anti-static bag.

Replace the Green Coherent Laser Controller

1. Position the new controller in place.
2. Replace the four screws that hold the controller to the standoffs.
3. Connect the cables to the controller board.
4. Turn on the analyzer.
5. Complete the optical alignment procedure. See *“Align the Optics for the Analyzer”*.
6. Replace the optics cover and replace the four screws that hold it in place.
7. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Optical Platform

FIGURE 44. Luminex® 200™ RoHS Optical Platform (Cover Screws)



This procedure involves servicing the laser components of the Luminex® 200™ RoHS system.



Laser radiation when open. Avoid direct exposure to beam

Remove the Optical Platform

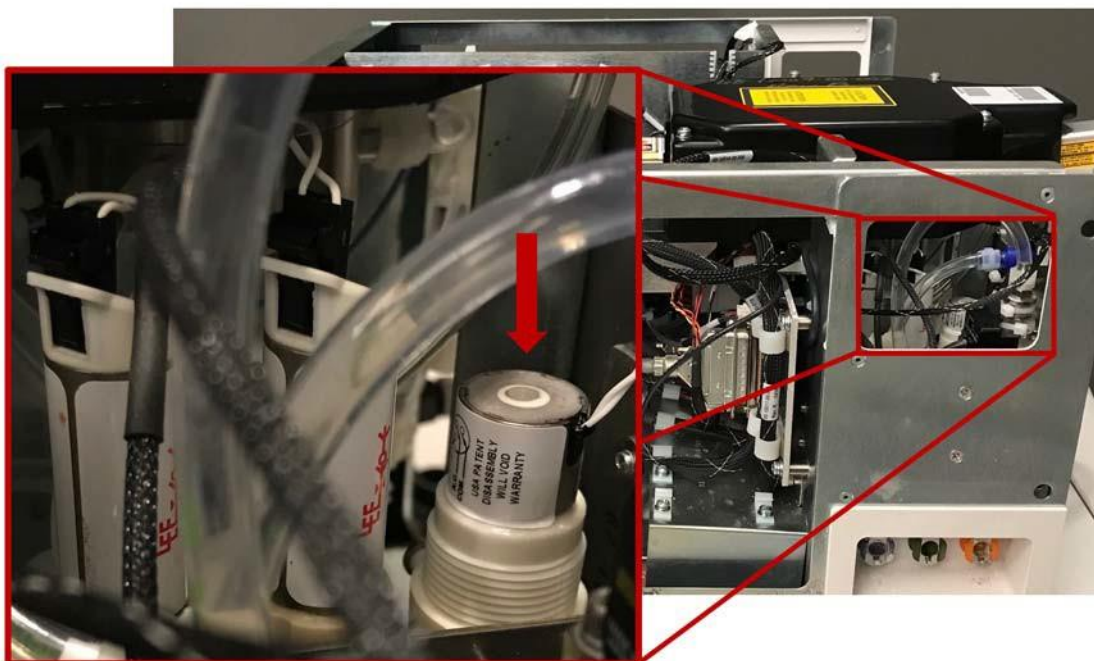
1. Perform decontamination procedure. See *"Maintaining the System"*.
2. Turn off and unplug the analyzer.
3. Remove the analyzer cover. See *"Remove the Analyzer Cover"*.
4. Remove the bezel. See *"Remove the Bezel"*.
5. Unscrew the four screws holding down the optics cover and lift the optics cover off of the analyzer.
6. Follow the tubing coming from the top of the cuvette and disconnect the waste line from the T-fitting.
7. Locate the line that runs from the backflush 3-way valve to the cuvette housing under the platform, and disconnect the end attached to the cuvette housing.
8. Disconnect the sheath line from the cuvette housing which is under the platform.
9. Disconnect the sample line from the bottom of the cuvette housing.
10. Remove the two cables on the left side of the green laser controller. Leave the ribbon cables attached.
11. Remove the four screws securing the controller box to the shelf.
12. Remove the two screws that secure the platform to the air duct.
13. Remove the four screws holding the platform to the test head bracket.
14. Disconnect the six cables from the L-block assembly, making notes of their positions.
15. Disconnect the red laser from the laser driver board at J3.
16. Remove the flexible temperature circuit from the analog board at J4 and release it from the cable clamps on the high voltage shelf and laser driver board.
17. Disconnect the PMT from the analog board at J7 and J3.
18. Remove the platform with the laser controller.

Replace the Optical Platform

1. Hold the new platform with the controller in place.
2. Replace the four screws that hold the platform to the test head bracket.
3. Install the two screws that secure the platform to the air duct.
4. Install the controller using the four screws previously removed. (Be certain the conductive film is placed between the controller and shelf.)
5. Reconnect the two cables on the left side of the controller.
6. Connect the PMT to the analog board at J7 and J3.
7. Install the flexible temperature circuit on the analog board at J4 and secure it in the cable clamps on the high voltage shelf and laser driver board.
8. Connect the red laser to the laser driver board at J3.
9. Connect the six cables to the L-block assembly.
10. Connect the sample line to the bottom of the cuvette housing.
11. Connect the sheath line to the front of the cuvette housing.
12. Connect the line that runs from the backflush 3-way valve to the cuvette housing.
13. Connect the waste line to the T-fitting.
14. Turn on the analyzer.
15. Complete the optical alignment procedure. See *"Align the Optics for the Analyzer"*.
16. Replace the optics cover and replace the four screws that hold it in place.
17. Replace the bezel. See *"Replace the Bezel"*.
18. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Backflush 3-Way Valve

FIGURE 45. Luminex® 200™ RoHS Backflush Valve



Release Pressure in the Luminex® Analyzer and Luminex® SD™

1. Perform Decontamination Procedure. See *"Maintaining the System"*.

NOTE: If you are not using an Luminex® SD™, turn off the system and open the sheath bottle cap to release the pressure in the analyzer.

2. Turn off and unplug the analyzer.
3. Remove the air tubing between the analyzer and the Luminex SD. (The air tubing has green quick-disconnects.)
4. Insert a screwdriver into the opening of the green quick- disconnect on each of the devices and depress the release valve.
5. Reconnect the air tubing.

Remove the Backflush 3-way Valve

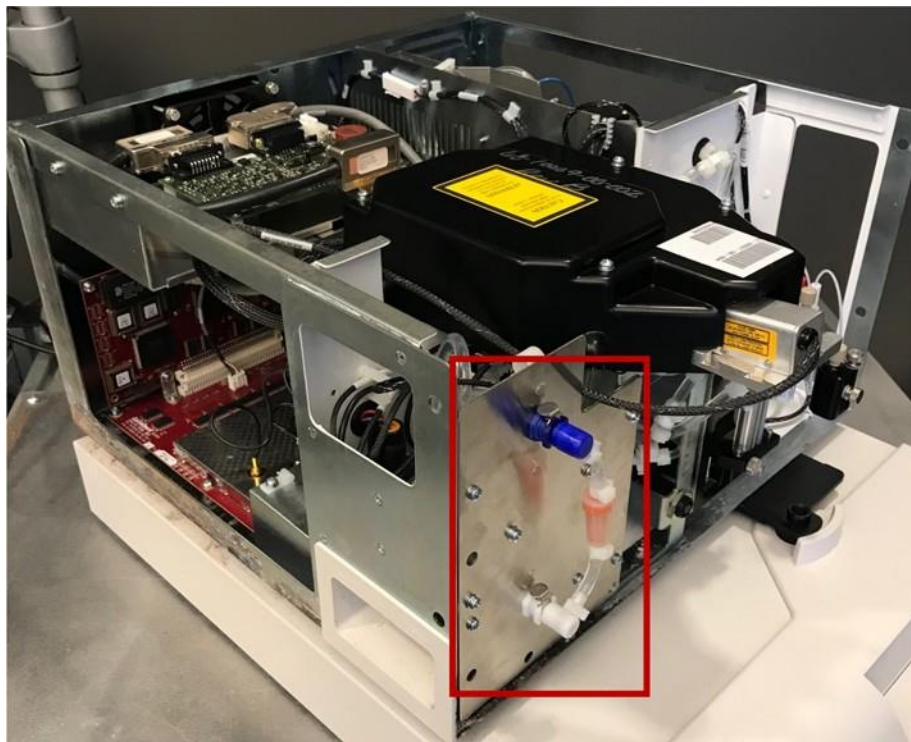
1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the bezel. See *“Remove the Bezel”*.
3. Remove the two screws that hold the fluidics panel to the side of the chassis.
4. Remove the two screws that hold the fluidics panel to the test head bracket.
5. Lean the fluidics panel forward.
6. Remove the three lines connected to the backflush 3-way valve. Note where each line attaches to the valve.
7. Disconnect the valve from J-12 on the Distribution board.
8. Remove the two screws securing the valve to the bracket.
9. Remove the backflush 3-way valve.

Replace the Backflush 3-way Valve

1. Hold the new valve in place.
2. Secure the valve to the bracket using the two screws previously removed.
3. Connect the valve to J-12 on the Distribution board.
4. Connect the three lines to the backflush 3-way valve in their appropriate locations.
5. Ease the fluidics panel in place and verify that the tubing lines are not crimped.
6. Replace the two screws that hold the fluidics panel to the test head bracket.
7. Replace the two screws that hold the fluidics panel to the side of the chassis.
8. Replace the bezel. See *“Replace the Bezel”*.
9. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Sheath Fluid Filter

FIGURE 46. Sheath Fluid Filter



Remove the Sheath Fluid Filter

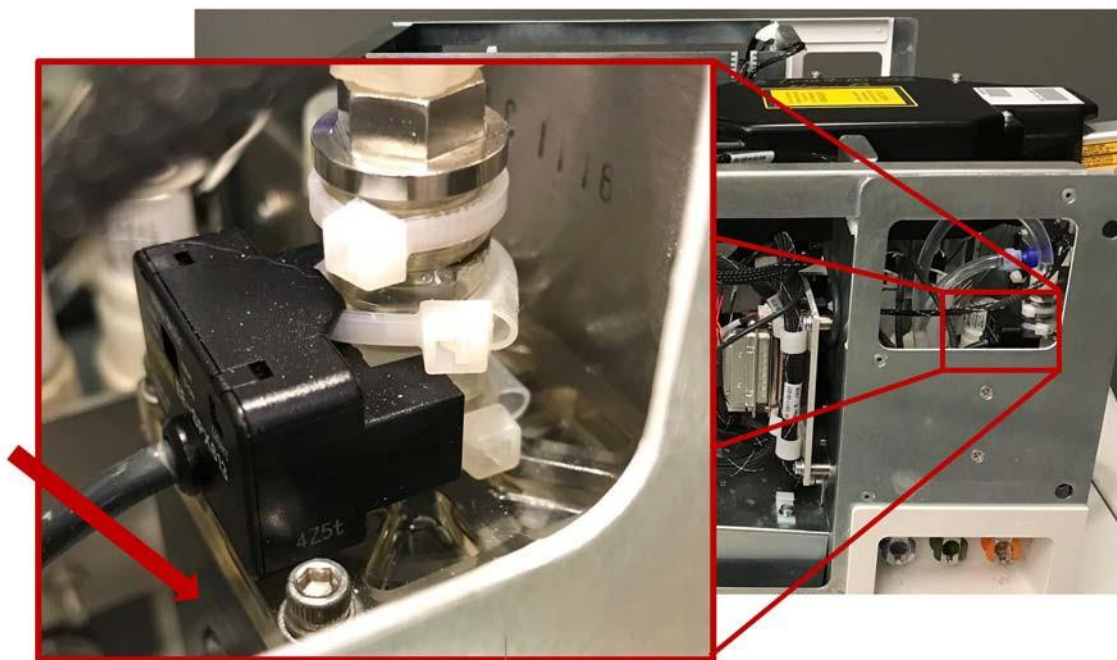
1. Turn off the analyzer but leave it plugged in.
2. Open the left front door on the analyzer.
3. Disconnect the quick-disconnect fittings from both ends of the sheath filter.
4. Remove the sheath fluid filter.

Replace the Sheath Fluid Filter

1. Hold the new sheath fluid filter in place and push in on each end of the filter until you hear a click. This ensures that the filter is in place.
2. Close the left front door.

Luminex® 200™ RoHS Bubble Trap

FIGURE 47. Luminex® 200™ RoHS Bubble Trap (Debubbler)



Release Pressure in the Luminex® Analyzer and Luminex® SD™

1. Perform Decontamination Procedure. See *“Maintaining the System”*.

NOTE: If you are not using an Luminex® SD™, turn off the system and open the sheath bottle cap to release the pressure in the analyzer.

2. Turn off and unplug the analyzer.
3. Remove the air tubing between the analyzer and the Luminex SD. (The air tubing has green quick-disconnects.)
4. Insert a screwdriver into the opening of the green quick- disconnect on each of the devices and depress the release valve.
5. Reconnect the air tubing.

Remove the Bubble Trap

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the bezel. See *“Remove the Bezel”*.
3. Remove the two screws that hold the fluidics panel to the side of the chassis.
4. Remove the two screws that hold the fluidics panel to the test head bracket.
5. Lean the fluidics panel forward.
6. Remove the two screws holding the bubble trap to the fluidics panel.
7. Pull the bubble trap away from the fluidics panel and clip the tie wraps holding the optical sensor to the bubble trap.
8. Disconnect the sensor from J10 on the distribution board and remove the sensor from the system.
9. Remove the three lines connected to the bubble trap. Note where each line attaches to the bubble trap.
10. Remove the bubble trap.

Replace the Bubble Trap

1. Hold the new bubble trap in place.
2. Connect the three lines to the bubble trap in their appropriate locations.
3. Replace the two screws holding the bubble trap to the fluidics panel.
4. Route sensor harness along the same path used previously and connect it to J10 on the distribution board.

5. Ease the fluidics panel in place and verify the tubing lines are not crimped.
6. Replace the two screws that hold the fluidics panel to the test head bracket.
7. Replace the two screws that hold the fluidics panel to the side of the chassis.
8. Replace the bezel. See *“Replace the Bezel”*.
9. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Debubbler Valve

FIGURE 48. Luminex® 200™ RoHS Debubbler Valve



Release Pressure in the Luminex® Analyzer and Luminex® SD™

1. Perform Decontamination Procedure. See *“Maintaining the System”*.
NOTE: If you are not using an Luminex® SD™, turn off the system and open the sheath bottle cap to release the pressure in the analyzer.
2. Turn off and unplug the analyzer.
3. Remove the air tubing between the analyzer and the Luminex SD. (The air tubing has green quick-disconnects.)
4. Insert a screwdriver into the opening of the green quick disconnect on each of the devices and depress the release valve.
5. Reconnect the air tubing.

Remove the Debubbler Valve

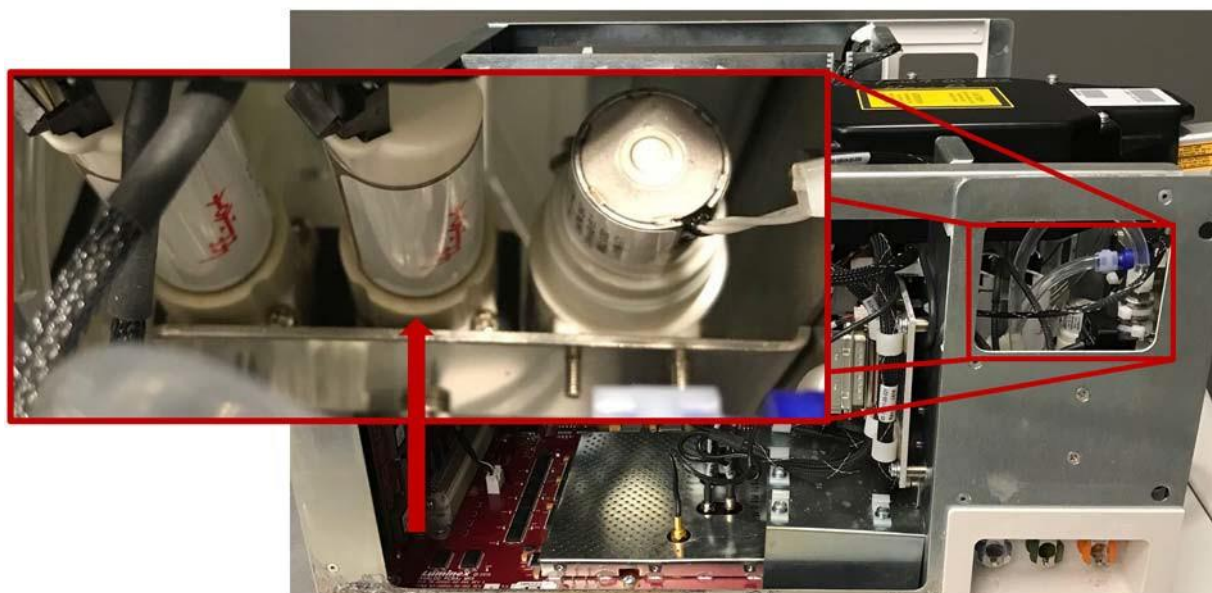
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the bezel. See *“Remove the Bezel”*.
4. Remove the two screws that hold the fluidics panel to the side of the chassis.
5. Remove the two screws that hold the fluidics panel to the test head bracket.
6. Lean the fluidics panel forward.
7. Remove the two lines connected to the debubbler valve.
8. Disconnect the debubbler valve from the 2-pin cable extension.
9. Remove the debubbler valve.

Replace the Debubbler Valve

1. Hold the new debubbler valve in place.
2. Connect the debubbler valve to the 2-pin cable extension.
3. Connect the two lines to the debubbler valve. (Locations are not specific.)
4. Ease the fluidics panel in place and verify that the tubing lines are not crimped.
5. Replace the two screws that hold the fluidics panel to the test head bracket.
6. Replace the two screws that hold the fluidics panel to the side of the chassis.
7. Replace the bezel. See *“Replace the Bezel”*.
8. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Sheath Valve

FIGURE 49. Luminex® 200™ RoHS Sheath Valve



Release Pressure in the Luminex® Analyzer and Luminex® SD™

1. Perform Decontamination Procedure. See *“Maintaining the System”*.
NOTE: If you are not using an Luminex® SD™, turn off the system and open the sheath bottle cap to release the pressure in the analyzer.
2. Turn off and unplug the analyzer.
3. Remove the air tubing between the analyzer and the Luminex SD. (The air tubing has green quick-disconnects.)
4. Insert a screwdriver into the opening of the green quick- disconnect on each of the devices and depress the release valve.
5. Reconnect the air tubing.

Remove the Sheath Valve

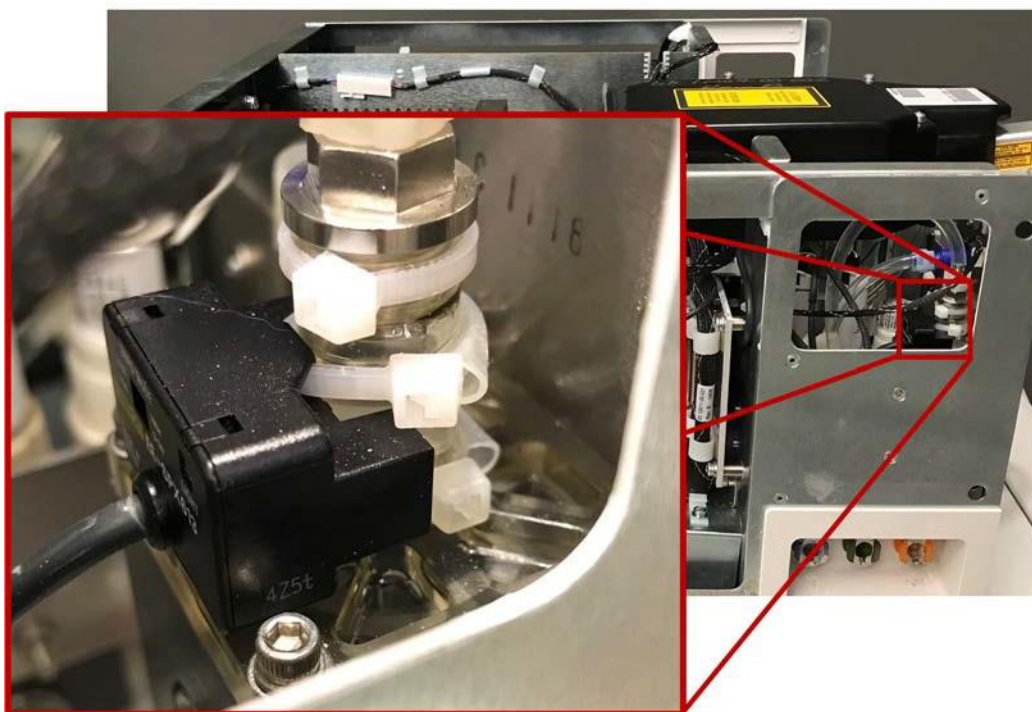
1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the bezel. See *“Remove the Bezel”*.
3. Remove the two screws that hold the fluidics panel to the side of the chassis.
4. Remove the two screws that hold the fluidics panel to the test head bracket; lean the fluidics panel forward.
5. Remove the two lines connected to the sheath valve.
6. Disconnect the sheath valve from the 2-pin cable extension.

Replace the Sheath Valve

1. Connect the sheath valve to the 2-pin cable extension.
2. Connect the two lines to the sheath valve. (Locations are not specific.)
3. Ease the fluidics panel in place. Verify that the tubing lines are not crimped.
4. Replace the two screws that hold the fluidics panel to the test head bracket.
5. Replace the two screws that hold the fluidics panel to the side of the chassis.
6. Replace the bezel. See *“Replace the Bezel”*.
7. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Bubble Trap Optical Sensor

FIGURE 50. Luminex® 200™ RoHS Bubble Trap Optical Sensor



Remove the Bubble Trap Optical Sensor

1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the bezel. See *“Remove the Bezel”*.
4. Remove the two screws that hold the fluidics panel to the side of the chassis.
5. Remove the two screws that hold the fluidics panel to the test head bracket, then lean the fluidics panel forward.
6. Remove the two screws holding the bubble trap to the fluidics panel.
7. Pull the bubble trap away from the fluidics panel and clip the tie wraps holding the optical sensor to the bubble trap.
8. Unplug the cable from J10 on the distribution board.
9. Remove the bubble trap optical sensor.

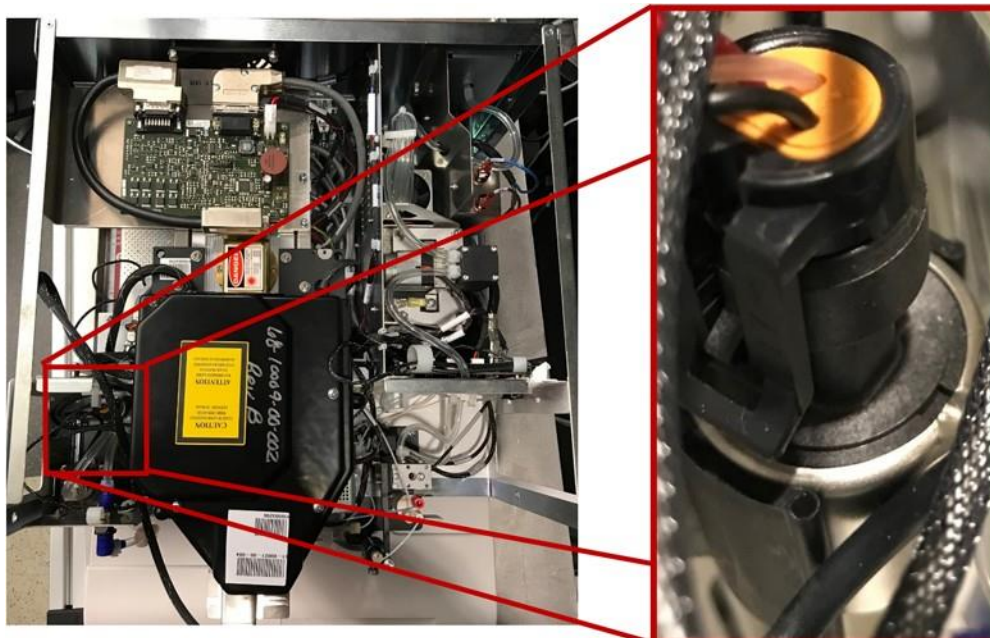
Replace the Bubble Trap Optical Sensor

1. Verify that the switch settings on the sensor are correct. (The top switch is to the right and bottom switch is to the left.)
2. Hold the new bubble trap optical sensor in place.
3. Attach the tie wraps holding the optical sensor to the bubble trap.

4. Replace the two screws holding the bubble trap to the fluidics panel.
5. Route the cable and plug it in to J10 on the distribution board.
6. Ease the fluidics panel in place. Verify that the tubing lines are not crimped.
7. Replace the two screws that hold the fluidics panel to the test head bracket.
8. Replace the two screws that hold the fluidics panel to the side of the chassis.
9. Replace the bezel. See *“Replace the Bezel”*.
10. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Pressure Transducer

FIGURE 51. Luminex® 200™ RoHS Pressure Transducer



Release Pressure in the Luminex® Analyzer and Luminex® SD™

1. Perform Decontamination Procedure. See *“Maintaining the System”*.
NOTE: If you are not using an Luminex® SD™, turn off the system and open the sheath bottle cap to release the pressure in the analyzer.
2. Turn off and unplug the analyzer.
3. Remove the air tubing between the analyzer and the Luminex SD. (The air tubing has green quick-disconnects.)
4. Insert a screwdriver into the opening of the green quick- disconnect on each of the devices and depress the release valve.
5. Reconnect the air tubing.

Remove the Pressure Transducer

1. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
2. Remove the bezel. See *“Remove the Bezel”*.
3. Remove the two screws holding the fluidics panel to the test head bracket.
4. Remove the two screws from the chassis.
5. Tilt the fluidics panel forward to give you access to the pressure transducer.
6. Remove the cap connector from the top of the pressure transducer, by lifting up on the tab on the front of the connector with a screwdriver.
7. Pull the pressure transducer out of its bracket.

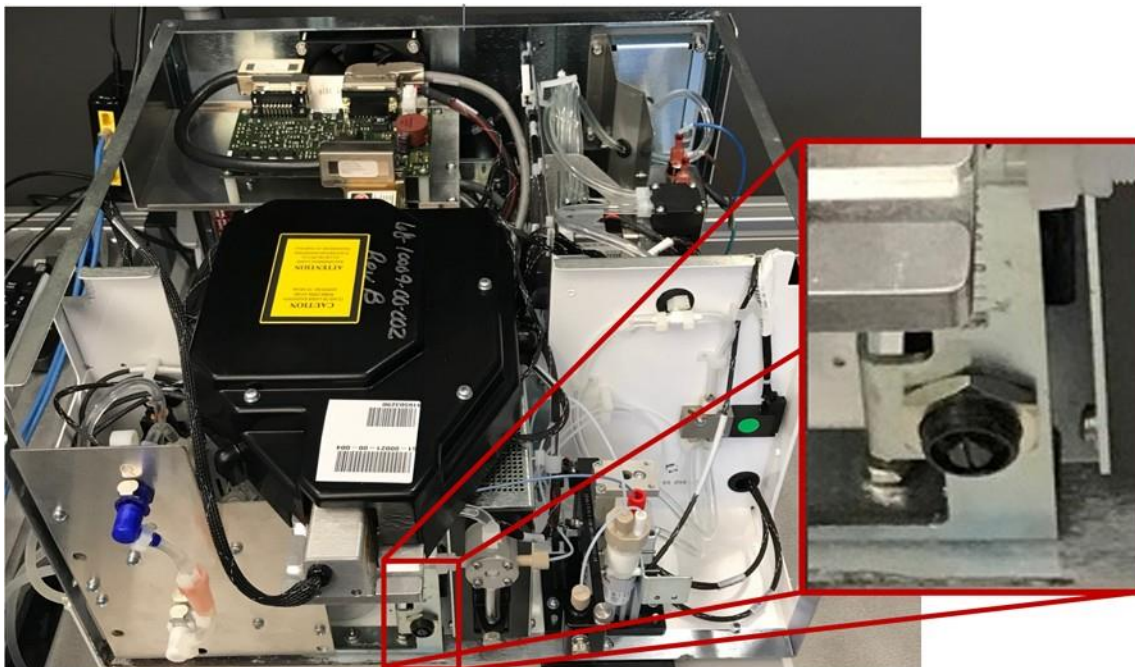
8. Unscrew the pressure transducer from the white fitting on the bottom.

Replace the Pressure Transducer

1. Apply 3 to 4 layers of Teflon tape to the pressure transducer threads, leaving the first couple of rows bare.
2. Attach the new pressure transducer onto the white fitting.
3. Reconnect the cap connector to the top of the pressure transducer.
4. Push the pressure transducer into its bracket.
5. Ease the fluidics panel back to its original position. Verify that the tubing is not crimped.
6. Replace the two screws that hold the fluidics panel to the chassis.
7. Replace the two screws that hold the fluidics panel to the test head bracket.
8. Replace the bezel. See *"Replace the Bezel"*.
9. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Pressure Regulator

FIGURE 52. Luminex® 200™ RoHS Pressure Regulator



Remove the Pressure Regulator

1. Turn off the analyzer but leave it plugged in.
2. Remove the bezel. See *"Remove the Bezel"*.
3. Remove the black nut which secures the regulator to the front of the test head bracket.
4. Pull out the regulator.
5. Disconnect the line from the back of the regulator by cutting the tubing close to the regulator. Save the clamp.

Replace the Pressure Regulator

1. Remove the tubing that comes with the regulator and set aside the yellow device, as it serves as a spare part for a portion of the pneumatic path.
2. Put the clamp on the tubing.
3. Connect the line to the new regulator. Slide the clamp as close to the regulator as possible.
4. Push the regulator back into position.

5. Replace the black nut on the front of the test head bracket.
6. Verify the flow rate and pressure settings.
 - For systems using bottles, verify the flow rate and adjust the regulator on the Luminex 200 RoHS as necessary. (Always release the system pressure after making adjustments. In addition, let the pressure rebuild prior to subsequent testing.)
 - For systems using Luminex® SD™, turn the Luminex® 200™ RoHS regulator on, ensuring the knob is all the way to the right. Use the SD regulator for making any necessary adjustments to the flow rate. (Always release the system pressure after making adjustments. In addition, let the pressure rebuild prior to subsequent testing.)
7. Replace the bezel. See *“Replace the Bezel”*.
8. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Check Valves

NOTE: If you are not using an Luminex® SD™, turn off the system and open the sheath bottle cap to release the pressure in the analyzer.

Release Pressure in the Luminex® Analyzer and Luminex® SD™

1. Perform Decontamination Procedure. See *“Maintaining the System”*.

NOTE: If you are not using an Luminex® SD™, turn off the system and open the sheath bottle cap to release the pressure in the analyzer.
2. Turn off the analyzer, but keep it plugged in.
3. Remove the air tubing between the analyzer and the Luminex SD. (The air tubing has green quick-disconnects.)
4. Insert a screwdriver into the opening of the green quick- disconnect on each of the devices and depress the release valve.
5. Reconnect the air tubing.

Remove a Check Valve

1. Perform decontamination procedure. See *“Maintaining the System”*.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the bezel. See *“Remove the Bezel”*.
4. There are three check valves in the analyzer. If you are replacing a check valve that is in the line coming from the top of the syringe pump, continue to Procedure A. If not, continue to Procedure B.

Procedure A



Check valve from
syringe pump

1. Note the orientation of the check valve. Cut the tubing close to the valve, on both sides.

2. Remove and save the clamps.
3. Remove the check valve.

Procedure B



*Check valve on
inside of chassis*

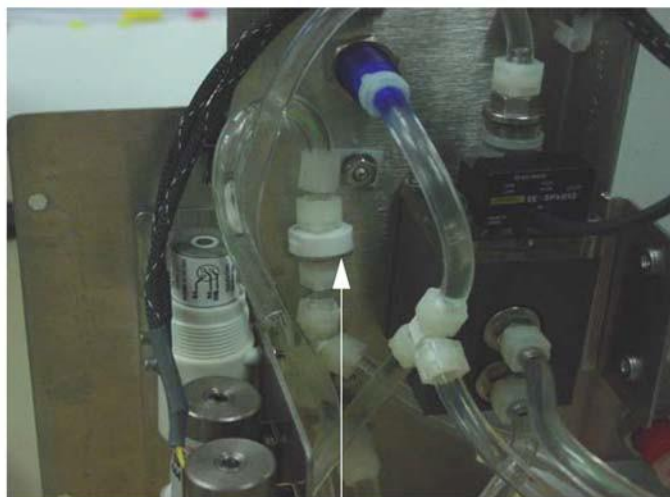
1. Remove the two screws that hold the fluidics panel to the side of the chassis.
2. Remove the two screws that hold the fluidics panel to the test head bracket.
3. Lean the fluidics panel forward.
4. Note the orientation of the check valve. Cut the tubing close to the valve, on both sides.
5. Remove and save the clamps.
6. Remove the check valve.

Replace a Check Valve

Procedure A

1. Install the clamps onto the end of the tubing.
2. Put the check valve into position. Install it in the same orientation as the original check valve.
3. Attach the tubing to the check valve.
4. Slide the clamps toward the check valve until they are past the barb on the tubing.
5. Replace the bezel. See *"Replace the Bezel"*.
6. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Procedure B

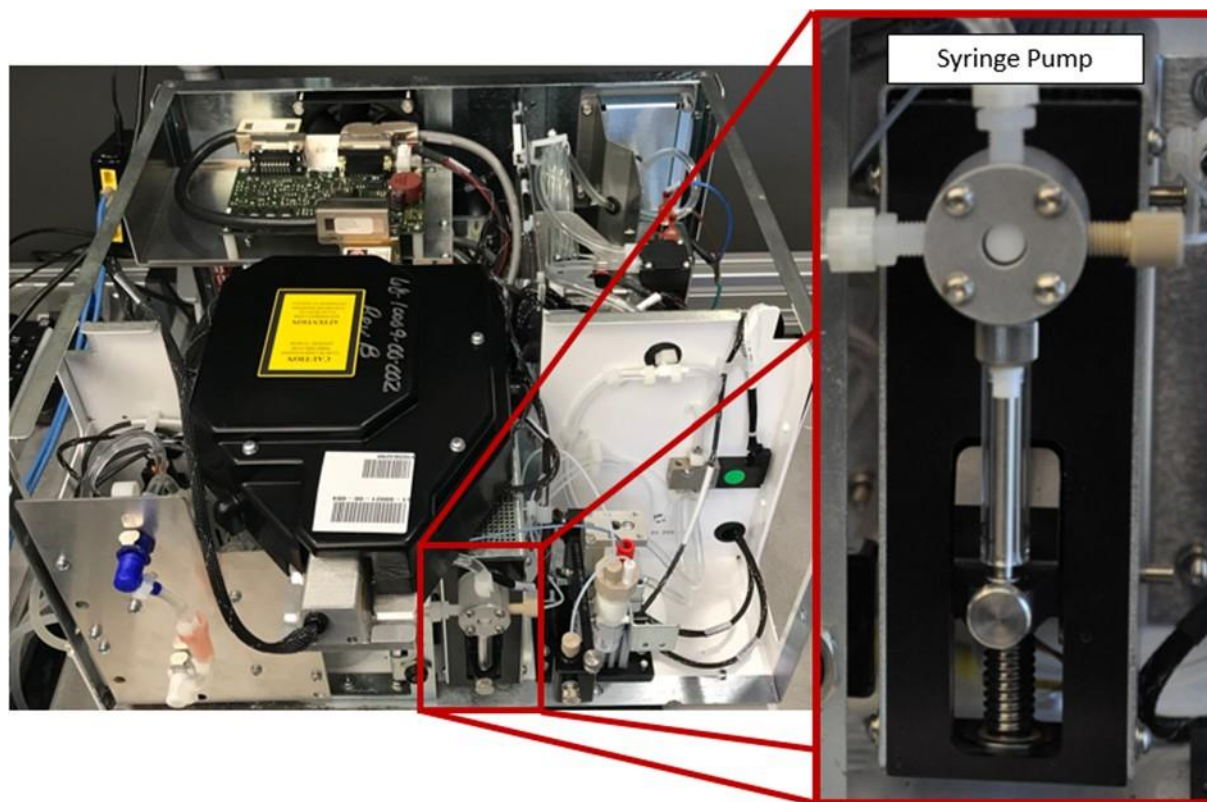


*Check valve on back of
fluidics panel*

1. Install the clamps onto the end of the tubing.
2. Put the check valve into position. Install it in the same orientation as the original check valve.
3. Attach the tubing to the check valve.
4. Slide the clamps toward the check valve until they are past the barb on the tubing.
5. Replace the fluidics panel.
6. Replace the two screws that hold the fluidics panel to the test head bracket.
7. Replace the two screws that hold the fluidics panel to the side of the chassis.
8. Replace the bezel. See *"Replace the Bezel"*.
9. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Syringe Pump

FIGURE 53. Luminex® 200™ RoHS Syringe Pump



Release Pressure in the Analyzer and Luminex® SD™

1. Perform Decontamination Procedure. See *"Maintaining the System"*.

NOTE: If you are not using an Luminex® SD™, turn off the system and open the sheath bottle cap to release the pressure in the analyzer.

2. Turn off and unplug the analyzer.
3. Remove the air tubing between the analyzer and the Luminex SD. (The air tubing has green quick-disconnects.)
4. Insert a screwdriver into the opening of the green quick- disconnect on each of the devices and depress the release valve.
5. Reconnect the air tubing.

Remove the Syringe Pump

1. Remove the analyzer cover. See *"Remove the Analyzer Cover"*.
2. Remove the bezel. See *"Remove the Bezel"*.
3. Remove the four nuts holding the pump's bracket to the chassis.
4. Unplug the connector from P2 on the distribution board.
5. Slowly pull the syringe pump forward until there is enough clearance to disconnect the cable on the back side. (You may need to give it some slack by pushing the cable that was removed from the distribution board through the opening on the inner chassis wall.) Make sure the end of the cable is still visible.
6. When there is enough slack in the cable, disconnect the cable from the back of the syringe pump. Leave the cable in place.
7. Remove the eight screws that secure the syringe pump to the mounting bracket.

Replace the Syringe Pump

NOTE: When replacing a Syringe Pump, ensure that the Digital Board firmware is upgraded to 2.8.0 to avoid introducing disconnect issues.

1. Install new fittings at the 9 and 12 o'clock positions of the new syringe pump.
2. Replace the connector to the back of the syringe pump.
3. Slowly slide the syringe pump in as you feed the cable back to the distribution board.
4. Plug in the connector to P2 on the distribution board.
5. While holding the new syringe pump in position, replace the four nuts that hold the syringe pump to the chassis.
6. Install the 50 inch line and blue light extension cable to their respective clamps located on the side of the syringe pump.
7. Connect the three tubes to the syringe pump.
8. Replace the bezel. See *"Replace the Bezel"*.
9. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Syringe

Remove the Syringe

1. Turn off the analyzer but leave it plugged in.
2. Loosen the thumb screw at the bottom of the syringe arm.
3. Push syringe arm down. (The plunger should stay up.)
4. Unscrew syringe from pump head.
5. Remove syringe.

Replace the Syringe

1. Screw the new syringe into the syringe head until it is finger tight.
2. Lift syringe arm up to bottom of syringe.
3. Tighten thumb screw.

NOTE: The syringe seal can be replaced separately from the syringe. Recommended replacement interval is six months.

Luminex® 200™ RoHS Blue LED Assembly

FIGURE 54. Luminex® 200™ RoHS Blue LED Housing



Remove the LED Assembly

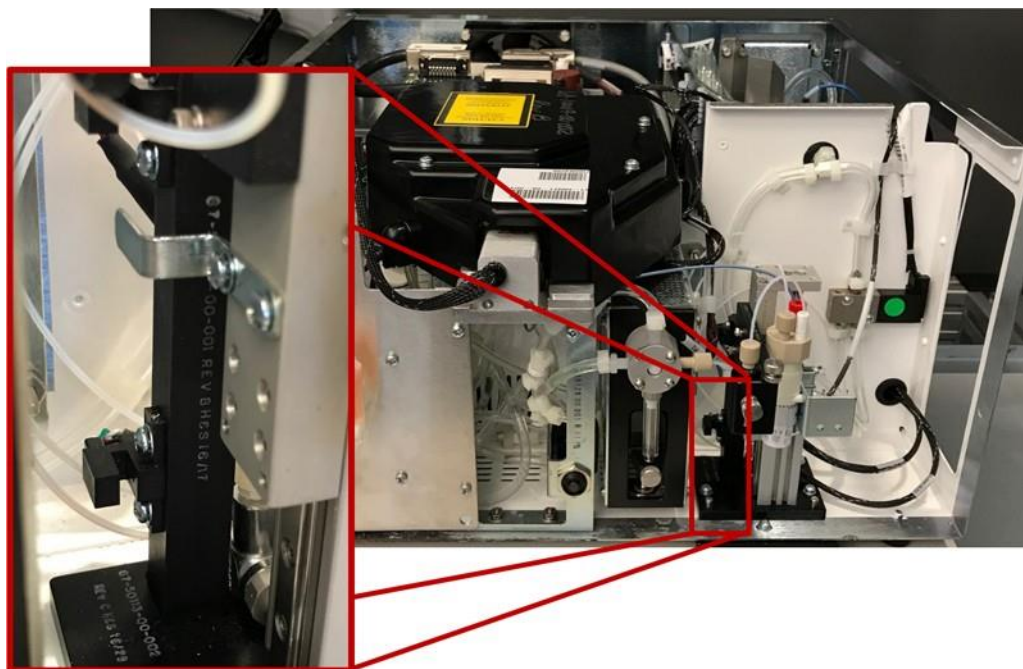
1. Turn off and unplug the analyzer.
2. Pull the LED housing out of the V-housing bracket and disconnect it from its extension.
3. Remove the blue LED housing.

Replace the LED Assembly

4. Reconnect the V-housing bracket to its extension.
5. Replace the LED housing into the V-housing bracket.
6. Turn on the analyzer.

Luminex® 200™ RoHS Pickup Arm Position Sensor

FIGURE 55. Luminex® 200™ RoHS Pickup Arm Position Sensor



Remove the Pickup Arm Position Sensor

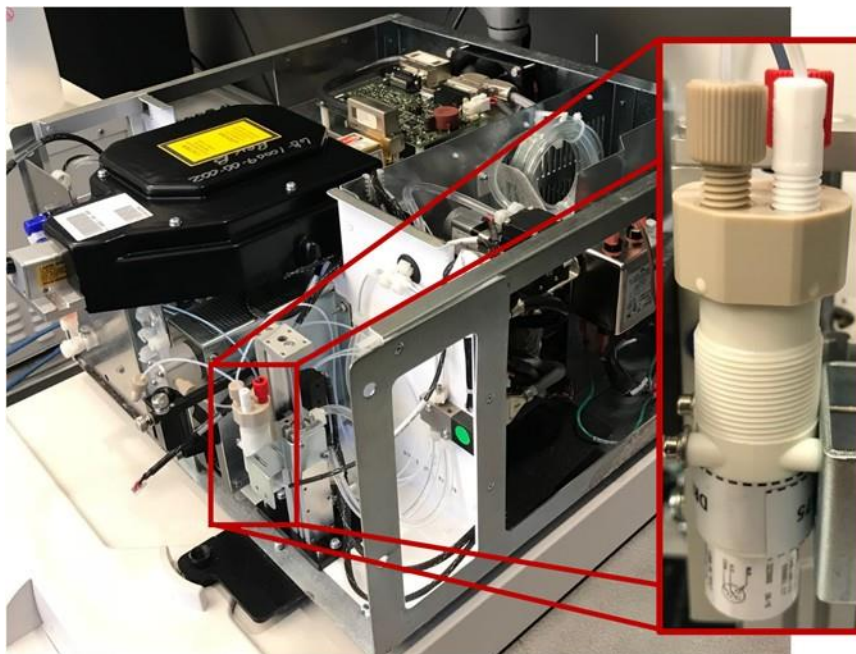
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the bezel. See *“Remove the Bezel”*.
4. Remove the 10-inch and 50-inch sample lines from the 3-way sample valve.
5. Remove the three screws that secure the Bimba cylinder mounting plate to the chassis.
6. Carefully lift up on the Bimba cylinder and pull it forward enough to gain access to the sensor.
7. Remove the two screws securing the sensor.
8. Remove the sensor.

Replace the Pickup Arm Position Sensor

1. Install the sensor and replace the two screws that secure it to the Bimba cylinder.
2. Install the 10-inch and 50-inch sample line to the 3-way valve.
3. Carefully place the Bimba cylinder back in place. It should align with the mounting pins on the chassis.
4. Move the Bimba cylinder up and down to verify that there is proper clearance between the flag and the sensor. If not, reposition the sensor and re-verify clearance.
5. Once the sensor is positioned correctly, install the three screws to secure the Bimba cylinder mounting plate to the chassis.
6. Plug in the sensor onto J7 on the Distribution board.
7. Replace the bezel. See *“Replace the Bezel”*.
8. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Sample 3-Way Valve

FIGURE 56. Luminex® 200™ RoHS Sample 3-Way Valve



Remove the Sample 3-way Valve

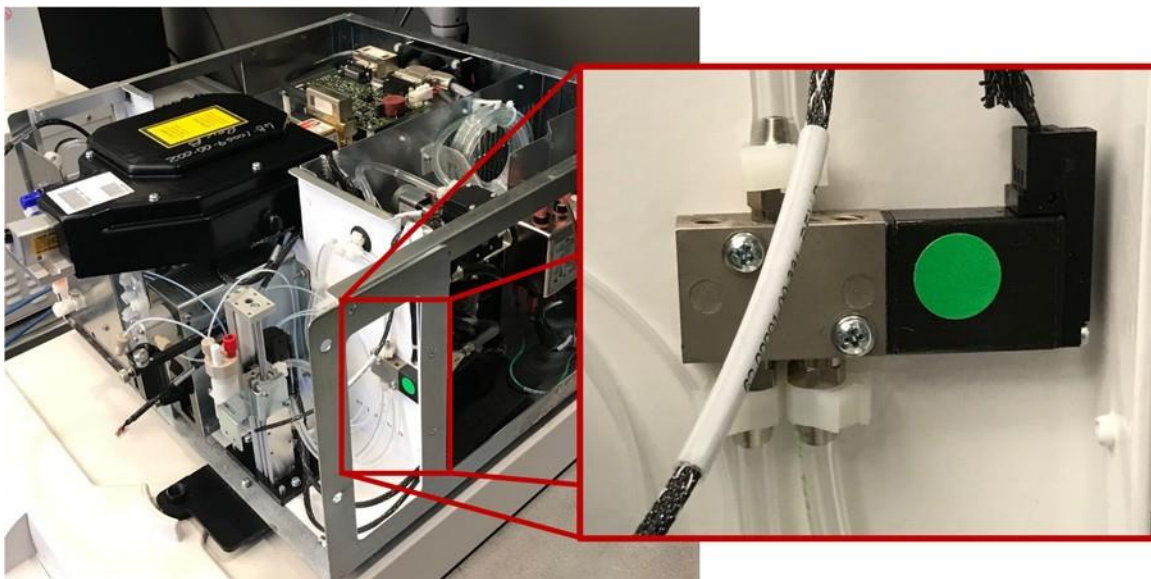
1. Perform decontamination procedure. See *“Maintaining the System”*.
2. Turn off and unplug the analyzer.
3. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
4. Remove the bezel. See *“Remove the Bezel”*.
5. Disconnect the valve cable from J11 on the distribution board.
6. Unscrew the three lines going into the sample 3-way valve, making note of their locations.
7. Remove the two screws that secure the valve mounting bracket to the air cylinder.
8. Remove the two screws that secure the valve to the mounting bracket.
9. Remove the sample 3-way valve.

Replace the Sample 3-way Valve

1. Install the two screws that secure the valve to the mounting bracket.
2. Install the two screws that secure the valve mounting bracket to the air cylinder.
3. Connect the cable of the new sample 3-way valve to position J11 of the distribution board.
4. Attach the three lines going into the sample 3-way valve in their appropriate locations.
5. Replace the bezel. See *“Replace the Bezel”*.
6. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Air Solenoid

FIGURE 57. Air Solenoid



Remove the Air Solenoid

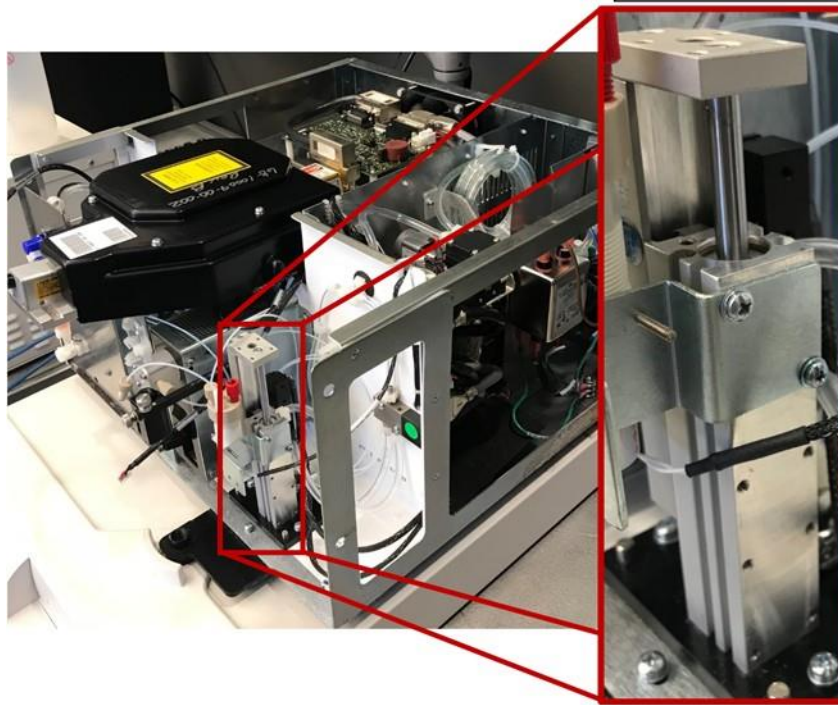
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *"Remove the Analyzer Cover"*.
3. Remove the bezel. See *"Remove the Bezel"*.
4. Using pliers, remove the three tubes from the air solenoid. Make notes of where each line goes.
5. Remove the two screws holding the air solenoid to the chassis.
6. Remove the solenoid connector from position J6 of the distribution board.
7. Remove the air solenoid.

Replace the Air Solenoid

1. Connect the cable of the new air solenoid to position J6 of the distribution board.
2. Cut approximately 1/4" of tubing off the end of each of the three tubes to ensure a tight fit on the new air solenoid. Attach the three lines going into the air solenoid to their specific locations.
3. Replace the two screws that hold the air solenoid to the chassis.
4. Replace the bezel. See *"Replace the Bezel"*.
5. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® 200™ RoHS Bimba Air Cylinder

FIGURE 58. Luminex® 200™ RoHS Bimba Air Cylinder



Remove the Bimba Air Cylinder

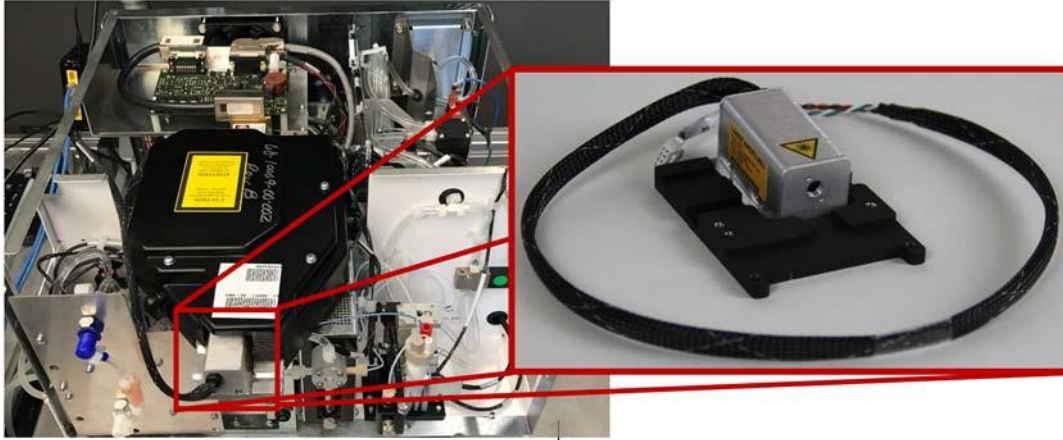
1. Turn off and unplug the analyzer.
2. Remove the bezel. See *“Remove the Bezel”*.
3. Remove the 5” sample line between the sample valve and sample needle holder.
4. Disconnect the remaining two sample lines from the sample valves, making note of their locations.
5. Remove the two screws which secure the sample valve bracket to the Bimba air cylinder.
6. Remove the sample valve and bracket. Set them aside.
7. Remove the three screws from the underside of the analyzer that secure the Bimba air cylinder to its base.
8. Remove the two tubing lines on the back of the Bimba air cylinder. Set the clamps aside.
9. Remove the sample needle holder mounting bracket from the Bimba air cylinder, making note of its location.
10. Remove the position sensor flags from the Bimba air cylinder, making note of their location.

Replace the Bimba Air Cylinder

1. Install the new fittings with Teflon tape onto the Bimba air cylinder.
2. Replace the position sensor flag on the Bimba air cylinder.
3. Replace the sample needle holder mounting bracket on the Bimba air cylinder.
4. Install the two tubing lines on the back of the Bimba air cylinder.
5. Reconnect the three screws from the underside of the analyzer that secure the Bimba air cylinder to its base.
6. Install the sample valve bracket on the Bimba air cylinder with two previously removed screws.
7. Install the sample valve in the bracket.
8. Reconnect the three sample lines to the sample valve.
9. Reconnect the 5” sample line to the sample needle holder.
10. Replace the bezel. See *“Replace the Bezel”*.
11. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS Red Laser

FIGURE 59. Luminex® 200™ RoHS Red Laser



This procedure involves servicing the laser components of the Luminex® analyzer.



Laser radiation when open. Avoid direct exposure to beam.

Remove the Red Laser

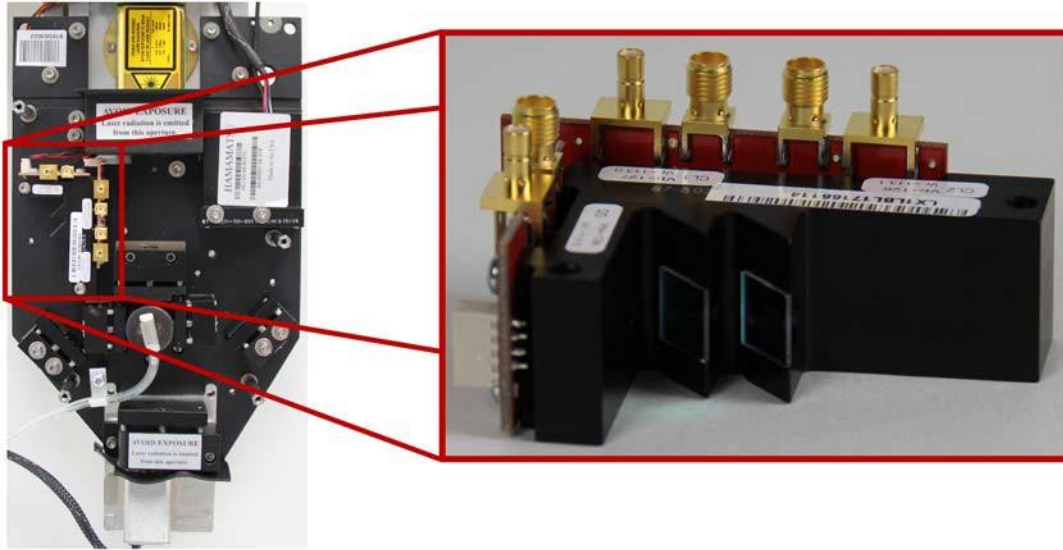
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *“Remove the Analyzer Cover”*.
3. Remove the bezel. See *“Remove the Bezel”*.
4. Unscrew the four screws holding down the optics cover and lift the optics cover off of the analyzer.
5. Disconnect the red laser from J14 on the Laser Driver PCBA.
6. Remove the two screws on the top and the one screw on the underside that secures the red laser baseplate to the optics platform.
7. Remove the red laser head by unscrewing the four small screws on the underside of the baseplate.
8. Remove the red laser head from the baseplate.

Replace the Red Laser

1. Place the new red laser head into place on the baseplate.
2. Screw in the four small screws previously removed from the underside that secure the red laser head to the baseplate.
3. Place the baseplate with the red laser head installed onto the optics platform, and screw in the two screws on the top and one screw on the underside that secures the baseplate to the optics platform. Leave the screws slightly loosened for coarse alignment.
4. Route the cable and connect the red laser to J14 on the Laser Drive PCBA.
5. Put on laser safety glasses and power on the analyzer to allow the lasers to come on.
6. Using the maintenance software, reset the Red Laser time to 0 seconds. Select **Write** to write the values to the digital board.
7. Perform red laser alignment. See *“Coarse-Align the Red Laser”* and *“Fine-Adjust the Red Laser”*.
8. Replace the optics cover and secure it with the four screws previously removed.
9. Replace the bezel. See *“Replace the Bezel”*.
10. Replace the analyzer cover. See *“Replace the Analyzer Cover”*.

Luminex® 200™ RoHS L-Block

FIGURE 60. Luminex® 200™ RoHS L-Block



Remove the L-Block

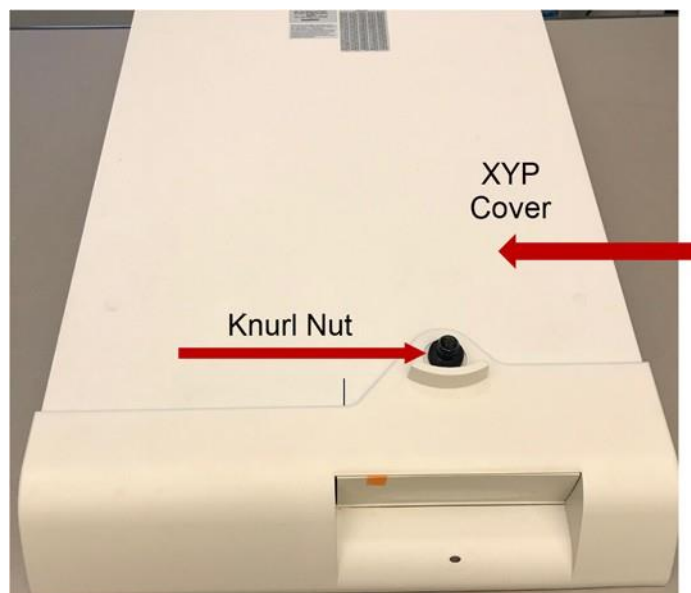
1. Turn off and unplug the analyzer.
2. Remove the analyzer cover. See *"Remove the Analyzer Cover"*.
3. Remove the bezel. See *"Remove the Bezel"*.
4. Unscrew the four Phillips head screws that hold the optics cover to the optics platform and remove the optics cover.
5. Remove the eight cables from the L-Block, making note of their locations.
6. Remove the two hex screws that hold the L-Block to the optics platform.
7. Remove the L-Block.

Replace the L-Block

1. Attach the new L-block to the optics platform using the two screws removed previously.
2. Connect the eight cables to their specific locations on the L-Block.
3. Turn on the analyzer.
4. Using the maintenance software, enter in the new L-Block values (V_b = Breakdown, V_r = Gain). Remember to select "Write" so that the values are written to the digital board.
5. Using the maintenance software, run Auto Offset for DD, CL1, CL2, and RP1.
6. Perform optics alignment.
7. Replace the optics cover on the optics platform and secure it with the four Phillips head screws removed previously.
8. Replace the bezel. See *"Replace the Bezel"*.
9. Replace the analyzer cover. See *"Replace the Analyzer Cover"*.

Luminex® XYP™ Cover

FIGURE 61. Luminex® XYP™ Cover



Remove the XYP Cover

1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP.
3. Remove the three screws on the back of the XYP.
4. Remove the knurl nut.
5. To remove the cover, lift slightly on the back end and then on the front end. Use this back-and-forth rocking motion until the cover comes off.

Replace the XYP Cover

1. Place the cover on the Luminex® XYP™ using the back-and-forth rocking motion you used to remove the cover. (If you can lift the front end of the cover, a corner of the cover is not seated correctly.)
2. Install the knurl nut.
3. Replace the three screws.

Luminex® XYP™ Fuses

FIGURE 62. Luminex® XYP™ Fuse



Remove the Fuse

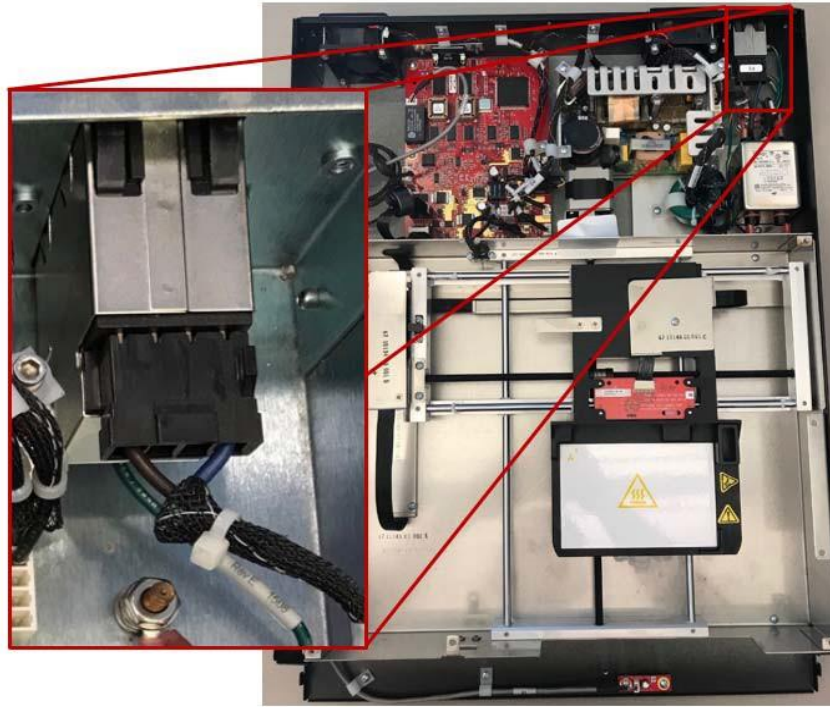
1. Turn off and unplug the Luminex® XYP™.
2. Locate the slot on the outside of the power inlet. It is at the top, right above a cut-out through which you can see the red fuse holder.
3. Insert a flathead screwdriver in the slot and pry the cover out.
4. Pull the cover forward.
5. Remove the fuse holder.
6. Remove the old fuses from the fuse holder.

Replace the Fuses

1. Insert the new fuses in the fuse holder. Refer to the label directly to the right of the power inlet for information about the correct replacement fuses.
2. Push the fuse holder back into place.
3. Snap the cover back into place.
4. Plug in and turn on the Luminex® XYP™.

Luminex® XYP™ AC Power Inlet

FIGURE 63. Luminex® XYP™ AC Power Inlet



Remove the XYP AC Power Inlet

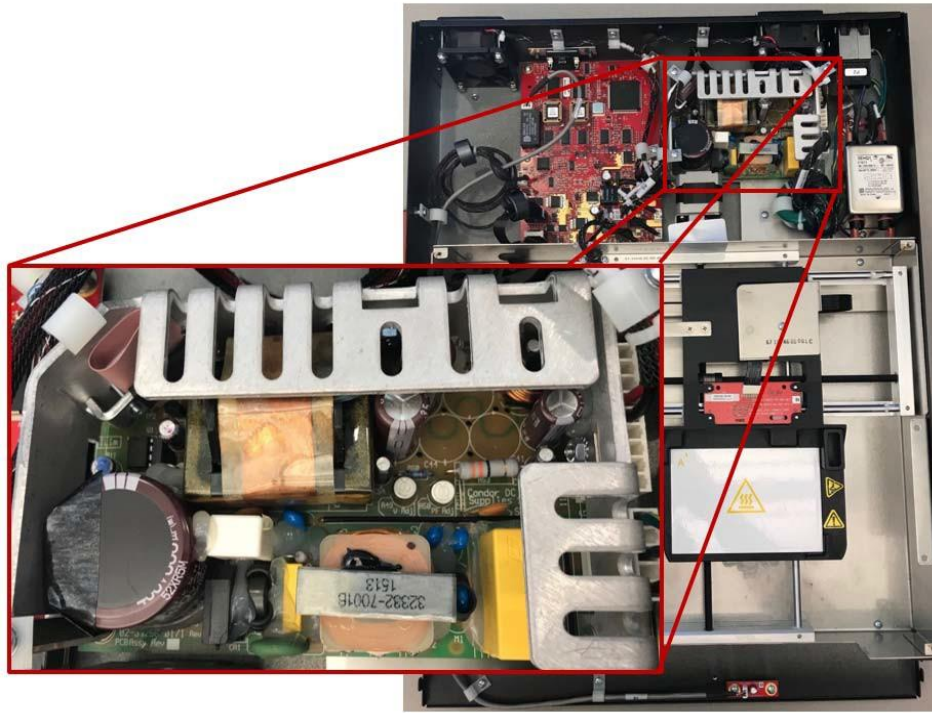
1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP. Allow the heater plate to cool for at least 15 minutes if the heaters were in use.
3. Remove the Luminex XYP cover. See *“Remove the XYP Cover”*.
4. Disconnect the plug on the back of the AC power inlet.
5. Remove the nut securing the green wire to the chassis ground stud and remove the wire.
6. Unscrew the two screws on the power switch.
7. Using pliers, push down on the clips holding the power inlet in place. Push down on both clips at the same time. Lift up on the bottom of the power inlet and push it firmly toward the outside of the analyzer.
8. Remove the power inlet.

Replace the XYP AC Power Inlet

1. Push the new power inlet into place until the clips catch.
2. Replace the two screws onto the power switch.
3. Replace the green wire on the ground stud and secure with the nut.
4. Reconnect the plug going into the back of the power inlet.
5. Replace the Luminex® XYP™ cover. See *“Replace the XYP Cover”*.

Luminex® XYP™ Power Supply

FIGURE 64. Luminex® XYP™ Power Supply



Remove the XYP Power Supply

1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP. Allow the heater plate to cool for at least 15 minutes if the heaters were in use..
3. Remove the Luminex XYP cover. See *“Remove the XYP Cover”*.
4. Disconnect J1 and J2 from the power supply.
5. Remove the four screws or standoffs that hold the power supply in place, making note of cable clamp positions.
6. Lift out the power supply.

Replace the XYP Power Supply

1. Hold the new power supply in position.
2. Replace the four standoffs and their washers. Make sure there is a washer between the circuit board and each stand-off.
3. Install the cable clamps.
4. Replace the four screws.
5. Connect J1 and J2 to the power supply.
6. Replace the Luminex® XYP™ cover. See *“Replace the XYP Cover”*.
7. Turn on the Luminex XYP.

Luminex® XYP™ Air Intake Fan

FIGURE 65. Luminex® XYP™ Air Intake Fan



Remove the XYP Air Intake Fan

1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP. Allow the heater plate to cool for at least 15 minutes if the heaters were in use.
3. Wipe the reservoir, plate holder, and all accessible surfaces with a 10% to 20% bleach solution.
4. Remove the Luminex XYP cover. See *“Remove the XYP Cover”*.
5. Remove the Luminex XYP power supply. See *“Remove the XYP Power Supply”*.
6. Remove the four screws on the back of the analyzer holding the fan in place.
7. Disconnect the fan from J8 on the control board.
8. Lift the fan out.

Replace the XYP Air Intake Fan

1. Connect the new fan to J8 on the control board.
2. Hold the fan in place and replace the four screws on the back of the analyzer.
3. Replace the Luminex® XYP™ power supply. See *“Replace the XYP Power Supply”*.
4. Replace the Luminex XYP cover. See *“Replace the XYP Cover”*.
5. Turn on the Luminex XYP and verify fan operation.

NOTE: Luminex recommends to replace the air intake fan filter every six months.

Luminex® XYP™ Exhaust Fan

FIGURE 66. Luminex® XYP™ Exhaust Fan



Remove the XYP Exhaust Fan

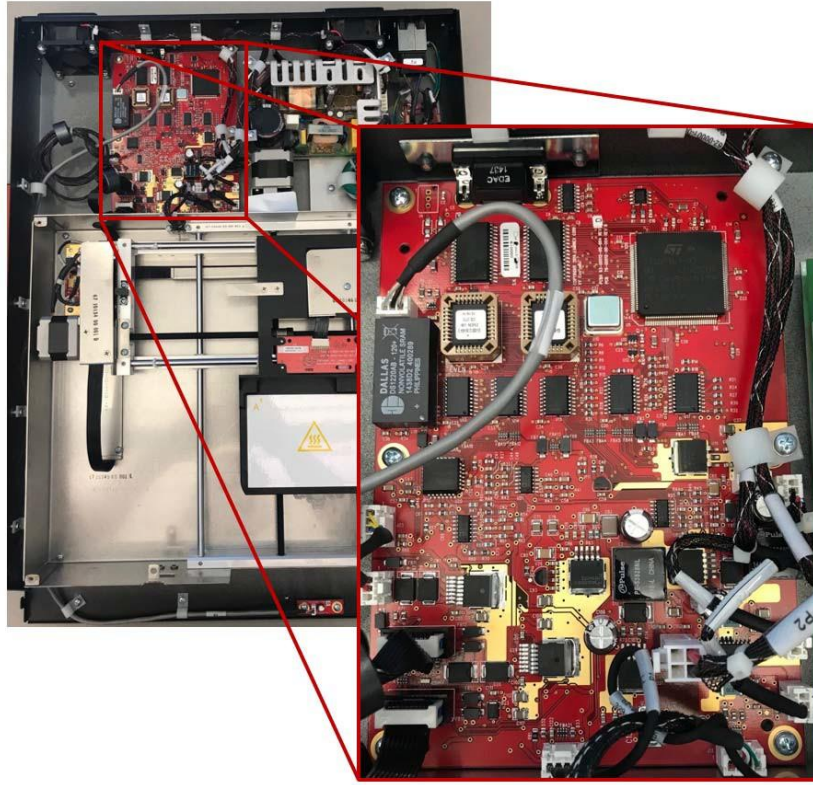
1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP and allow the plate to cool for 15 minutes..
3. Remove the Luminex XYP cover. See *“Remove the XYP Cover”* .
4. Remove the four screws on the back of the analyzer holding the fan in place.
5. Disconnect the fan from J7 on the control board.
6. Lift the fan out.

Replace the XYP Exhaust Fan

1. Connect the new fan to J7 on the control board.
2. Hold the fan in place and replace the four screws on the back of the analyzer.
3. Turn on the Luminex® XYP™ and verify fan operation.
4. Replace the Luminex XYP cover. See *“Replace the XYP Cover”* .

Luminex® XYP™ Controller Board

FIGURE 67. Luminex® XYP™ Controller Board



NOTE: If the Luminex® XYP™ controller board is replaced in the field without the necessary calibration device, the HTR information may no longer be accurate.

Remove the XYP Controller Board

1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP. Allow the heater plate to cool for at least 15 minutes if the heaters were in use.
3. Remove the Luminex XYP cover. See *“Remove the XYP Cover”*.
4. Remove all cables on the controller board.
5. Remove the six screws.
6. Remove the three stand-offs and washers.
7. Remove the two screws that hold the D-connector to the chassis. The D-connector bracket must be installed on the new board.
8. Remove the top cover plate of the inner chassis by removing the four screws holding it in place.
9. Remove the three screws holding the top heater in place and set the heater to the side.
10. Remove the four screws holding the inner chassis box in place. There is one in each corner of the inner chassis box.
11. Lift the chassis box to lift the controller board out.

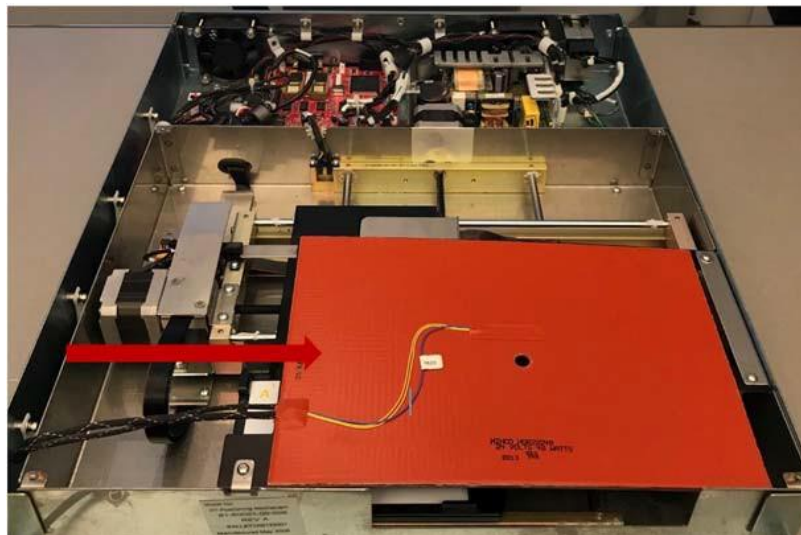
Replace the XYP Controller Board

1. Install the D-connector bracket on the new board and replace the screws.
2. Put the new controller board in place.
3. Lower the inner chassis box.
4. Replace the top heater and screws that secure it.
5. Replace the top cover plate of the inner chassis and four screws.

6. Reinstall the cables on the controller board. See *“XYP Controller PCBA”*.
7. Replace the two screws holding the D-connector to the chassis.
8. Replace the three stand-offs and washers.
9. Enter the serial number and revision into the board using the maintenance software.
10. Replace the six screws.
11. Turn on the Luminex® XYP™ and check performance.
12. Verify that the firmware and DSP are correct for the software version. If they are not, install them using the maintenance software.
13. Calibrate the XY Offsets. Refer to *“Calibrate the XY Offsets”*.
14. Replace the Luminex XYP cover. See *“Replace the XYP Cover”*.

Luminex® XYP™ Top Heater

FIGURE 68. Luminex® XYP™ Top Heater



Remove the XYP Top Heater

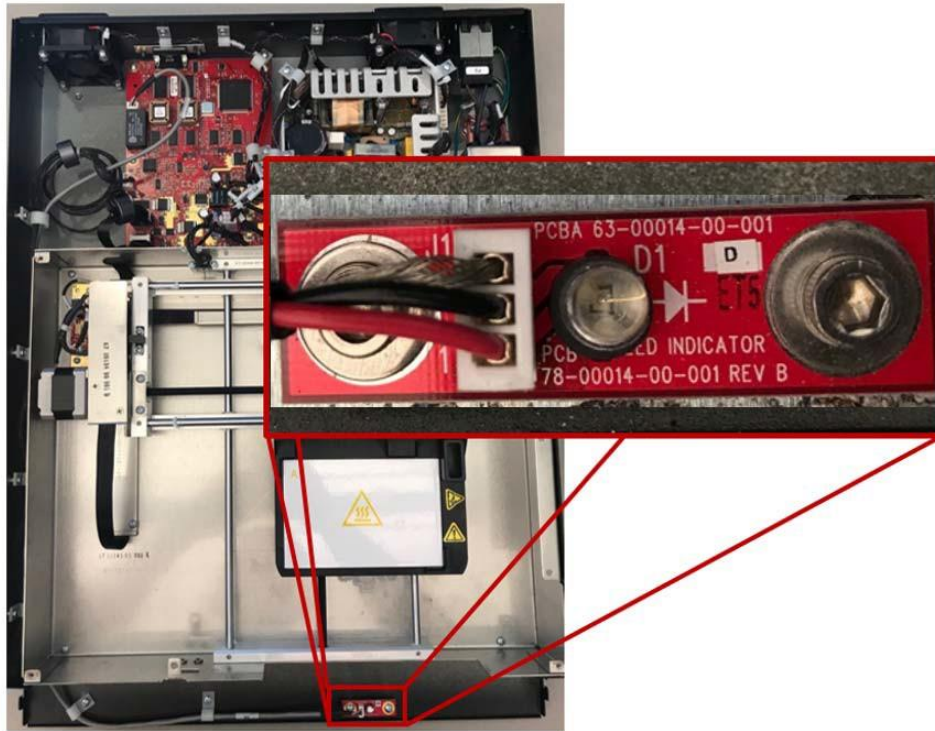
1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP. Allow the heater plate to cool for at least 15 minutes if the heaters were in use.
3. Remove the Luminex XYP cover. See *“Remove the XYP Cover”*.
4. Remove the inner chassis cover.
5. Remove the three screws holding the heater plate to the chassis.
6. Disconnect the heater cable from J22 of the controller board.
7. Remove the cable from the cable clamps.
8. Lift the top heater out.

Replace the XYP Top Heater

1. Install the cable for the new top heater into the cable clamps.
2. Reconnect the cable to J22 on the controller board.
3. Replace the three screws that hold the heater plate to the chassis.
4. Replace the inner chassis cover.
5. Replace the Luminex® XYP™ cover. See *“Replace the XYP Cover”*.
6. Turn on the Luminex XYP.

Luminex® XYP™ LED Board

FIGURE 69. Luminex® XYP™ LED Board



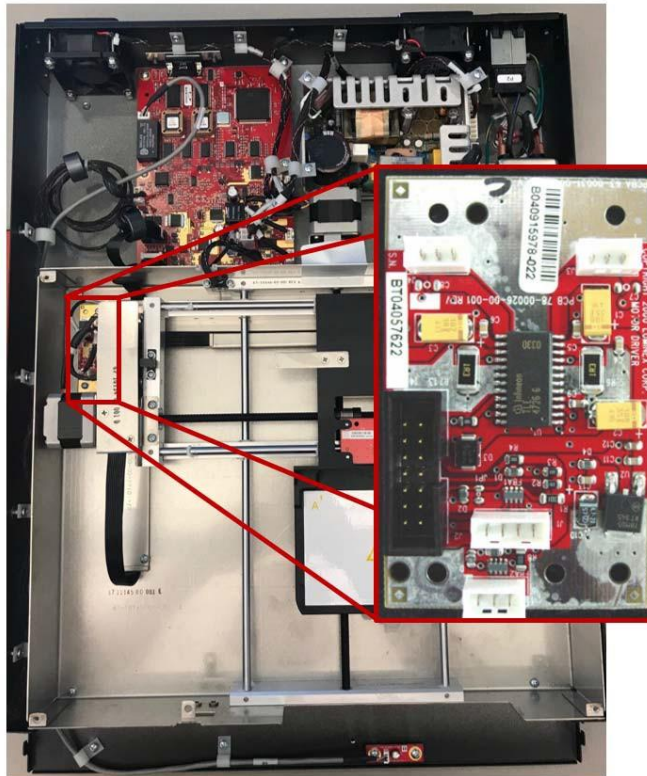
Remove the XYP LED Board

1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP. Allow the heater plate to cool for at least 15 minutes if the heaters were in use.
3. Remove the Luminex XYP cover. See *“Remove the XYP Cover”*.
4. Unplug J1 on the LED board.
5. Remove the two screws holding the LED board in place.
6. Lift the LED board out.

Replace the XYP LED Board

1. Hold the new LED board in place.
2. Replace the two screws holding the LED board in place.
3. Plug in J1 on the LED board.
4. Replace the Luminex® XYP™ cover. See *“Replace the XYP Cover”*.
5. Turn on the Luminex XYP.

FIGURE 70. Luminex® XYP™ Motor Driver Board



Remove the XYP Motor Driver Board

1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP. Allow the heater plate to cool for at least 15 minutes if the heaters were in use.
3. Remove the Luminex XYP cover. See *“Remove the XYP Cover”*.
4. Remove the inner chassis cover.
5. Remove the Luminex XYP top heater. See *“Remove the XYP Top Heater”*.
6. Disconnect all cables from the motor driver board.
7. Remove the four screws holding the motor driver board to the inner chassis.
8. Lift out the motor driver board.

Replace the XYP Motor Driver Board

1. Attach the new motor driver board to the inner chassis.
2. Replace the four screws holding the motor driver board to the drive chassis.
3. Reattach all cables to the new motor driver board.
4. Replace the Luminex® XYP™ top heater. See *“Replace the XYP Top Heater”*.
5. Replace the inner chassis cover.
6. Turn on the Luminex XYP and verify operations.
7. Calibrate the XY Offsets. Refer to *“Calibrate the XY Offsets”*.
8. Replace the Luminex XYP cover. See *“Replace the XYP Cover”*.

Luminex® XYP™ Inner Chassis

FIGURE 71. Luminex® XYP™ Inner Chassis



Remove the XYP Inner Chassis

1. Decontaminate the Luminex® XYP™ by wiping the reservoir, the plate holder, and all accessible surfaces with a 10% to 20% household bleach solution.
2. Turn off and unplug the Luminex XYP. Allow the heater plate to cool for at least 15 minutes if the heaters were in use.
3. Remove the Luminex XYP cover. See *"Remove the XYP Cover"*.
4. Remove the inner chassis cover.
5. Remove the three screws holding the top heater plate to the chassis. Lift the top heater plate and lay it out of your way.
6. Disconnect J13, J15, J16, J18, J20, and J21 from the controller board.
7. Remove the four screws in the corners of the inner chassis.
8. Remove the inner chassis.

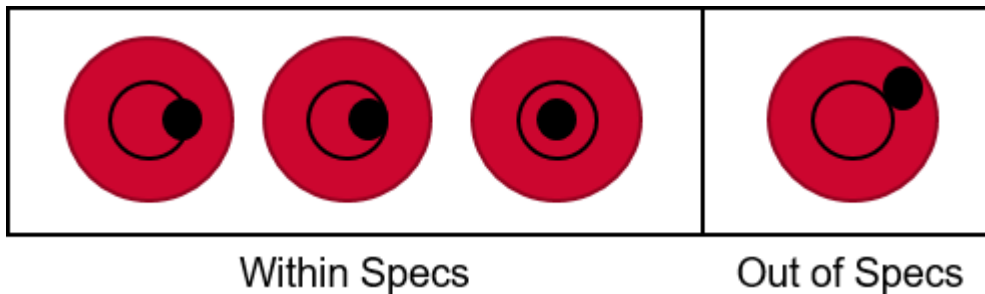
Replace the XYP Inner Chassis

1. Put the new inner chassis in place and replace the four screws in the corners.
2. Reconnect J13, J15, J16, J20, and J21 to the controller board.
3. Reinstall the top heater plate to the chassis.
4. Replace the three screws holding the top heater plate to the chassis.
5. Replace the inner chassis cover.
6. Turn on the Luminex® XYP™ and verify operations.
7. Calibrate the XY offsets. Refer to *"Calibrate the XY Offsets"*.
8. Replace the Luminex XYP cover. See *"Replace the XYP Cover"*.

Calibrate the XY Offsets

NOTE: Removal of the inner cover will require re-calibration of XY offsets. Ensure all covers are on when performing this procedure.

1. Press a sheet of aluminum foil over a standard 96-well microtiter plate.
2. Insert the plate and send the plate carrier to its 'Home' position. (Default: X = 1696 Y = 830)
3. Send the carrier plate to location D6.
4. Send the probe down, and then up. Eject the carrier plate.
5. Inspect the hole location. The hole should be centered in the well for location D6.
6. See the figure below to determine if the puncture is within specifications, and make adjustments as necessary. (Half a well radius is ~29 steps):



7. If adjustments are necessary, go to the **Movement Position Calibration** tab in LxMaintenance.
 - X-direction: Move left - increase X steps. Move right - decrease X steps.
 - Y-direction: Move up - increase Y steps. Move down - decrease Y steps.
8. Choose a different well location and send the probe down, and up once again.
9. Continue this procedure until the puncture is centered and within specifications. (Ensure you use a different well location for each puncture).

NOTE: You must cycle power to the XYP in order for the changes to be saved.

Luminex® SD™

Remove and Replace SDS Top Cover

Remove the SDS Top Cover

1. Turn off and unplug the Luminex SDS.
2. Remove the four (only two shown) hex screws that secure the Top Cover on the SDS unit. Remove the Top Cover.



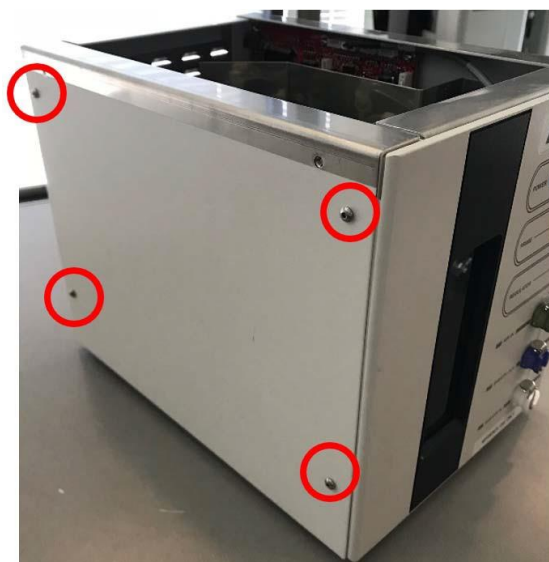
Replace the SDS Top Cover

1. Put the Top Cover back into place, and screw in the four hex screws that you removed previously.
2. Power on the SDS and allow it to initialize.

Remove and Replace SDS Left Side Panel

Remove the SDS Left Side Panel

1. Remove the Top Cover using the *“Remove the SDS Top Cover”*.
2. Remove the four hex screws that secure the Left Side Panel on the SDS unit. Remove the Left Side Panel.



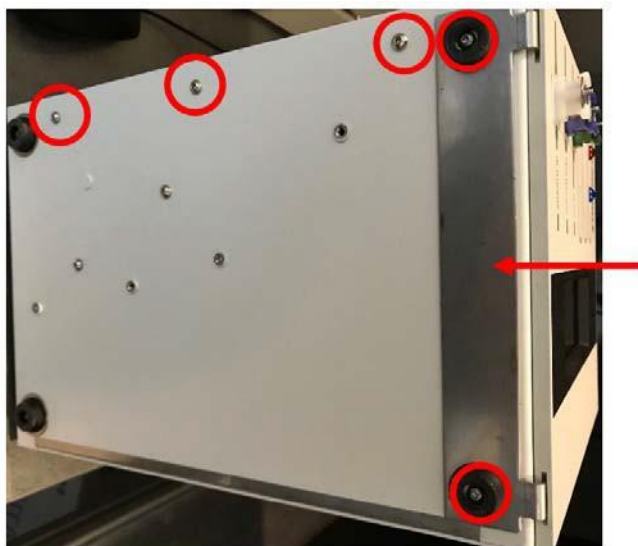
Replace the SDS Left Side Panel

1. Put the Left Side Panel back into place, and screw in the four hex screws that you removed previously.
2. Replace the Top Cover using the *“Replace the SDS Top Cover”*.
3. Power on the SDS and allow it to initialize.

Remove and Replace SDS Right Side Panel

Remove the SDS Right Side Panel

1. Remove the Top Cover using *“Remove the SDS Top Cover”*.
2. Lay the SDS Unit on its side and remove the three hex screws that secure the Right Side Panel to the SDS unit.
3. Remove the two front rubber feet and bottom face plate retainer by unscrewing the two hex screws that secure them to the bottom of the SDS unit.

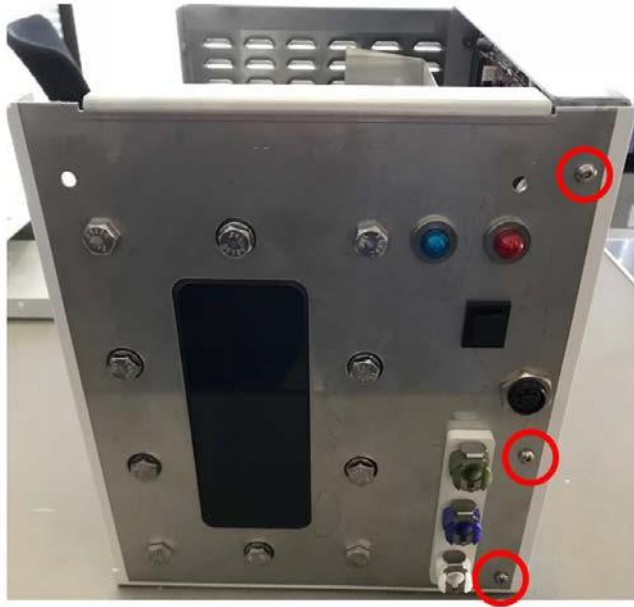


NOTE: The face plate of the SDS Unit must be removed in order to access three of the hex screws that secure the Right-Side Panel to the SDS Unit.

4. Remove the two hex screws that secure the Face of the SDS unit. Remove the face plate.



5. With the face plate removed, unscrew the three hex screws that secure the Right Side Panel to the SDS Unit.



6. Remove the two hex screws on the right side of the SDS unit that secure the barrier to the Right-Side Panel.



7. On the back of the SDS unit, remove the five hex screws that secure the Right Side Panel.



8. Disconnect all cables from the Controller PCBA, and Power Supply making not of their locations.
9. Remove the Right-Side Panel.

Replace the SDS Right Side Panel

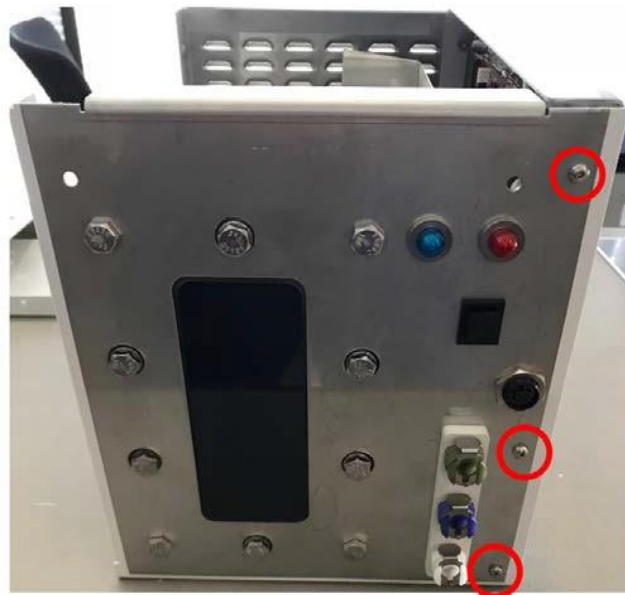
1. Put the Right-Side Panel back into place and re-connect all Controller PCBA and Power Supply cables to their proper locations.
2. On the back of the SDS unit, screw in the five hex screws that you removed previously.



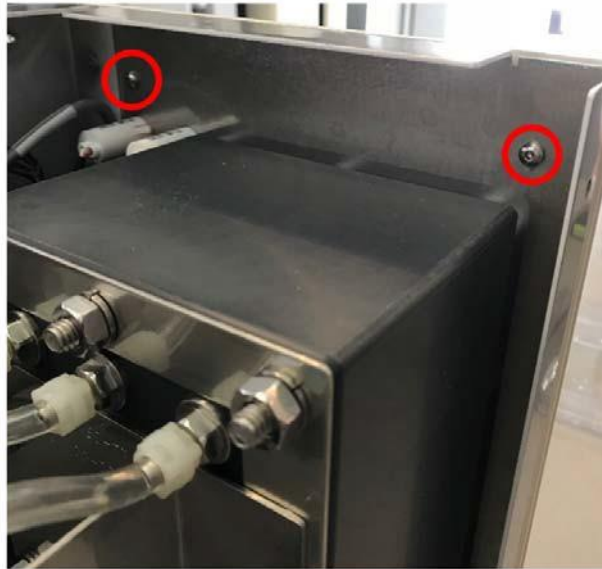
3. Screw in the two hex screws on the right side of the SDS unit that secure the barrier to the Right Side Panel that you removed previously.



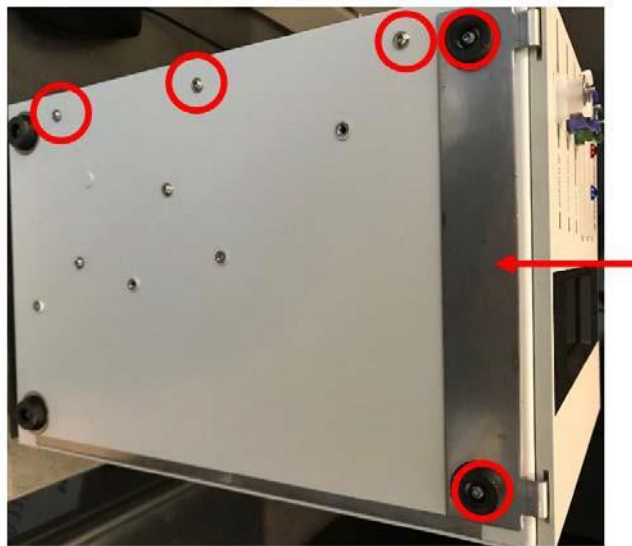
4. On the front on the SDS unit, screw in the three hex screws that you removed previously.



5. Put the face plate back into place, and screw in the two hex screws that you removed previously that secure the face plate to the SDS unit.



6. Lay the SDS Unit on its side, and screw in the three hex screws that secure the Right-Side Panel to the SDS unit.
7. Put the bottom face plate retainer and two rubber feet back into place and secure them by screwing in the two hex screws that secure them to the bottom of the SDS unit that you removed previously.



8. Carefully place the SDS unit back on its feet.
9. Replace the Top Cover using *"Replace the SDS Top Cover"*.

Remove and Replace the Fuses

Remove the Fuses

1. Turn off and unplug the Luminex® SD™.
2. Locate the slot on the outside of the power inlet. It is located at the top, right about a cut-out through which you can see the red fuse holder.
3. Insert a flathead screwdriver in the slot and pry the cover out.
4. Pull the cover forward.

5. Remove the fuse holder.
6. Remove the old fuses from the fuse holder.

Replace the Fuses

1. Insert the new fuses in the fuse holder. Refer to the label directly to the right of the power inlet for information about the correct replacement fuses.
2. Push the fuse holder back into place.
3. Snap the cover back into place.
4. Plug in and power on the Luminex® SD™.



Remove and Replace the AC Power Inlet

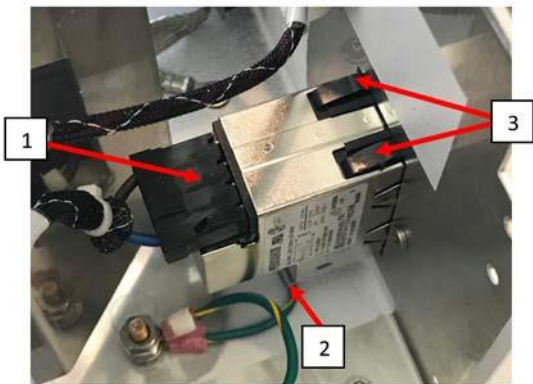
Remove the AC Power Inlet

1. Turn off and unplug the Luminex® SD™.
2. Remove the Top Cover using the “Remove the SDS Top Cover”.
3. Remove the Right Side Panel using the “Remove and Replace SDS Right Side Panel”.

- 4. Unscrew the two hex screws on the power switch.



- 5. Disconnect the power connection on the back of the AC power inlet.
- 6. Disconnect the green ground wire from the AC power inlet.
- 7. Using pliers, push down on the four clips on the top and bottom that hold the power inlet in place. Lift up on the bottom of the power inlet and push it firmly toward the outside of the SDS.



1	Power connection
2	Ground Wire
3	AC power inlet clips (only top clips shown)

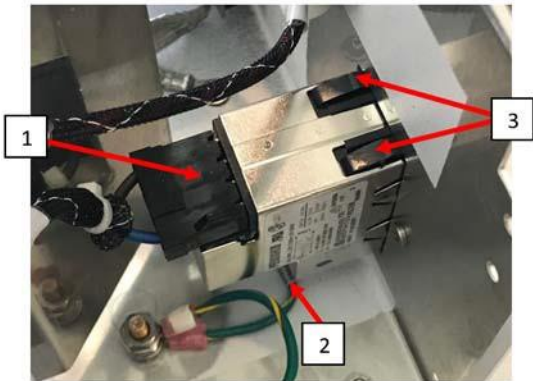
NOTE: You must use force when removing the power inlet. If the clips break when removing the power inlet it's OK, because you will be replacing the power inlet.

- 8. Remove the power inlet.

Replace the AC Power Inlet

- 1. Push the new power inlet into place until the top and bottom clips catch.
- 2. Re-connect the green ground wire to the AC power inlet.

3. Reconnect the power connection on the back of the AC power inlet.



1	Power connection
2	Ground wire
3	AC power inlet clips

4. Replace the two screws onto the power switch that you removed previously.



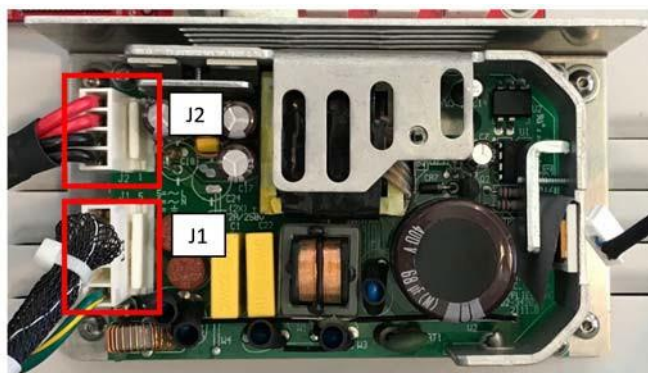
5. Replace the Right-Side Panel using the *“Remove and Replace SDS Right Side Panel”* .
6. Replace the Top Cover using the *“Replace the SDS Top Cover”* .
7. Plug in and power on the Luminex SDS.

Remove and Replace the Power Supply

Remove the Power Supply

1. Turn off and unplug the Luminex® SD™ .
2. Remove the Top Cover using the *“Remove the SDS Top Cover”* .
3. Remove the Right-Side Panel using the *“Remove and Replace SDS Right Side Panel”* .

4. Disconnect J1 and J2 from the power supply.



5. Disconnect the sheath pump cable from J11 on the Controller PCBA to allow for more room to remove the power supply.
6. Remove the four screws or standoffs that hold the power supply in place, making note of cable clamp positions.



7. Lift out the power supply.

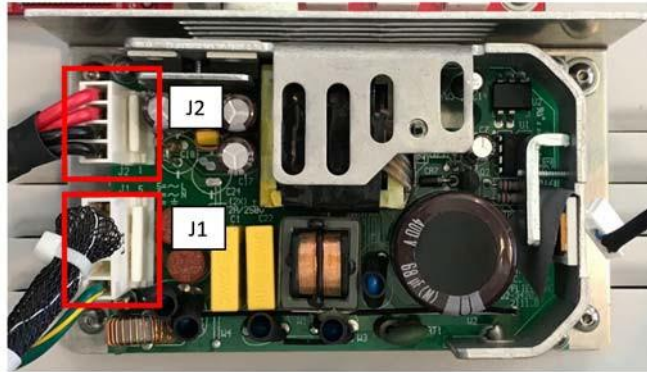
Replace the Power Supply

1. Put the new power supply into place.
2. Replace the four screws or standoffs that you previously removed to secure the power supply.



3. Re-connect the sheath pump cable J11 to the Controller PCBA.

4. Connect J1 and J2 to the power supply.

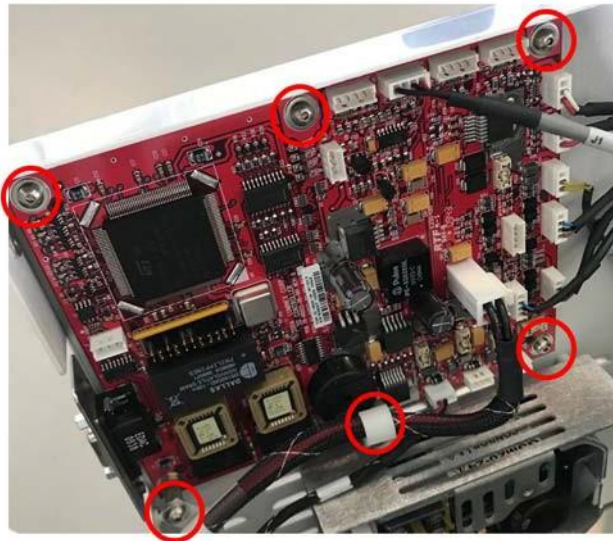


5. Replace the Right-Side Panel using the *“Remove and Replace SDS Right Side Panel”*.
6. Replace the Top Cover using the *“Replace the SDS Top Cover”*.
7. Plug in and power on the Luminex SDS.

Remove and Replace the Controller PCBA

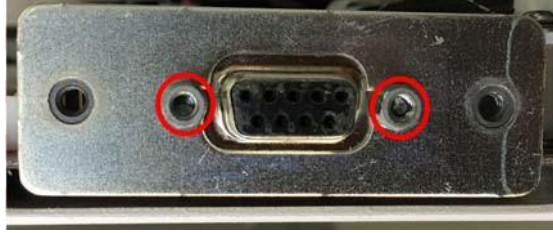
Remove the Controller PCBA

1. Turn off and unplug the Luminex® SD™.
2. Remove the Top Cover using the *“Remove the SDS Top Cover”*.
3. Remove the Right-Side Panel using the *“Remove and Replace SDS Right Side Panel”*.
4. Disconnect all cables from the Controller PCBA making note of their locations.
5. Remove the six screws and standoffs that hold the Controller PCBA in place.



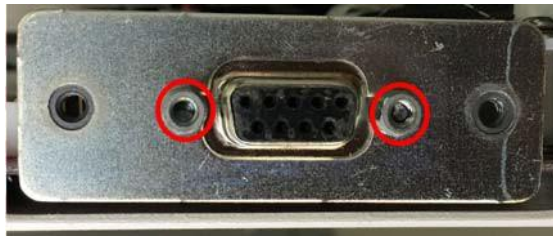
6. Lift out the Controller PCBA.

7. Remove the existing DB9 access plate from the Controller PCBA by removing the two jack screws and washers that secure it to the PCBA. Set aside and re-use on the new PCBA.

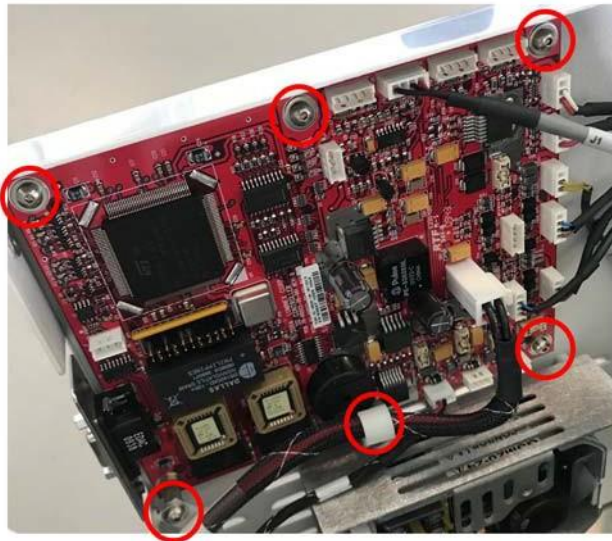


Replace the Controller PCBA

1. Put the DB9 access plate that you removed previously into place on the new Controller PCBA. Place the washers into place and secure it using the two jack screws that you removed previously.



2. Put the new Controller PCBA into place.
3. Screw in the six screws or standoffs that hold the Controller PCBA in place using the screws you removed previously.
4. Reconnect all cables to their specified locations.



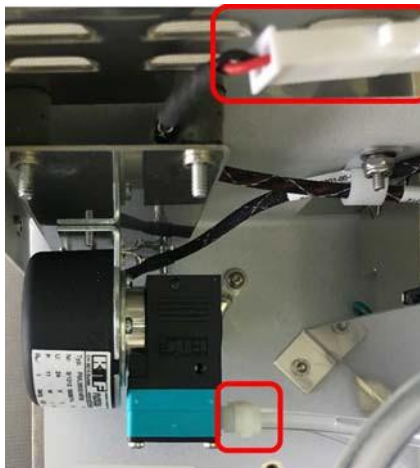
5. Replace the Right-Side Panel using the *“Remove and Replace SDS Right Side Panel”*.
6. Replace the Top Cover using the *“Replace the SDS Top Cover”*.
7. Plug in and power on the Luminex SDS.

Remove and Replace the Sheath Pump

Remove the Sheath Pump

1. Turn off and unplug the Luminex SD.
2. Drain the Luminex SD.
3. Remove the Top Cover using the *“Remove the SDS Top Cover”*.

4. Remove the Left Side Panel using the *“Remove and Replace SDS Left Side Panel”* .
5. Disconnect the clamps and tubes from the two barb fittings on the sheath pump and disconnect the cable connection shown.



6. On the back of the SDS unit, remove the four hex screws from the back of the sheath pump mounting plate.



7. Remove the sheath pump.

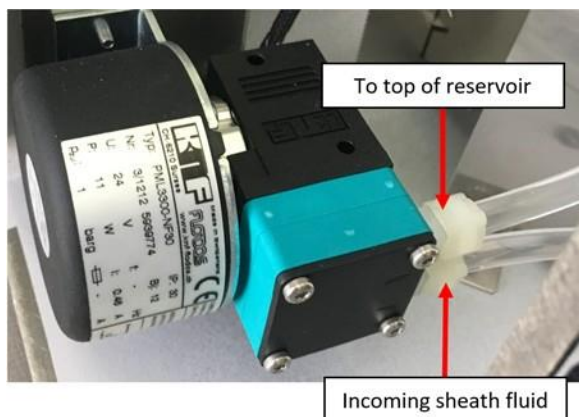
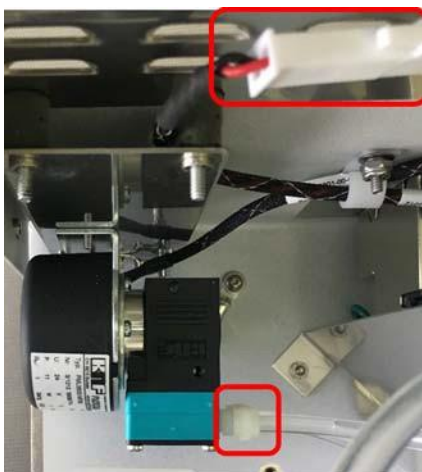
Replace the Sheath Pump

1. Put the new Sheath Pump into place.

- On the back of the SDS unit, screw in the four hex screws removed previously.



- Re-connect the cable connection, as well as the tubes and clamps on the barb fittings ensuring a good seal. (Ensure you verify that you connect the tubes to their proper location, as shown).



- Plug in and power on the Luminex SDS. Check for leaks.
NOTE: If leaks are present from the barb fittings, power off the SDS, drain the reservoir and ensure that the barb fittings are tightened. If the barb fittings are tight, then it may be necessary to trim 1/4" of tubing to ensure a proper seal over the barb fitting.
- If no leaks are present, power off and replace the Left Side Panel using the *"Remove and Replace SDS Left Side Panel"*.
- Replace the Top Cover using *"Replace the SDS Top Cover"*.

Remove and Replace the Sight Glass

Remove the Sight Glass

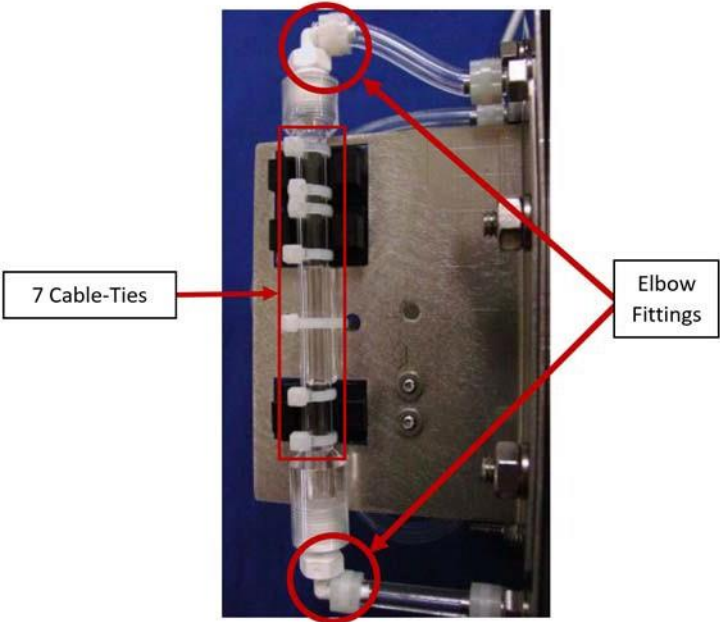
- Turn off and unplug the Luminex SD.
- Drain the Luminex SD. Remove the Top Cover using the *"Remove the SDS Top Cover"*.
- Remove the Left Side Panel using the *"Remove and Replace SDS Left Side Panel"*.
- Remove the clamps and tubing from the top and bottom elbow fittings. DO NOT remove the tubing from the reservoir.
NOTE: Ensure there is no sheath fluid in the tubing. If sheath fluid is present, have paper towels or wipes ready.
- Unscrew the top and bottom elbow fittings using a 7/16" open-end wrench.
- Cut and remove the 7 cable-ties from the sight glass.
- Remove the sight glass.

Replace the Sight Glass

- 1. Place the new sight glass into place.
- 2. Fasten the sight glass using 7 new cable-ties in the locations shown in figure below.
- 3. Wrap the threads on the top and bottom elbow fittings with 7-inches of Teflon tape. DO NOT cover the first two threads to ensure it catches when screwing it in the sight glass.
- 4. Screw the elbow fittings into the sight glass using a 7/16" open-end wrench. DO NOT over-tighten.
- 5. Re-connect the tubes to the elbow fittings and secure with the clamps.
- 6. Plug in and power on the Luminex SDS and allow the reservoir to fill. Check for leaks.

NOTE: If leaks are present on the elbow fittings, power off the SDS, drain the reservoir and ensure that the elbow fittings are tight. If the elbow fittings are tight, then another layer of Teflon tape may be required to ensure a good seal.

- 7. If no leaks are present, power off and replace the Luminex SDS covers. See *"Replace the SDS Top Cover"* and *"Replace the SDS Left Side Panel"* .



Remove and Replace the Level Sensors

Remove a Level Sensor

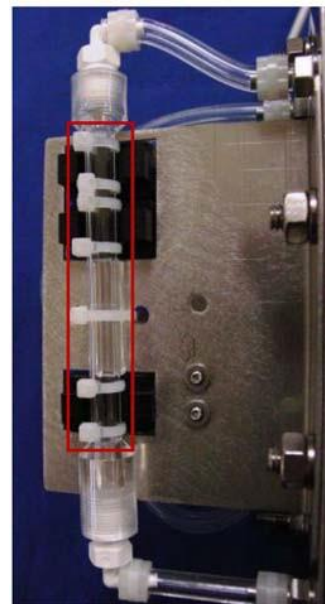
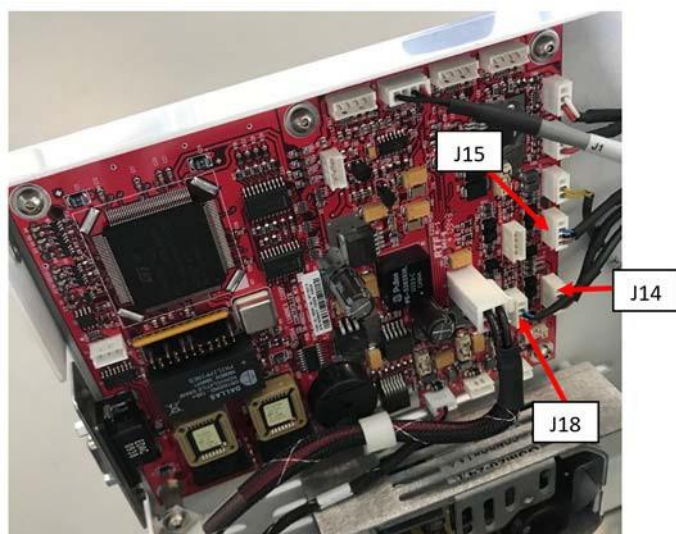
- 1. Turn off and unplug the Luminex® SD™ .
- 2. Remove the Top Cover using the *"Remove the SDS Top Cover"* .
- 3. Remove the Left Side Panel using the *"Remove and Replace SDS Left Side Panel"* .
- 4. Disconnect the cable that corresponds to the Level Sensor you are removing from the Controller PCBA.

J14	Full (Normal) Level Sensor
J15	High Level Sensor
J18	Empty Level Sensor

- 5. Cut and remove the cable-tie that secures the Level Sensor you are removing to the site glass.
- 6. Lift out the Level Sensor.

Replace a Level Sensor

1. Put the new Level Sensor into place.
2. Secure the Level Sensor with a cable-tie in the location that was removed previously.
3. Re-connect the cable to the Controller PCBA where you removed the previous Level Sensor cable.
4. Plug in and power on the Luminex SDS and allow the reservoir to fill. Verify the new Level Sensor operates properly.
5. Power off the SDS unit and replace the Luminex SDS covers. See *“Replace the SDS Top Cover”* and *“Replace the SDS Left Side Panel”*.



Recommended Tools

Image	Description
	Techni-Tool Spring Hook, Double End, Pull and Life, 8"
	Carica Telescoping Pen Magnet, Clip-on
	General Tools Needle File Set, Swiss Pattern, 5 1/2", 4 Pc.
	Eclipse Tools Alignment Tool Kit, 3 Pc. ESD
	Xcelite Slotted Screwdriver, 1/8", Pocket Style, 5 1/4" OAL
	Xcelite Slotted Screwdriver, 3/32", Pocket Clip, 5 1/4" OAL
	Crescent Slotted Screwdriver, Dura-Driver, 1/4" x 4"
	Crescent Slotted Screwdriver, Dura-Driver, 3/16" x 3"
	Xcelite Phillips Screwdriver #1, Stubby, 3 1/4" OAL
	Klein Socket Set, Ratchet, 3/8" Dr., 6 and 12 Pt. w/Box, 12 Pc.

	Xcelite Ball End Hex Blade Set 0.05" - 3/16", 99 Series, 11 PC.
	Techni-Tool L-Wrench Hex Key Set, Ballpoint, 0.5"-5/16", 12 Pc.
	Techni-Tool L-Wrench Hex Key Set, Ballpoint, 1.5mm-5mm, 6 Pc.
	Bondhus Star Wrench Set, T6-Y25, Metal, Fold-Up, 8 Pc.
	Techni-Tool Electrician's Knife, 2 Blade w/ Screwdriver
	Crescent Plier Groove Joint 1/ 1/8" Jaw, 7" OAL
	Klein Plier, Long Nose, Heavy Duty w/ Side Cutter, 8 5/16"
	Xcelite Plier, Long Chain Nose, Serrated Jaw, 5" OAL
	Xcelite Plier, Long Nose w/ Cutter Serrated 5 1/2" OAL

	Xcelite Cutter, Diagonal, Slim Tapered, Flush, 4" OAL
	Techni-Tool Cut/Crimp/Strip Service Tool 22-10 AWG 7 1/2"
	Crescent Wrench, Adjustable, 1/3" Jaw, Red Grip, 4" OAL
	Ratcheting Combination Wrench, 5/16"
	Greenlee Ratcheting Combination Wrench 7/16"
	Techni-Tool Holder, Straight, Ratchet, Lock Handles, 5 1/2" OAL
	Techni-Tool Tweezer Economy Reverse Action Type 932
	Proedge Knife #1 w/ #11 Aluminum Blade w/ Safety Cap
	Ullman Inspection Mirror 7/8" Dia., 7 1/2" OAL
	Techni-Tool Scribe Machinists Straight/90 Degree Ends 8" OAL
	Coast Mini Tac Torch LED Light 2.15/16" OAL

	Eclipse Tools Magnifier, Round Handheld, 3x - 2" diameter																
	Bausch & Lomb Inspection Loupe 10X Magnification 1" Focal																
	Loctite Threadlocker Adhesive, Assure 425, Surface Curing																
 <td>Tec-Tuff Tool Case w/ Wheels, 17.75" x 14.5" x 10" Black</td>	Tec-Tuff Tool Case w/ Wheels, 17.75" x 14.5" x 10" Black																
	Fluke 115 Multimeter TRMS Current AC/DC Frequency Cap																
	Oscilloscope with the following specifications: 100MHz 2-Channel																
	Skewdriver Pro Kit Model SPK-4040 https://spectools.com/shop/skewdriver-pro-kit/																
	Handheld Laser Power Meter (RoHS) #1098293 https://www.coherent.com/measurement-control/measurement/lasercheck																
	Laser Safety Glasses with the following specifications: AGR: EN207 Certified. Luminous transmittance: 31% Grey. <table><tr><th>λ (nm)</th><th>L-Rating</th><th>λ (nm)</th><th>OD</th></tr><tr><td>0.01W 2*10E-6J 532 + 633-640</td><td>RB1</td><td>190-400</td><td>5+</td></tr><tr><td>5-2.5 NOIR 1 S</td><td>CE</td><td>532</td><td>1.5+</td></tr><tr><td></td><td></td><td>633-640</td><td>1+</td></tr></table>	λ (nm)	L-Rating	λ (nm)	OD	0.01W 2*10E-6J 532 + 633-640	RB1	190-400	5+	5-2.5 NOIR 1 S	CE	532	1.5+			633-640	1+
λ (nm)	L-Rating	λ (nm)	OD														
0.01W 2*10E-6J 532 + 633-640	RB1	190-400	5+														
5-2.5 NOIR 1 S	CE	532	1.5+														
		633-640	1+														

Chapter 7: Final Testing Procedures

Safety

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Always wear personal protective equipment suitable for the activity.
- Follow all warnings marked on the instrument.
- Follow all biological safety instructions
- Follow all safety, warnings, cautions, and notes

NOTE: Reference Chapter 12 for Install, PM, Service checklist and complete required test depending on the service type.

Perform Final Test After a Service Call

1. Turn on the LX200 instrument. Observe that the following startup actions take place:
 - All Modules Visually Inspected
 - Blue light illuminates
 - Fan turns on
 - Syringe Pump initializes
 - Compressor turns on
 - Observe drip to waste
2. Run a Self-Test and confirm that it passes.
3. Check/Adjust Flow Rate to within specification (10.2-11.4 mL / 2 minutes).
4. Verify Probe travels up and down smoothly
5. Perform a Probe Height Adjustment.
6. In the service software, set the XY heater temperature target to 55°C, and verify the heaters reach the target temperature.

NOTE: Allow up to 13 minutes for the heaters to reach the target temperature.

NOTE: No heater block required for heater verification

7. Run repeating Well Test and ensure it Completes w/o Error
8. Complete XY well alignment
9. Verify SD Fills and Stops at Full Level
10. Perform Calibration & Verification (Reference Calibrate and Verify the System section for instructions)
11. Obtain a Support Utility File, Calibration/Verification report, or a picture of Calibration/Verification report and attach to work order.

NOTE: Support Utility File is preferred but if it can't be obtained minimum requirement is attached calibration/verification report (Calibration/verification report can be PDF or Image).

NOTE: If you removed the analyzer cover or bezel for the final test, ensure you replace the cover and bezel.

Calibrate and Verify the System

Calibrator xMAP® microspheres are used to normalize the settings for the reporter channel, both classification channels, and the doublet discriminator channel. Control xMAP microspheres are used to verify the calibration and optical integrity for the system.

NOTE: Do not dilute calibrators or controls.

Each step in the calibration procedure usually takes less than one minute. If you are using any version of xPONENT® software, you must run xMAP controls after each calibration. Once calibrated, the calibration values remain until you calibrate again. You can track system calibration and verification results through the Calibration Trend report and the System Control Trend report.

Import CAL or VER Kits

1. Navigate to **Maintenance** page > **Lot Management** tab.
2. Click **Import Kit**. The **Import Calibration or Performance Kit** dialog box opens.
3. Select the kit you want to import, then click **Open**.

Run Calibration and Verification

Run the **Calibration/Performance Verification** routine as part of your weekly maintenance routine:

1. On the **Home** page, click **System Initialization** under **Daily Activities**.
2. Click the **Calibration Verification** button under **Automated Maintenance** options.
3. Add the appropriate reagents to the off-plate reservoir and the strip well, using the diagram in the **Reagents** pane of the **Auto Maint** tab to guide you.
4. Click **Run**.

Run the Performance Verification Routine

Run the Performance Verification routine as part of your daily startup routine.

1. On the **Home** page, click **System Initialization** under **Daily Activities**.
2. On the **Auto Maint** tab, click **Performance Verification**.
3. Add the appropriate reagents to the off-plate reservoirs and the strip well, using the diagram in the **Reagents** pane of the **Auto Maint** tab to guide you.
4. Click **Run**.

Delete CAL and VER Kit Information

1. Navigate to **Maintenance** page > **Lot Management** tab.
2. In the **Active Reagents** section, select the kit you want to delete from the **Calibration Kit** or **Performance Verification Kit** lists.
3. Click **Delete Kit**.

Chapter 8: Troubleshooting

Safety

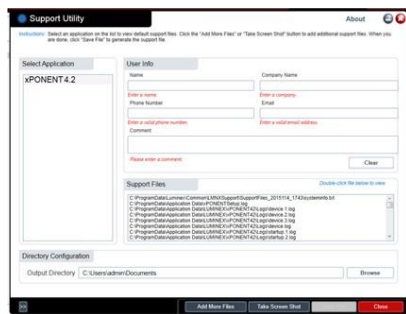
The non-observance of safety instructions may result in serious personal injuries and material damages.

- Always wear personal protective equipment suitable for the activity.
- Follow all warnings marked on the instrument.
- Follow all biological safety instructions
- Follow all safety, warnings, cautions, and notes

Use the information in this chapter to troubleshoot problems with the Luminex® 200™ RoHS system.

Generate and Send the Support Utility File

1. Power on the instrument and ensure that it is connected.
2. Open the xPONENT® software and navigate to the **Maintenance** Tab > **Support Utility** subtab.
 - a. If you want to include a batch file, select it and check **Include Batch Info**.
3. Click **Support**. The **Support Utility** launches.
4. Input the following information into their respective fields:
 - Enter your name in the **Name** field.
 - Enter your company name in the **Company Name** field.
 - Enter your phone number in the **Phone Number** field.
 - Enter your email address in the **Email** field.
 - In the **Comment** field, type a detailed description about the problem you are experiencing.



5. In the **Directory Configuration** section, verify the location where you want to store the file in the **Output Directory** field. To change the location, click **Browse**, then navigate to the new folder and click **OK**.
6. Once the above information is entered, click **Save File**. The saved file includes date and time information.
7. Send an email to support@luminexcorp.com and attach the **Support Utility File** to the email.

Luminex® 200™ RoHS General Troubleshooting

1. Power on the Luminex® 200™ RoHS analyzer.
 - Does the air compressor come on? If not, see *Figure 72*.
 - Does the syringe pump move? If not, see *Figure 76*.
2. Start xPONENT® software.
 - Is the sheath pressure between 6.0 and 9.0 PSI? If not, see *Figure 75*.
 - Is the air pressure between 6.0 and 9.0 PSI? If not, see *Figure 74*.

NOTE: You may need to run two **Prime** cycles to stabilize the sheath and air pressure.

3. Does Calibration pass? Reference *"Calibration and Verification the System"*.
4. Do the controls pass? Reference *"Calibration and Verification the System"*. If not, see *Figure 75*.

NOTE: If the answers to the above questions were "yes," the system should operate properly.

FIGURE 72. Luminex® 200™ RoHS Air Compressor Does Not Come On

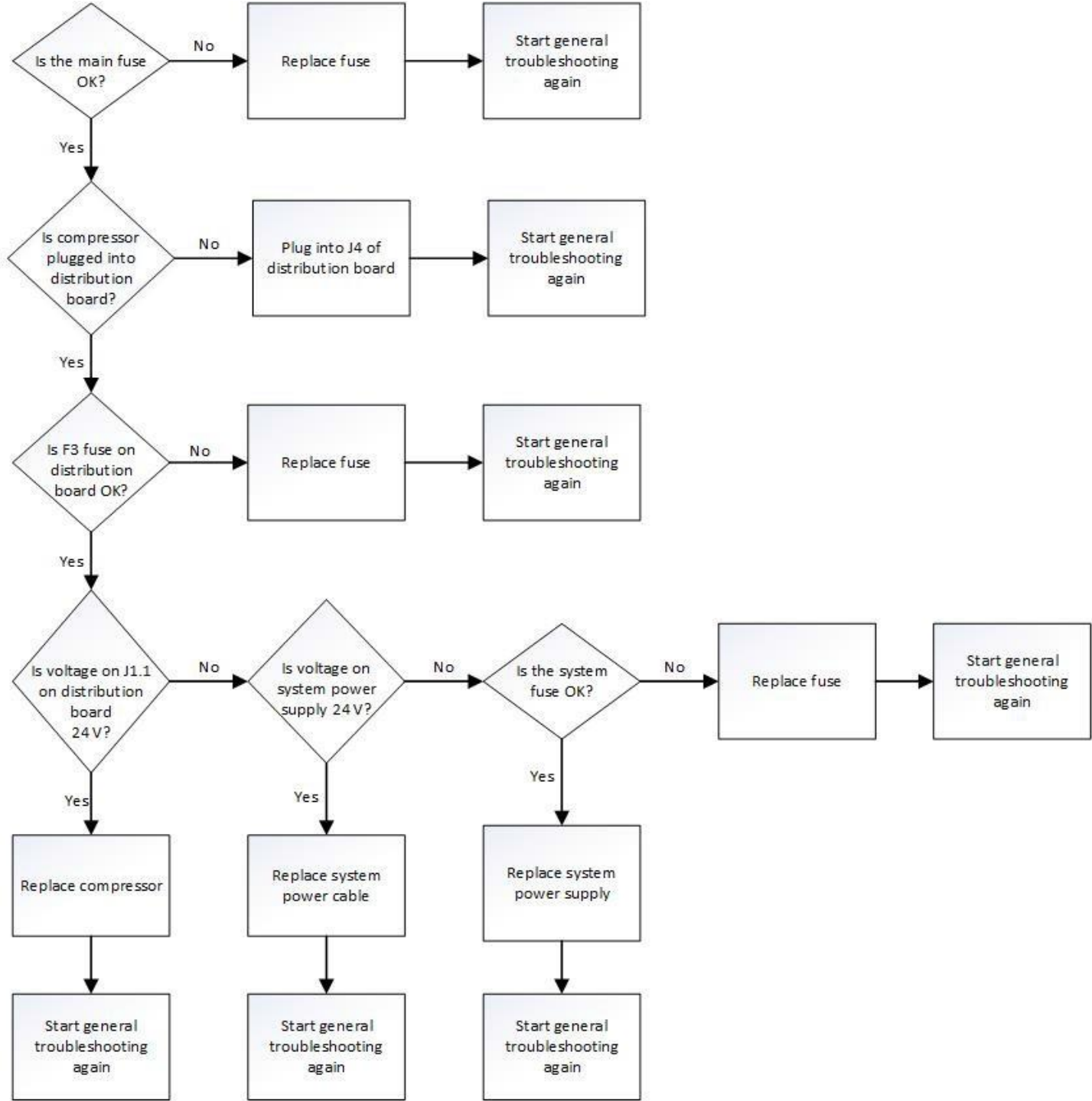


FIGURE 73. Luminex® 200™ RoHS Sheath Pressure Out of Range

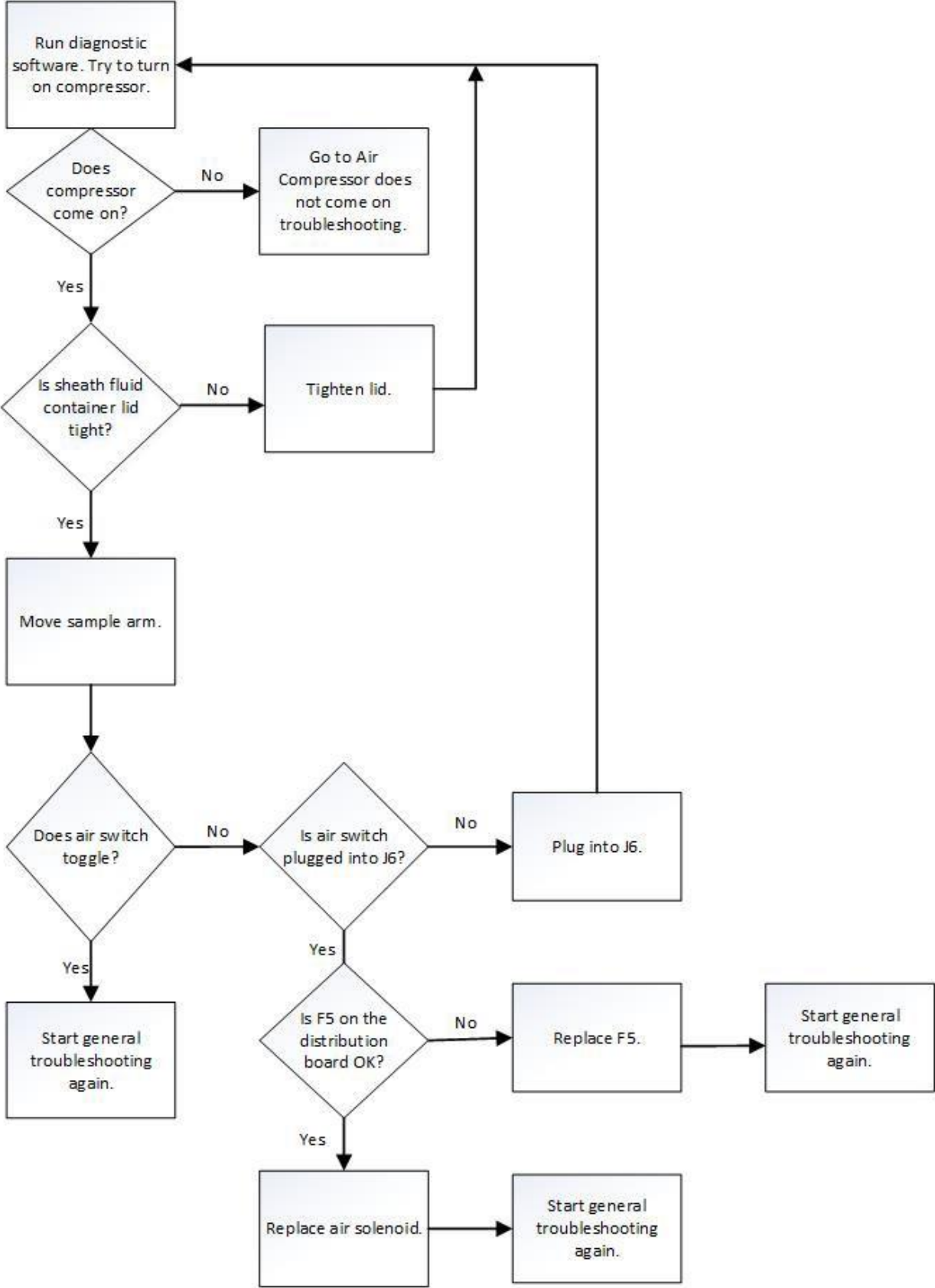


FIGURE 74. **Luminex® 200™ RoHS Analyzer Does Not Calibrate**

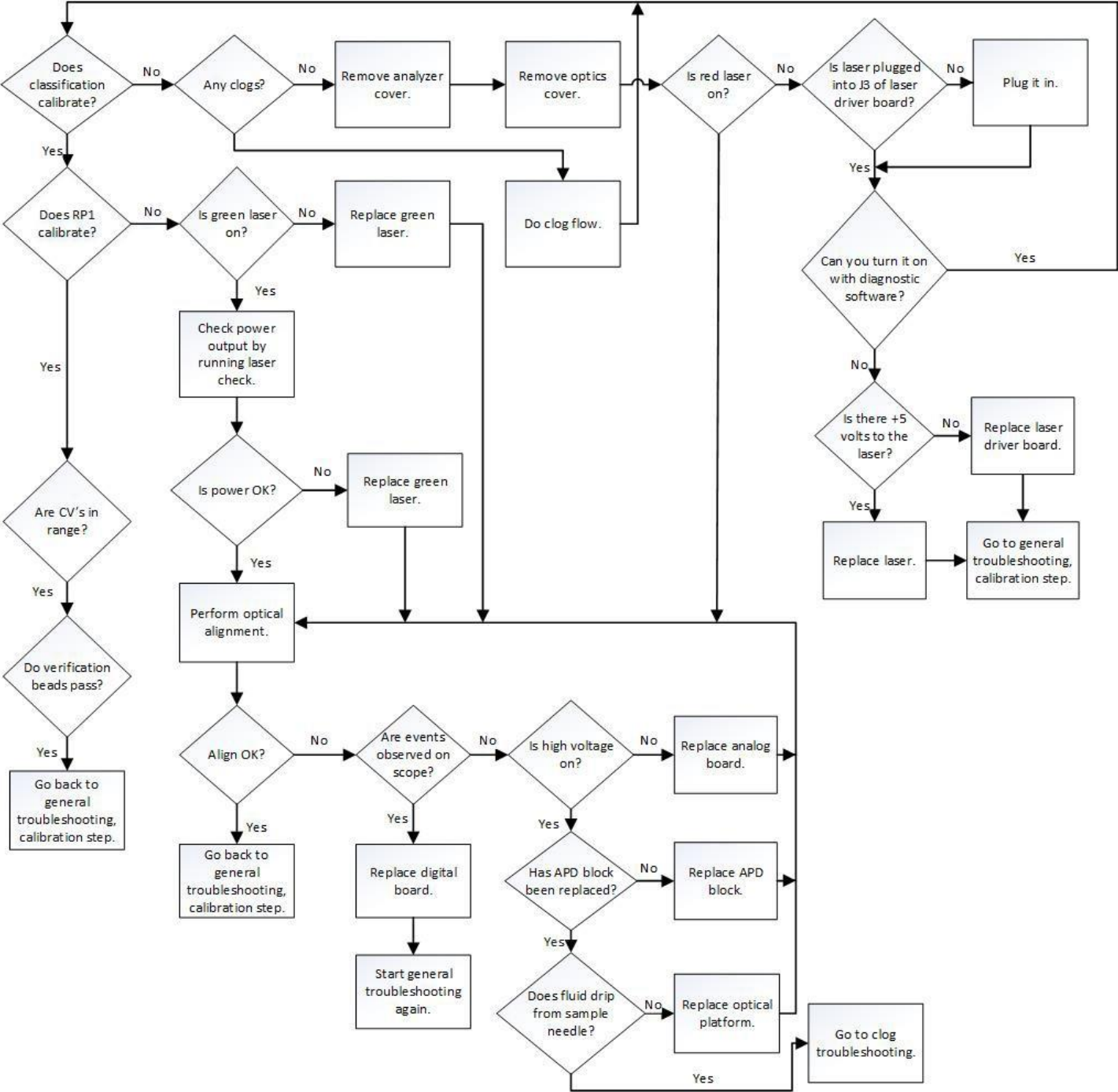


FIGURE 75. **Luminex® 200™ RoHS Optical Alignment Troubleshooting**

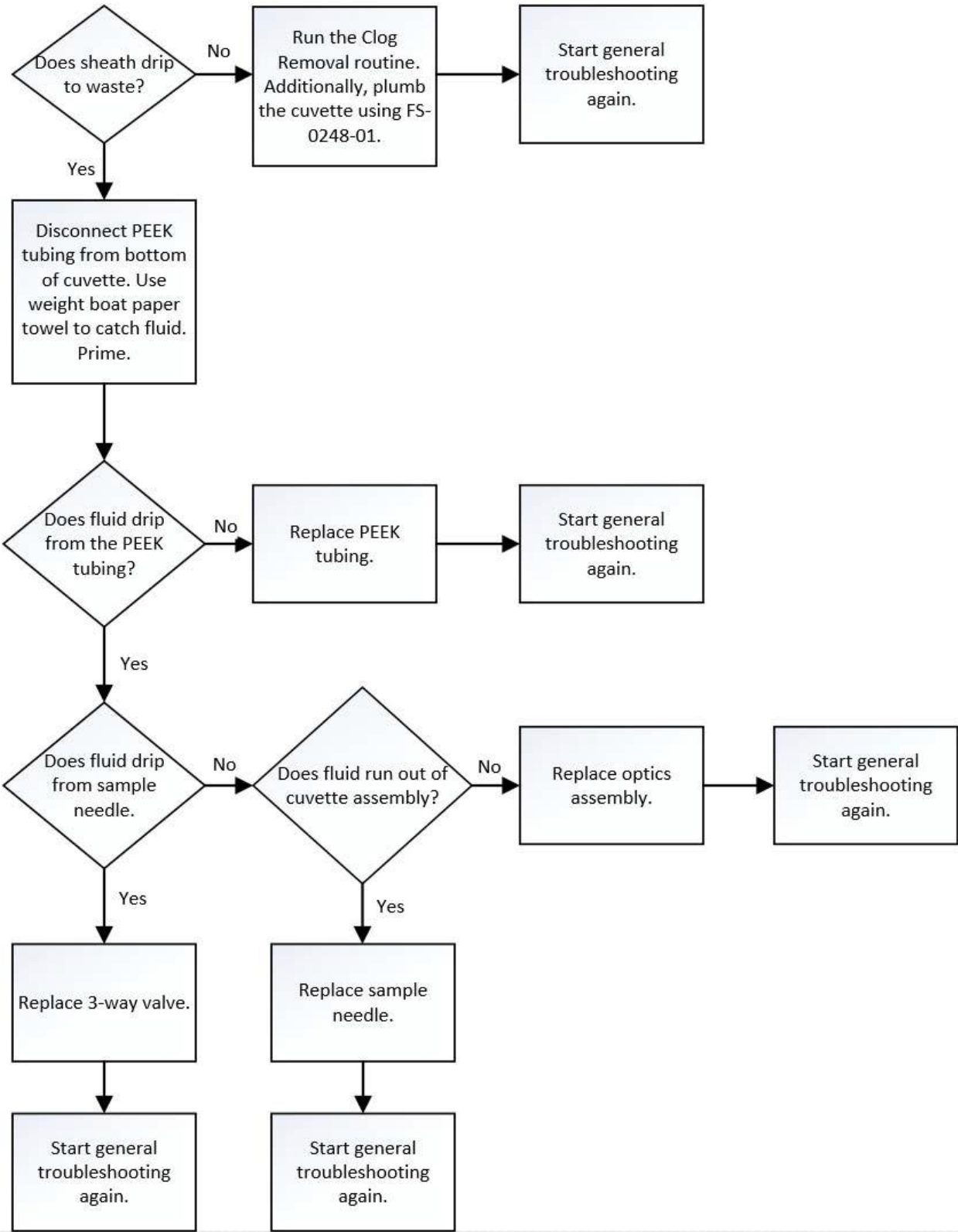


FIGURE 76. **Luminex® 200™ RoHS Syringe Pump Does Not Move**

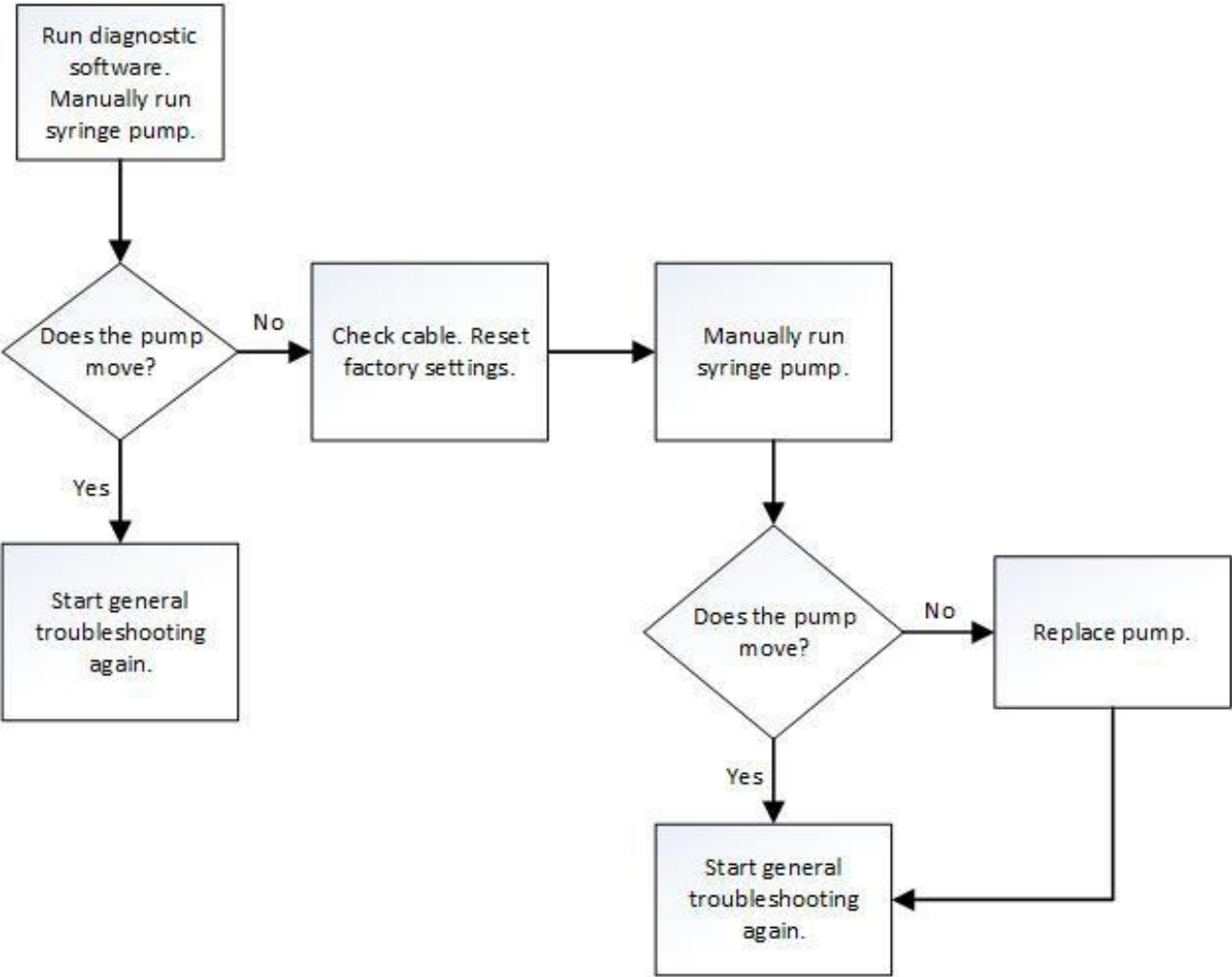
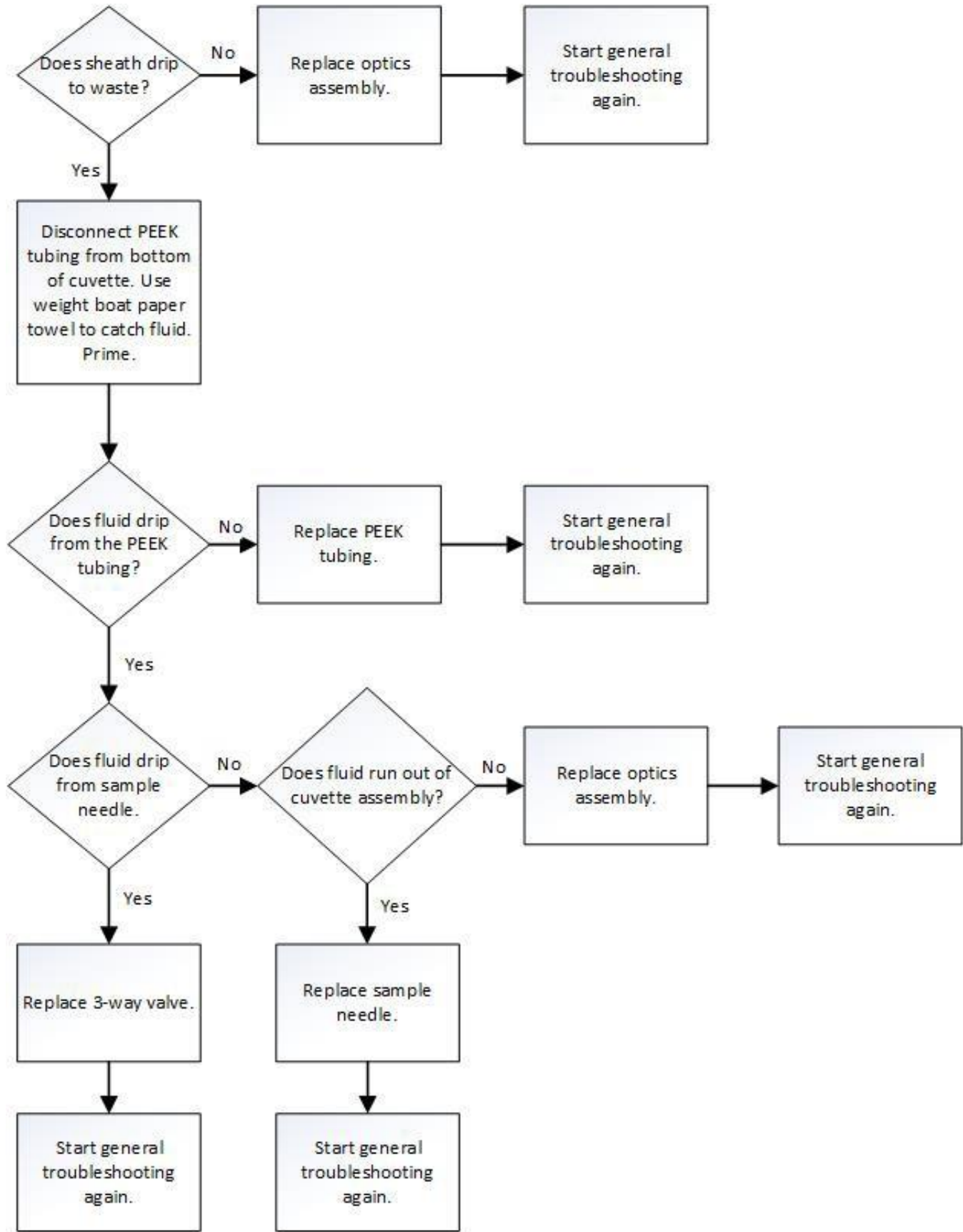


FIGURE 77. **Clog in Luminex® 200™ RoHS System**



Luminex® XYP™ Troubleshooting

To troubleshoot problems with the Luminex® XYP™:

1. Turn on the power.
 - Does the unit home? If not, see *Figure 78*.
 - Does the plate holder eject? If not, see *Figure 79*.
 - Does the plate holder retract? If not, see *Figure 80*.
 - Does the unit travel to the well specified? If not, see *Figure 81*.

NOTE: If the answers to all of the above questions were “yes,” the system is OK.

FIGURE 78. Luminex® XYP™ Does Not Home

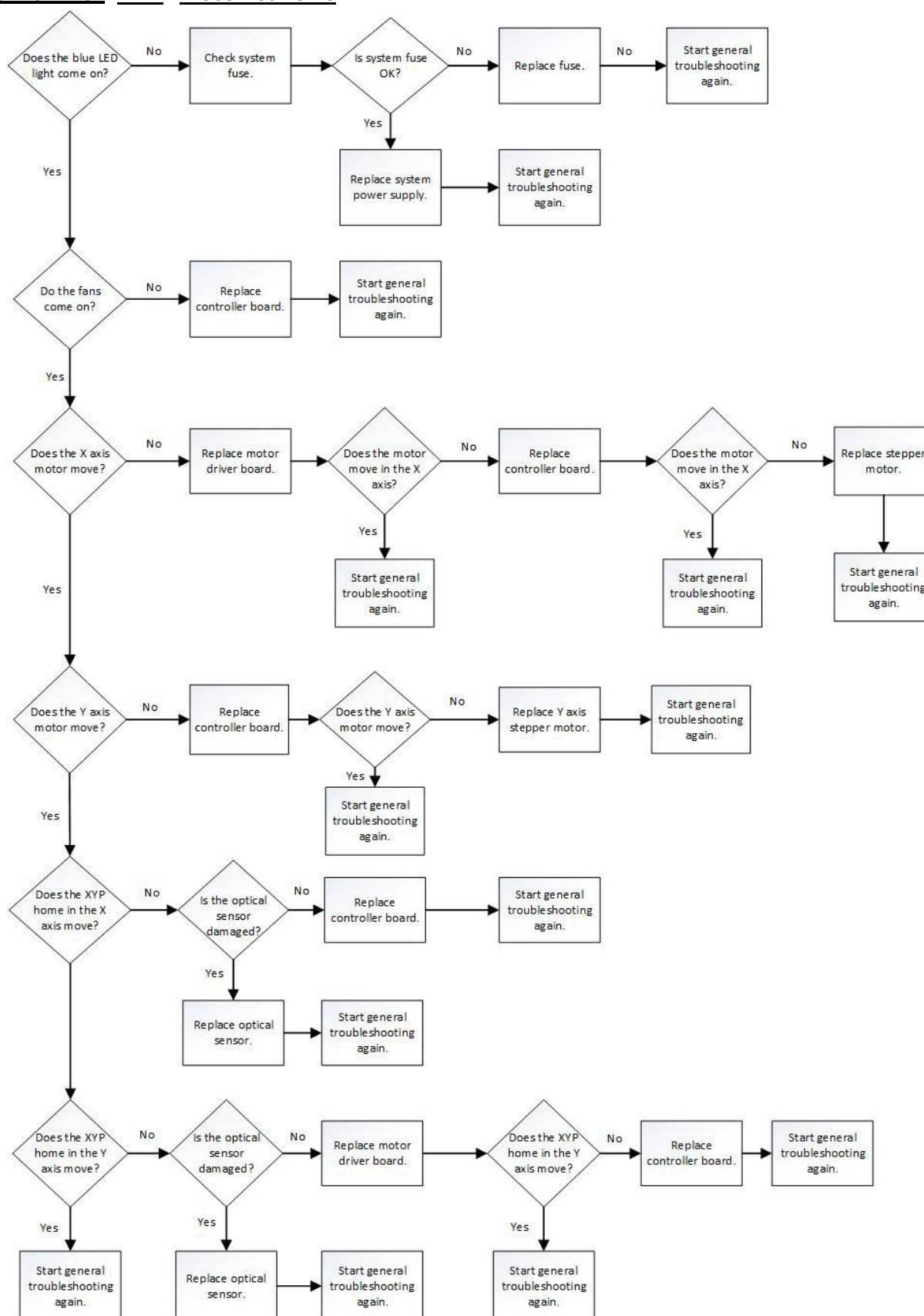


FIGURE 79. Luminex® XYP™ Plate Holder Does Not Eject

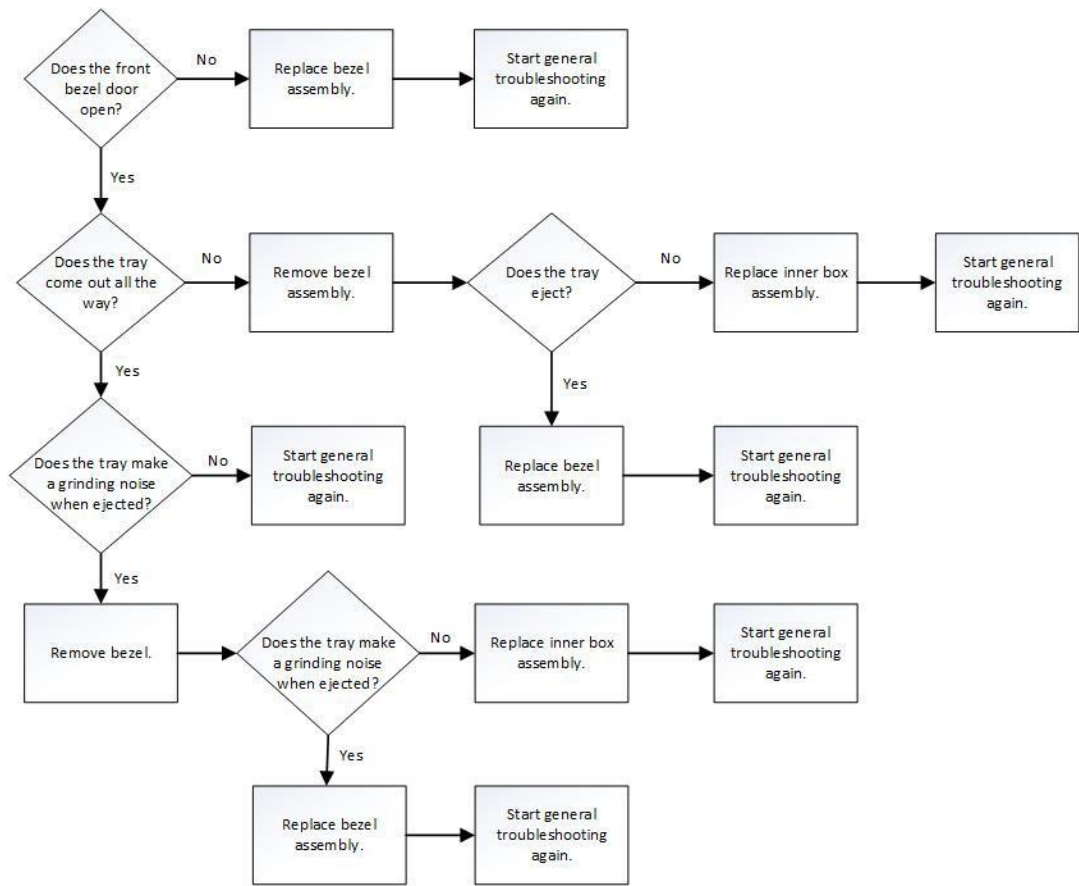


FIGURE 80. Luminex® XYP™ Plate Holder Does Not Retract

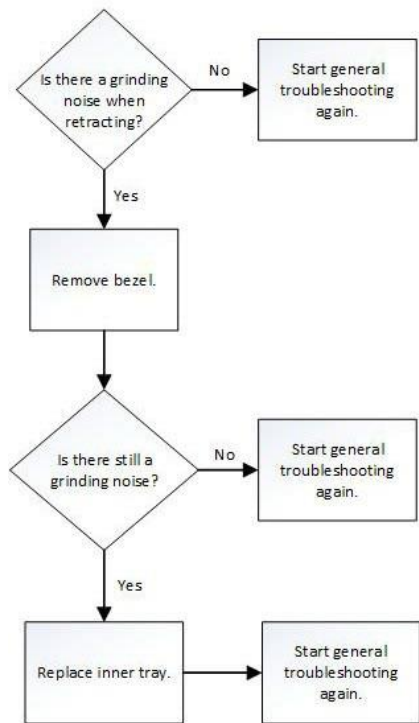
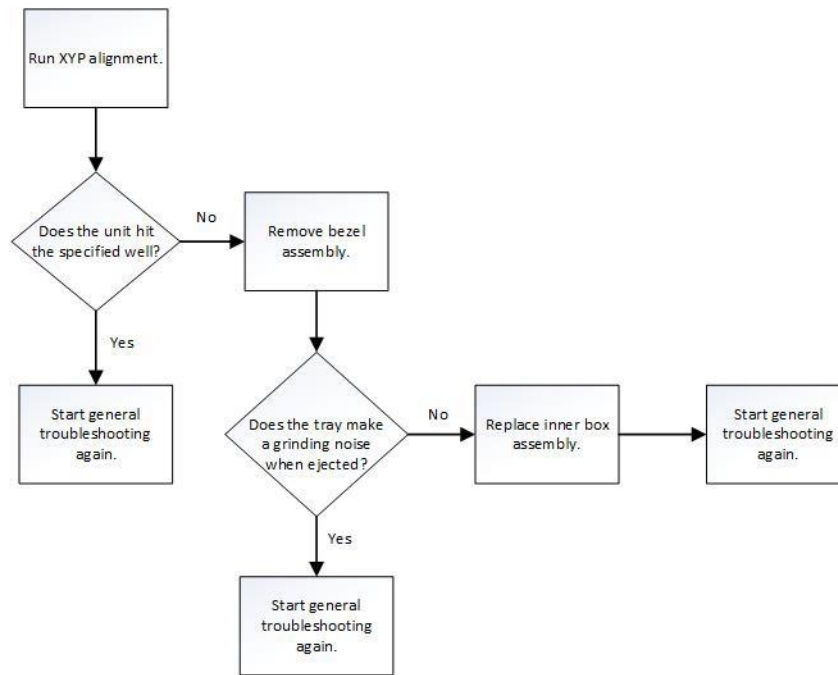


FIGURE 81. Luminex® XYP™ Does Not Travel to Correct Well

Luminex® 200™ RoHS Installation Troubleshooting

Luminex® 200™ RoHS Does Not Connect to LxMaint After New Driver Installation

1. Open the Device Manager.
2. Look for a device named **Lx100(COMx)** under **Ports** (where “x” is the COM number used by the Luminex® 200™ RoHS analyzer to communicate with the Luminex® XYP™).
3. If the device is not found in the **Ports** section but shows up as an LxDevice, the new driver was not installed correctly.
 - a. Reinstall the new driver while the device is still plugged in.
 - b. Reboot PC.
 - c. Check the Device Manager again to verify if the device shows up.

NOTE: If the device shows up as **Lx100(COMx)** but the “x” value is 10 or more, Luminex 200 RoHS will not connect to the Luminex XYP. This is because the driver was reinstalled and the device moved from one USB port to another too many times. Windows® increments the COM port number each time the device is plugged into a new USB port or when the driver is reinstalled.

Mitigation for COM Port Number Issue

1. Open the command prompt.
2. Type **regedit**. The registry editor opens.
3. Find the key **HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\COM Name Arbiter**.
4. Locate the value **ComDB** and modify it.
5. When the **Edit Binary Value** window opens, **replace** (do not add) the first two bytes with **0D 00**.
6. Close regedit.
7. Reboot the PC.
8. Reinstall the driver.
9. Make sure the **Lx100(COMx)** device shows up with a value for “x” that is less than 10.
10. Reboot the PC.
11. Wait for the PC to establish a connection to the XYP and detector on LxMaint.

Debug an Luminex® 200™ RoHS XYP Connection Issue

In some cases, when the Luminex® 200™ RoHS analyzer does not connect to xPONENT®, or when the Luminex® XYP™ does not connect even if the drivers, ports, and firmware are correct, the issue might be with the **LxService.ini** file. This file designates the device it connects to: either **LxXYP** or **LxSD** (sheath delivery). In LxMaint and LxFirmware, you can change this device.

Error condition 1: When, using **LxFirmware**, you switch from XYP to SD and the application crashes or is killed from Task Manager, the **LxService.ini** file will continue to point to LxSD. xPONENT will never connect to the analyzer, and LxMaint will never connect to XYP.

Error Condition 2: When, using **LxMaint**, you use one of the panels to switch the connection to SD from the default XYP and do not change the selection back to XYP. The **LxService.ini** file will continue to point to LxSD unless it is changed back. Even after rebooting, the analyzer will not connect to xPONENT, and XYP will not connect to LxMaint.

Solution: In both scenarios:

1. Open LxMaint and change a panel to **Sheath Delivery**.
2. Change the selection to **Default XYP**.
3. Wait for a connection, reboot everything, and verify the connections.

If LxMaint is not installed:

1. Open the file **c:\Programfiles (x86)\Luminex\LXR\DataFiles\Lxservice.ini** or **c:\Programfiles\Luminex\LXR\DataFiles\Lxservice.ini**.
2. Search for the string **LxSD.dll** and replace it with **LxXYP.dll**.
3. Save the file and restart the PC.

Updated Luminex® 200™ RoHS (3-Port Communications) Does Not Connect to LxMaint or xPONENT® Upon Initial Installation

1. Power on the PC and wait for the system to boot up.
2. Power on the analyzer.
3. Try unplugging the USB cable from the PC. Wait a moment, then plug it back in. This action will cause the operating system to enumerate the new USB connection. In most cases, this will allow the system to connect.

Updated Luminex® 200™ RoHS (3-Port Communications) Does Not Connect to LxMaint or xPONENT® After New Driver Installation

1. Open the Device Manager.
2. Look for a device named **Lx100 (COMx)** under **Ports**, where “x” is the COM number used by the Lx200 analyzer to communicate with the XYP.

NOTE: If the device is not found in the ports section but shows up as an LxDevice, the new driver was not installed properly.

3. Power off the analyzer.
4. Reinstall the new driver
5. Reboot the PC.
6. Check the Device Manager again to verify that the device displays properly.

NOTE: If the device shows up as **Lx100(COMx)** but the “x” value is 10 or more, the analyzer will not connect to XYP. This is because the driver was reinstalled and the device was moved from one USB port to another too many times. Windows® increments the COM port number each time the device is plugged into a new USB port or the driver is reinstalled.

Mitigation for Windows Driver Issue

If the new Luminex® Windows® driver is installed but the Device Manager shows that the older Luminex driver is being used, the driver can be updated by completing the following steps:

1. Go to **Add/Remove Programs** in the Control Panel and uninstall all Luminex drivers.

2. Restart the PC and the analyzer.
3. Run the new Luminex driver installer once the PC and analyzer are done restarting. Open the Device Manager and check for the name of the device.
 - a. If the name is **Lx100(COMx)** under **Ports**, where the com port is under COM10, you should be able to proceed.
 - b. If the device shows up as **LxDevice**, Windows® is still trying to use the old driver. Complete the following:
 - i. Right-click **LxDevice** in the Device Manager and choose **Update Driver Software**.
 - ii. Click **Browse my computer for driver software**.
 - iii. Click **Browse**.
 - iv. Navigate to the following path: **C:\program file (x86)\Luminex\LuminexWinDrivers\Free**.
 - v. Click **OK**.
 - vi. Select the **Include subfolders** checkbox.
 - vii. Click **Next**.
 - viii. Wait for the driver to be installed.
 - ix. Restart both the analyzer and PC.
 - x. Check the Device Manager to make sure that the reader shows up as **Lx100(COMx)** under **Ports**.
 - xi. Wait for 1 to 2 minutes and launch xPONENT.
 - c. If the above process does not work, re-image the PC and re-install the xPONENT software and drivers.

Mitigation for Windows Driver Installation on XP OS

1. Open the Device Manager.
2. Under **Ports**, look for the device named **Lx100(COMx)**, where “x” is the COM number used by the updated Lx200 analyzer to talk to the XYP.
 - a. If the device is not displayed in **Ports**, if the “x” value is **between 5 and 9**, power cycle the analyzer and launch Luminex application that is installed. Wait 2 to 3 minutes for the analyzer to connect. If it still does not connect, contact Luminex Technical Support.

NOTE: If the “x” value is **10 or more**, LXR will not connect to the XYP. This is because the driver was reinstalled too many times and the device was moved from one USB port to another several times. Contact Luminex Technical Support and advise them that **Mitigation for COM Port Issue >= 10** is required.
3. If the device is not found in the **Ports** section but shows up as an **LxDevice** under **Luminex Devices** or **Other Devices**, the new driver was not installed properly. Reinstall the driver by doing the following:
 - a. In the **Device Manager** tree, right-click **LX Device** and select **Update Driver**.
 - b. At **Hardware Update Wizard**, select **Yes, this time only**. Click **Next**.
 - c. Select **Install from a list or specific location (Advanced)**. Click **Next**.
 - d. Select **Search for the best driver in these locations**. Check only the box that says **Include the location in the search**. Click **Browse** to navigate to the path: **C:\ProgramFiles\Luminex\LuminexWinDrivers\freelx86_xp**.
 - e. Click **Next**.
 - f. A **Hardware Installation** dialog box will appear, stating that the software being installed for the Lx100 Device has not passed Windows® Logo testing. Click **Continue Anyway**.
 - g. In the **Completing the Hardware Update Wizard** dialog box, click **Finish**.
 - h. A **System Settings Change** dialog appears prompting you to restart your PC. Click **Yes**.
4. Power cycle the analyzer.
5. Reboot the PC.
6. Log into the PC as an administrator.
7. Launch the Luminex application that is installed. The analyzer should connect successfully.

Luminex SDS Troubleshooting

Audible Alarm and Fault LED

- If the alarm sounds and beeps quickly, the Sheath Delivery System has overfilled and needs draining back to the normal fill level. See *“Remove and Replace the PEEK, Sample, and Sample Loop Tubing”*.

NOTE: It is possible that the overfill alarm is due to condensation buildup in the sight-glass at or around the overfill level sensor. For condensation buildup, you will need to disconnect the SDS and flip it upside-down to allow sheath to fill the top of the sight-glass and remove the condensation. Once the condensation has been removed, re-connect the SDS to its original configuration.

- If the alarm sounds and beeps slowly, the Sheath Delivery System timed out when attempting to fill. This usually happens when the bulk container (sheath bottle or container) is empty and needs to be replaced. Alternately, the level in the container may have dropped below the filter on the end of the sheath line and need to be repositioned.

Sheath Pressure Out of the 5% Tolerance

1. Verify that the SDS installation procedure was conducted properly. (If not, install the SDS using *“Install the SD System”*).
2. Verify that the Air Pressure Regulator on the Analyzer is fully opened (clockwise). (If not, fully open the Air Pressure Regulator on the Analyzer).
3. Verify and update SDS firmware, if necessary.
4. Verify that the Air Cylinder (Bimba) is operating properly.
5. Verify operation of the Debubbler Valve and check its fuse.
6. Check the Sheath Valve for leaks.

Air/Sheath Pressure Reads Zero or Incorrect

1. Verify that the sheath bottle or container is plugged in and the cap is fastened securely.
2. Check that the metal cover on the Analog Board is not shorting out the heat sinks next to it.
3. Check the sheath bottle or SDS for a broken plastic fitting. (Unplug the sheath bottle or SDS from the Analyzer and see if air pressure rises).
4. Check the green quick-disconnect from the sheath bottle or SDS for a damaged or missing O-Ring. (Unplug the sheath bottle or SDS from the Analyzer and see if air pressure rises).
5. Verify proper operation of the Air Compressor and its fuse.
6. Verify that the Air Pressure Regulator on the Analyzer is fully opened (clockwise). (If not, fully open the Air Pressure Regulator on the Analyzer).
7. Verify that the Syringe Pump takes a full stroke during start up. (This allows the Air Compressor to turn on).
8. Check the yellow restrictor between the Air Compressor and Air Pressure Regulator for clogs.

Other Possible Luminex SDS Issues

- Sheath Pump failure
- Fuses
- Controller PCBA failure
- Contamination in the Reservoir (Floaties)

Drain the Luminex® SD™ Reservoir

If you need to ship the Luminex® SD™ system back to Luminex Corporation, drain the reservoir before you pack the system.

1. Perform a warmup to pressurize the system.
2. Leave the green air tubing connected between the Luminex® 200™ analyzer and the Luminex SD system.
3. On the Luminex SD front panel, disconnect the blue tubing from the intake labeled **Sheath Out** and disconnect the white tubing from the intake labeled **Sheath In**.
4. Insert the white tubing into the intake labeled **Sheath Out** and insert the blue tubing into the intake labeled **Sheath In**.
5. Turn the unit off and then on again.
6. Press the **Prime** button on the front panel of the Luminex SD system.
7. Sheath fluid will pump from the Sheath Delivery System reservoir into the 20 L Sheath Box.

Chapter 9: Field Definitions

Safety

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Always wear personal protective equipment suitable for the activity.
- Follow all warnings marked on the instrument.
- Follow all biological safety instructions
- Follow all safety, warnings, cautions, and notes

Field Definitions

TABLE 3. Field Definitions

Field Name	Field Value Description
Program, "value"<,CC>	The name of the Luminex application that created the OUTPUT.csv file. If the file is generated by a non-US operating system, the country code (in hex) is appended.
Build, "value"	The version of the Luminex application that created the OUTPUT.csv file.
Date, "date", "time"	The date & time that the OUTPUT.csv file was created. This field is not related to batch execution time. Note that the date and time values are separated into distinct, adjacent fields ("Date", "04/14/2003", "02:46:45 PM") to maintain compatibility with previous versions of software.
SN, "value"	The serial number of the Luminex 200 RoHS device with which the batch was executed.
Session, "value"	The name of the batch. The term "Session" was used here to maintain compatibility with previous versions of software. Limited to 30 characters.
Operator, "value"	The name contained in the Current User field on the General tab of the Options dialog in the IS software.
TemplateID, "value"	(Optional) The database ID that is unique to the template used to create the batch.
TemplateName, "value"	(Optional) The name of the template used to create the batch. Limited to 30 characters.
TemplateVersion, "value"	(Optional) The version of the template used to create the batch. Limited to 10 characters.
TemplateDescription, "value"	(Optional) The description of the template used to create the batch. Limited to 200 characters.
TemplateDevelopingCompany, "value"	(Optional) The name of the company that developed the template used to create the batch. Limited to 30 characters.
TemplateAuthor, "value"	(Optional) The name of the person who created the template used to create the batch. Limited to 30 characters.
SampleVolume, "value"	(Optional) The sample volume defined in the template for sample acquisitions. Units = microliters.
DDGate, "value"	(Optional) The DD gate defined in the template for sample acquisitions
SampleTimeout, "value"	(Optional) The sample timeout defined in the template for sample acquisitions. Units = seconds.
BatchAuthor, "value"	(Optional) The name of the person who created the batch. Limited to 30 characters.
BatchStartTime, "date time"	(Optional) The date & time that the batch was started.

Field Name	Field Value Description
BatchStopTime, "date time"	(Optional) The date & time that the batch was finished.
BatchDescription, "value"	(Optional) The batch description. Limited to 200 characters.
BatchComment, "value"	(Optional) Comment entered after the batch has run.
CALInfo:	(Optional) Indicates the start of the CAL1 & CAL2 machine calibration information logged just prior to and anytime during batch acquisition
CONInfo:	(Optional) Indicates the start of the CON1 & CON2 machine verification information logged just prior to and anytime during batch acquisition
AssayLotInfo:	(Optional) Indicates the start of the Lot information for any standards and/or controls associated with the batch
Samples,"num", MinEvents,"0"	Samples indicate the number of samples run in the batch. The MinEvents field is unused and will always be zero. Note that this field is adjacent to the Samples field ("Samples","28","Min Events","0"), rather than below it to maintain compatibility with previous versions of software.
Min Events	See the Samples field above.
Results	This field has no associated value. It is used to indicate the beginning of the statistical results section of the OUTPUT.csv file.
DataType:,"type"	This field is the name of the statistic represented in the Sample field (versus the Test data block immediately below this field). Possible values for DataType include: Median, Result, Count, Mean, %CV, Peak, Std Dev, Trimmed Count, Trimmed Mean, Trimmed %CV, Trimmed Peak, Trimmed Std Dev, and Avg Result. See <i>Table 4</i> for more information.
CRC Entry	(Optional) CRC indicator for the file data. Used to detect external changes to the file.

Statistics Definitions

Statistical calculations are performed for each test in each sample. The number of events to collect for each test in a sample is defined in the template from which the batch was created.

In any equation listed below, "N" indicates the number of events that were collected for an individual test in a single sample. The trimmed distribution represents the events that were collected for an individual test in a single sample with the lowest 5% and highest 5% of the data points removed to help eliminate outliers.

TABLE 4. Statistics Definitions

Statistic	Description
Median	The middle value in the distribution of data.
Result	The final test result based on a qualitative or quantitative analysis. This value could have units associated with it, as defined in the template. It may also indicate some error condition in the analysis, such as the sample was beyond the range of the curve fit, a divide by zero error occurred, etc. Some example include: - "Invalid" - the user invalidated this sample, or a single test within a sample - "<10 pg/mL" - the result could not be calculated because it fell outside the valid range of the curve fit - ">10000 pg/mL" - the result could not be calculated because it fell outside the valid range of the curve fit - "ERROR" - some mathematical error occurred, such as an MFI value that does not intersect the concentration curve. - N/A - a result is not applicable for this sample (i.e., a background sample)
Count	The number of data points in the distribution (N). The number of gated events that fell within the test's specified region.
Mean	(Optional) The sum of the data points in the distribution divided by the number of data points. $\text{Mean} = \sum x_i / N$
%CV	(Optional) The measure of relative dispersion within the distribution. $\%CV = 100 \times \text{Std Dev} / \text{Mean}$
Peak	(Optional) The value that is equal to the largest number of data points within the distribution. For example in data set {1,2,2,3,3,3,4,5}, 3 is the peak because it occurs the most number of times in the distribution list.
Std Dev	(Optional) The measure of dispersion within the distribution. $\text{Std Dev} = ((N \sum x_i^2 - \sum x_i)^2 / N (N-1))^{1/2}$
Trimmed Count	(Optional) The number of data points in the trimmed distribution (N_t).
Trimmed Mean	(Optional) The sum of the data points in the trimmed distribution divided by the number of data points. $\text{Trimmed Mean} = \sum x_i / N_t$
Trimmed %CV	(Optional) The measure of relative dispersion within the trimmed distribution. $\text{Trimmed \%CV} = 100 \times \text{Trimmed Std Dev} / \text{Trimmed Mean}$
Trimmed Peak	(Optional) The value that is equal to the largest number of data points within the trimmed distribution.
Trimmed Std Dev	(Optional) The measure of dispersion within the trimmed distribution. $\text{Trimmed Std Dev} = ((N_t \sum x_i^2 - \sum x_i)^2 / N_t (N_t - 1))^{1/2}$
Avg Result	(Optional) The average of any replicate samples' final test results based on a qualitative or quantitative analysis.

Statistics Column Definitions

The blocks of statistical data are displayed such that the first row of data represents the column headers and the following rows represent the samples that were acquired for the batch.

TABLE 5. Statistics Column Definitions

Column Name	Description
Location	The location of the sample in terms of the command list sequence (1, 2, 3,...), the well placement (A1, B1, C1, ...) or both (1(A1), 2(B1), 3(C1), ...)
Sample	The name of the sample as defined in the batch setup. Limited to 30 characters.
Test1, Test2, ... , TestN	The number of test columns following the Sample column will depend on the number of tests defined in the template used to create the batch. Each of the test columns will contain the test name for any given test. Therefore, if the template has three tests named Protein A , Protein B , and Protein C , these names will appear in the three test column headers in the OUTPUT.csv file. Limited to 30 characters.
Total Events	The number of events that fell within the defined DD gate and into one of the defined regions for a test in the batch. For example, if a template had three tests defined, and the batch had counts of 100, 102, and 105 for the tests, then the total count would be 307, even though more events may have been detected that did not fall within the DD gate or one of the defined regions.
Notes	Sample notes.

Chapter 10: Decontaminate the system

Safety

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Always wear personal protective equipment suitable for the activity.
- Follow all warnings marked on the instrument.
- Follow all biological safety instructions
- Follow all safety, warnings, cautions, and notes

Decontaminate the System



Some service procedures involve working with fluidics lines that may need to be cut. To avoid exposure to biohazard, perform this decontamination procedure before disconnecting or cutting any fluidics lines.

To decontaminate:

1. Wear appropriate biohazard safety attire.
2. Check the analyzer for leaks. If you see a leak, fix the leak before going on to the next step.
3. Use a 10% to 20% bleach solution to wipe down the outside of the analyzer, Luminex® XYP™, and Luminex® SD™. Be sure to wipe all surfaces, including the rear and bottom of each system component.
4. If the system uses an Luminex SD, connect the sheath bottle.
5. Replace the fluid in the sheath bottle with a solution of 10% to 20% household bleach and water.
6. Eject the XY carrier tray. Remove all specimens, disposables, reagents, and plates from the Luminex XYP.
7. Using a solution of 10% to 20% household bleach and water, sanitize the system using the **Sanitize** function. Follow this by washing twice with distilled water.
8. Disconnect the system from AC power by turning off the power switch on the rear of the analyzer, then unplugging the analyzer power cord from the wall source.
9. Disconnect the sheath fluid and waste containers.
10. Drain the sheath fluid and waste containers.
11. Rinse the waste container with 10% to 20% household bleach solution and drain.

Chapter 11: Fixture storage and shipping

Storage Conditions

All Luminex 200 field service tools and fixtures must be stored in a protective pouch or container in accordance with manufacturing instructions. If storage instructions are not available, please contact the manufacturer to ensure the items are stored properly.

Shipping

All Luminex 200 field service tools and fixtures must be packaged and shipped in accordance with manufacturing instructions. If shipping instructions are not available, please contact the manufacturer to ensure the items are packaged and shipped properly.

Chapter 12: Field Service checklists

Install checklist

LX200 Install
Initial Setup
All Modules Visually Inspected
Chassis Fan Start Up Indicator
Compressor Start Up Indicator
Blue Light Start Up Indicator
Syringe Pump Start Up Indicator
Prime & Drip to Waste Start Up Indicator
System Functions
Self-Test Results
System Specifications
Flow Rate 2 Min (10.2-11.4 mL)
Classification/Verification Results
Laser Alignment Required?
Calibration Results
Verification/Control Results
XYP Information
Probe travels up and down smoothly
Probe height adjusted with plate
Heater Reaches 55 C within 13 minutes
Repeating Well Test Completes w/o Error
Well Alignment (96 Well plate)
SD System
SD Fills and Stops at Full Level

PM Checklist

LX200 PM
Initial Setup
All Modules Visually Inspected
System Functions
Self-Test Results
System Specifications
Flow Rate 2 Min (10.2-11.4 mL)
Classification/Verification Results
Laser Alignment Required?
Calibration Results
Verification/Control Results
XYP Information
Probe travels up and down smoothly
Probe height adjusted with plate
Heater Reaches 55 C within 13 minutes
Repeating Well Test Completes w/o Error
Well Alignment (96 Well plate)
SD System
SD Fills and Stops at Full Level

Service checklist

LX200 Service
System Functions
Self-Test Results
System Specifications
Flow Rate 2 Min (10.2-11.4 mL)
Classification/Verification Results
Laser Alignment Required?
Calibration Results
Verification/Control Results

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